

Unit 0: Sneak Peak | 06.05.25 15:00-15:15

Welcome to the 2025 T/U(X)W Design Challenge!

2015 Cart Lab Design Challenge

Mapping the transnational hazardous waste trade

Interested in environmental justice? Have a little mapping experience? Use the first to look for patterns and stories in our dataset. Add a new map to your portfolio and compete for prizes in our one-day design event!

Over one million tons of hazardous waste are traded among the United States, Canada, and Mexico each year. What kinds of waste are traded, and where do they go? What happens to it upon arriving at its destination? What does it mean for communities that are major hazardous waste importers?

In this design challenge, UW-Madison students are invited to participate in a mapping competition to begin to answer these questions. Participants will be the first to look at a previously unreleased dataset assembled from two Freedom of Information Act Requests to the U.S. Environmental Protection Agency on U.S. hazardous waste imports.

To participate, you must:

- Be a student
- Register by **5pm February 6th** via the form at bit.ly/yfG5wW. Name details and the dataset will be sent after registration.
- Have taken GIS/370 or possess equivalent experience
- Work individually or in groups of two
- Work in any medium (e.g., hand drawn, print, or interactive), but maps must be visible on powerpoint slides or work in browser

Register by 5pm February 6th at bit.ly/yfG5wW. Questions? Contact Robert Roth at reroth@wisc.edu.

\$800
in prizes
(\$500 top prize)

Meals
provided to
participants

**Saturday
Feb. 14,
9am-7pm**

Cartography
Lab, Science Hall
M390, 550 N.
Park St



DesignChallenge2020

SPEND THE DAY IN THE NEWSROOM

On **February 22, 9am-8pm** Geography students will convene in the UW Cartography Lab for the annual **Design Challenge**. If you are interested in an experience of mapping the news, please sign up by February 14 to participate. Also join our special guest, Lauren Tierney (Washington Post) for her Yi-Fu talk on Friday February 21.

expert guest:
Lauren Tierney
Graphics reporter and cartographer
Washington Post



Any questions should be directed to
Bob Roth (reroth@wisc.edu) or Tanya
Buckingham (t Buckingham@wisc.edu).
Please sign up so we know how
much food to order!
bit.ly/DesignChallenge2020

Design Challenge 2023

visualizing the evolution of the Great Plains

February 25 110am - 8pm | UW Cart Lab, M390 Science Hall



This year, we'll be collaborating with Geography Prof. Joe Mason on an NSF-funded project to tell visual stories about how landscapes and ecosystems are responding to a changing climate. You will gain experience collaborating on interdisciplinary teams of cartographers and Earth systems geographers to create compelling explanations of landscape evolution! The event will give you the chance to explore new terrain representation and hydrological flow mapping techniques, combine realistic and artistic visual styles to promote public engagement, and add a proof-of-concept design piece to your portfolio. **Meals provided throughout the event to all participants.**

Sponsored by the National Science Foundation and the University of Wisconsin Cartography Lab. For questions, please contact Robert Roth at reroth@wisc.edu.

Register by Feb. 17 here:

tinyurl.com/23uaejms



SCAN ME

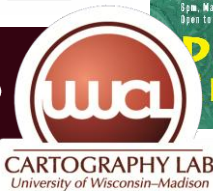
VISUALIZE SPATIO-TEMPORAL FOSSIL DATA

SHRIMP IN A TRANSFORMING ENVIRONMENT: RESEARCH
SIGN UP: bit.ly/DesignChallenge2023

Sat. Feb. 20
Design 8:30am-5pm
UW Cartography Lab
Meals provided
Presentations:
5pm-7pm Library
Open to the public

DESIGN CHALLENGE

What is a Design Challenge?



Design Challenge 2018

feminist cartographies and recidivism

February 17 | 9am-8pm | UW Cart Lab, Science Hall



Sponsored by: UW-Madison Cartography Lab, Geography, WIG, Geography and the Holtz Center for Science and Technology Studies

With Who? Catherine Dignazio (feminist data visualization expert) AND the United Way of Dane County, specifically the Journey Home Initiative. The Journey Home Initiative addresses recidivism or "return to prison rates" by providing housing, employment, support, and treatment programming for individuals looking for a fresh start.

What? This year's Design Challenge explores the geographies of recidivism in Madison through feminist mapping! Working with Catherine Dignazio, we'll learn about feminist visualization principles and then put theory into practice in an all-day mapping event. We'll break into teams and work with the Journey Home Initiative of the United Way to make maps that have a positive and productive impact on our community disrupting the cycle of recidivism in Madison.

Why? This workshop is for you if you're interested in maps, design, feminist theory, and social justice! You'll walk away with a portfolio piece, work on a real-world project, meet more people in the department and on campus, and develop feminist mapping practices in a supportive environment!

BREAKFAST, LUNCH, SNACKS, BEVERAGES, AND DINNER ARE PROVIDED!

RSVP: <https://tinyurl.com/DesignChallenge2018>
or email Meghan Kelly (mkelly22@wisc.edu)



Map provided by Catherine Dignazio, Robert D'Amico, Andrea DeGroot, and David Hays - Geography 212

2017 DESIGN CHALLENGE

MEANINGFUL WORK FOR YOUR CITY

hosted by the UW-Madison Cartography Lab
March 4, 2017 • 9am-8pm • UW-Madison Cartography Lab

Maps are powerful and persuasive visuals for tackling local needs. The 2017 Design Challenge will focus on mapping and community needs in Madison! Participants will create a comprehensive plan for Madison focused on community services (where are the services and who is being served), and create a focused map analysis of Bridge Lake Point Waunona Community Center (what does the center provide, and who's being served). The goal is to put our plans into action to make a difference within our community.

Join us in the Cart Lab if you have taken 395, 390, 377, or have expertise in city planning and you want to:

- Improve your design skills
- Work on a meaningful, real-world project
- Meet more people in the department



WATCH HERE FOR AN
ANNOUNCEMENT
ABOUT SPECIAL GUEST
CARTOGRAPHER

INFO & REGISTRATION: bit.ly/DesignChallenge2017

BREAKFAST, LUNCH, & DINNER WILL BE SERVED TO PARTICIPANTS

Design Challenge

Transforming Justice

Pre-Workshop Fieldtrip to Milwaukee
Saturday, February 9 | 9am-5:30pm

Kelly Lytle Hernandez, Yi-Fu Tuan Lecture
Friday, February 22 | 3:30-4:30pm

Transforming Justice Mapping Workshop
Saturday, March 2 | 9am-8pm

Join us in collaboration with Jenna Loyd, Kelly Lytle Hernandez, and Transforming Justice for the fifth annual Design Challenge in the UW-Cart Lab! We will explore everyday experiences of policing in Milwaukee. Drawing on video documentaries and workshop transcripts, we will explore how these experiences build on histories of segregation and how they continue to shape divisions in the city, access to shared spaces, and well-being.

For more information: <https://tinyurl.com/TransformingJusticeDC2019>

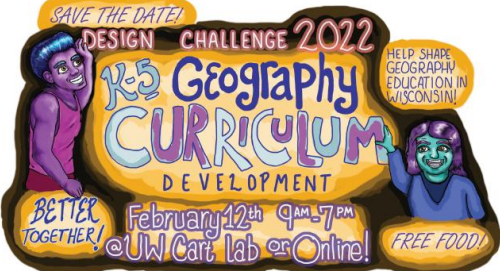
Sponsored by... Afro-American Studies, Center for Humanities, Chicana@ and Latina@ Studies, Department of History, Geography Department, Havens Center, Holtz Center, IACIS, Morgridge Center, and UW Cart Lab.

Illustration by Carmen Quintan



Save the Date!

THE UW CARTOGRAPHY LAB AND
THE CENTER FOR TEACHING, LEARNING & MENTORING PRESENT



COME FOR THE CURRICULUM DEVELOPMENT, STAY FOR THE FOOD!



A Design Challenge for Enhancing Cartographic Research, Education, & Outreach

Robert E. Roth ^{a,*}

^a *Cartography Lab, Department of Geography, University of Wisconsin–Madison – reroth@wisc.edu*

* Corresponding author

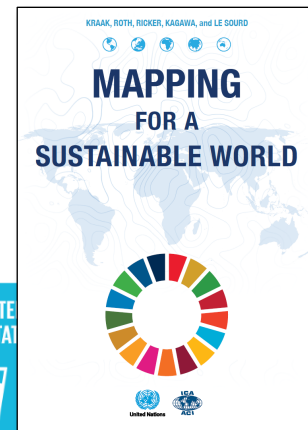
Keywords: design challenge, mapping workshop, pedagogy, community-engaged scholarship, cartography labs

Abstract:

The University of Wisconsin Cartography Lab (Cart Lab) *Design Challenge* is a day-long mapping workshop that brings together Cartography and Geography students, often in collaboration with campus or community partners, around a curated mapping theme (Moore et al. 2017, Bley et al. 2022). The Design Challenge is now an institution in the University of Wisconsin–Madison Geography Department, first running in 2015 and wrapping its 10th annual event this past February 2024. We seek to identify topics that are inclusive of the diverse breadth of expertise and interests in the Geography Department, oscillating between human/critical and environmental/geophysical case studies and between research versus community partners (Table 1). In total, the ten Design Challenges have involved 68 organizers and partners and participation from 198 students from UW-Madison undergraduate and graduate Cartography and Geography programs. Accordingly, there have been numerous contributors and individual experiences across the ten Design Challenges, and in this presentation, I reflect on the strategic role of the Design Challenge for expanding cartographic research, education, and outreach at UW–Madison from my perspective as the Cart Lab Director.

2025 Prompt:

SUSTAINABLE DEVELOPMENT GOALS



Kraak, Roth, Ricker Kagawa, Le Sourd (2020)

Mapping for a Sustainable World

Why Participate in a Design Challenge?

- ✓ Experience a real-world day in the life of a cartographer, and test your skills along the way to self-assess knowledge gaps



UNCERTAINTY / PATTERNS / INSIGHTS

CLARITY / FOCUS



RESEARCH

CONCEPT PROTOTYPE

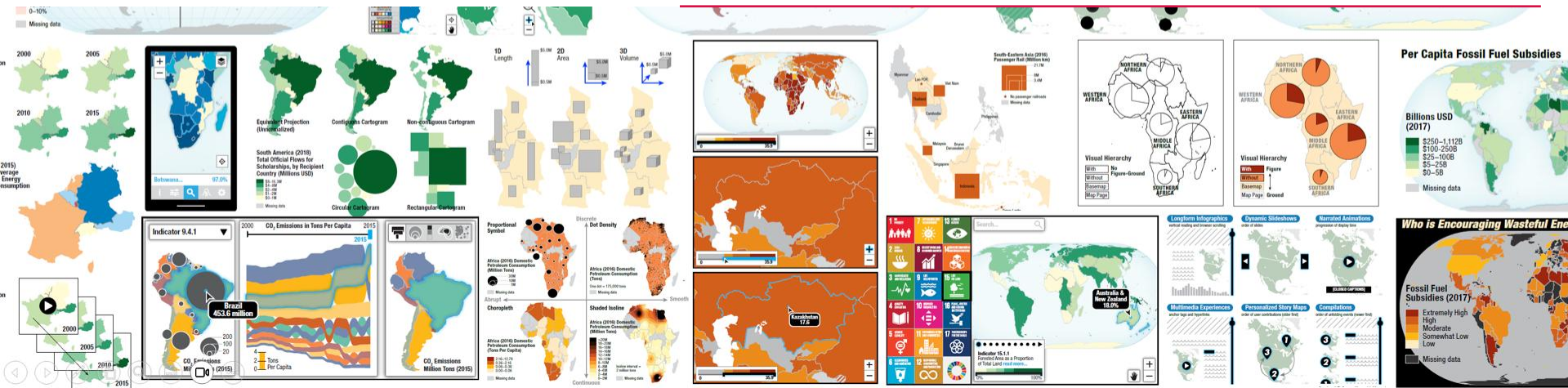
DESIGN

Why Participate in a Design Challenge?

- ✓ Experience a real-world day in the life of a cartographer, and test your skills along the way to self-assess knowledge gaps
- ✓ Learn about many of the most pressing challenges facing our world and how maps can help understand and solve them



Create a collective atlas of the SDGs telling stories of local & regional inequities from the perspectives of your lived experiences





Why Participate in a Design Challenge?

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- ✓ Learn about many of the most pressing challenges facing our world and how maps can help understand and solve them
- ✓ Create an original, professional-quality map as part of a collective atlas to boost your design portfolios



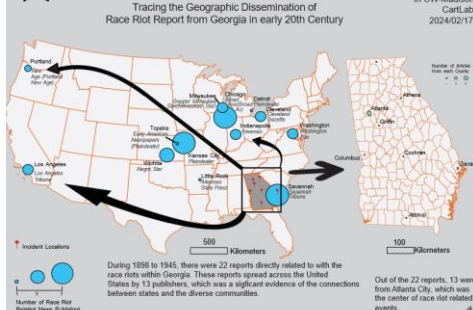
UW Cart Lab
Design Challenge 2024



The Sites of Three Reactionary Bombing Attacks and the Persistent Struggle for Black Freedom

[illegible]

Tracing the Geographic Dissemination of Riot Report from Georgia in early 20th Century



During 1898 to 1945, there were 22 reports directly related to with the race riots within Georgia. These reports spread across the United States by 13 publishers, which was a significant evidence of the connection between states and the diverse communities.

Out of the 22 reports, 13 were from Atlanta City, which was the center of communist-related

by Atlas Guo

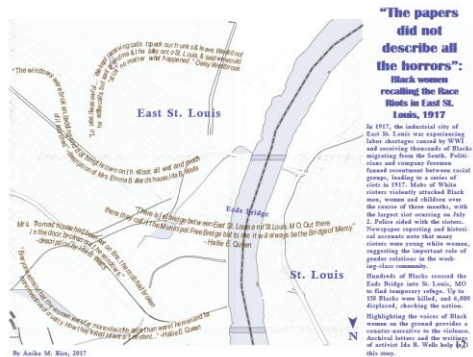
(UW Cartography Lab Design Challenge 2022)



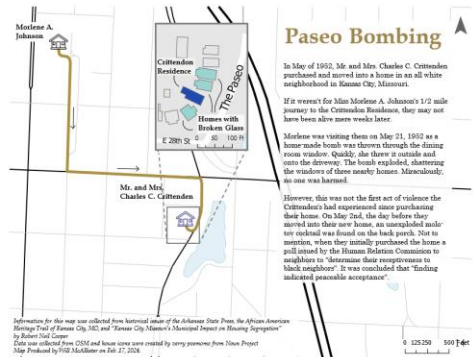
Harry T. Moore



Louis, 1917



.....



"...the dark and bloody

record of the South shows 728 Afro-Americans lynched during the past eight years"

In writing the pamphlet *Southern Horrors: Lynch Law in All Its Phases*, anti-lynching activist and journalist Ida B. Wells was influenced by the she-

number of incidents of white mob violence against black communities in the South and the pervasive stereotype of black men in the m

However, the pamphlet
not have been possi-
out his personal ex-
with white mob vio-
Memphis in 1892
and labor of black v

Lynching at the Curve
On March 9, 1992, there was a lynching in this same city of one of the best specimens of young since-the-war Afro-American manhood. They were peaceful, law-abiding citizens and energetic business men. They owned a flourishing grocery business in a thickly populated suburb of Memphis.

Three of these men, the president, the manager and clerk of the grocery — 'the leaders of the conspiracy' — were secretly taken from jail and lynched in a shockingly brutal manner. 'The Negroes are getting to be independent,' they say, 'we must teach them a lesson.'

— Ida B Wells

The Curve
"People's Grocery Store"

2024

"To the Afro-American women of New York and Brooklyn, whose race love, earnest zeal and unselfish effort at Lyric Hall, in the City of New York, on the night of October 5, 1892—made possible its publication, this pamphlet is gratefully dedicated by the author.

The Afro-American is not a bestial race. If this work can contribute in any way toward proving this, and at

prompt:

2024 Prompt:

Why Participate in a Design Challenge?

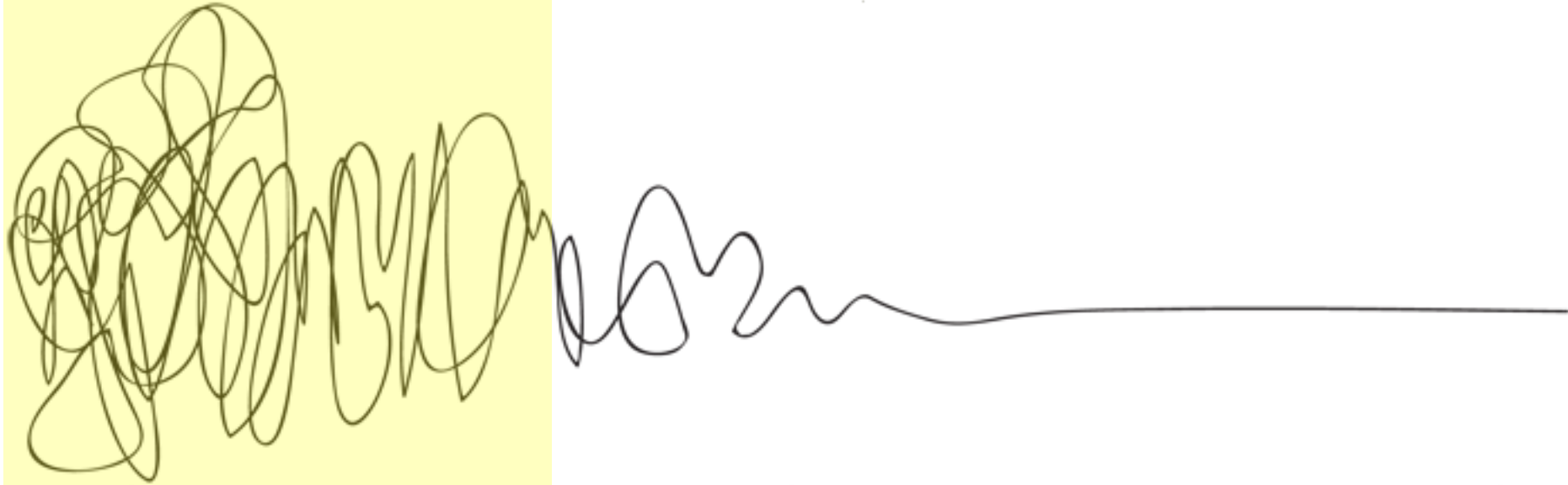
- ✓ Experience a real-world day in the life of a cartographer, and test your skills along the way to self-assess knowledge gaps
- ✓ Learn about many of the most pressing challenges facing our world and how maps can help understand and solve them
- ✓ Create an original, professional-quality map as part of a collective atlas to boost your design portfolios
- ✓ Provide constructive feedback and learn from your peers' mistakes and breakthroughs



Monday

UNCERTAINTY / PATTERNS / INSIGHTS

CLARITY / FOCUS



RESEARCH

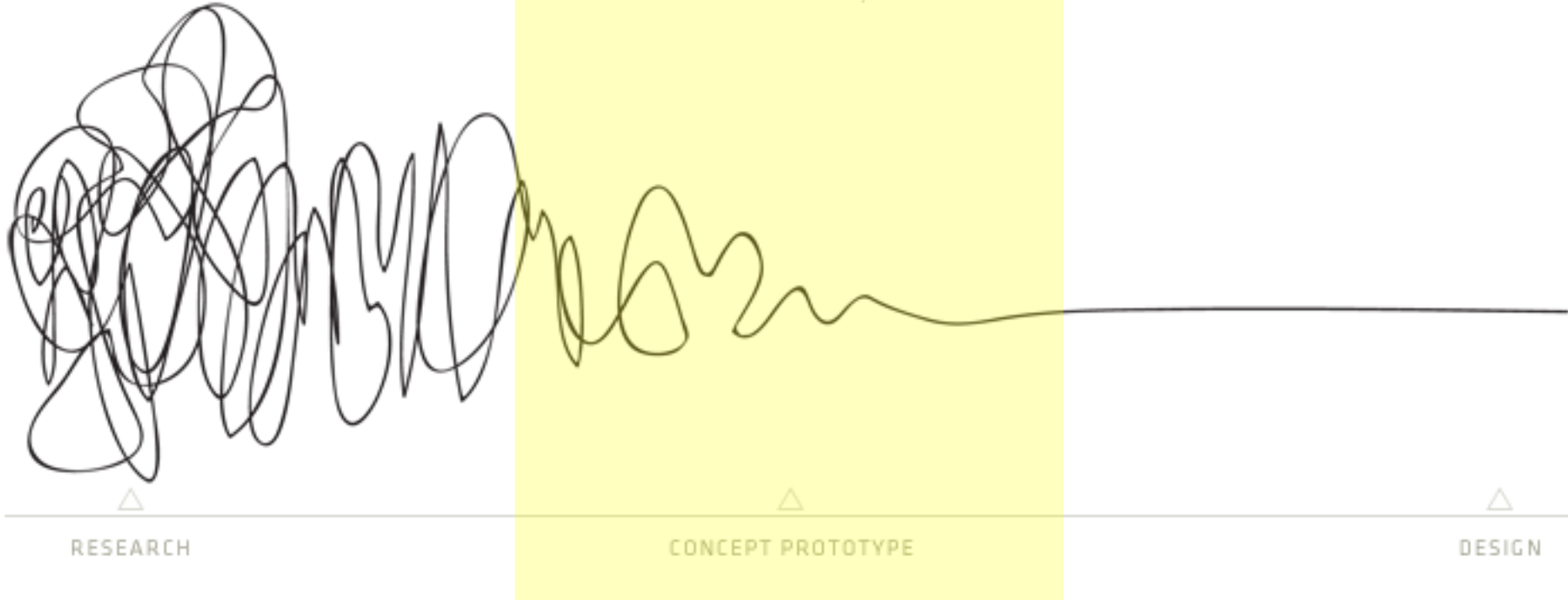
CONCEPT PROTOTYPE

DESIGN

Tuesday-Wednesday

UNCERTAINTY / PATTERNS / INSIGHTS

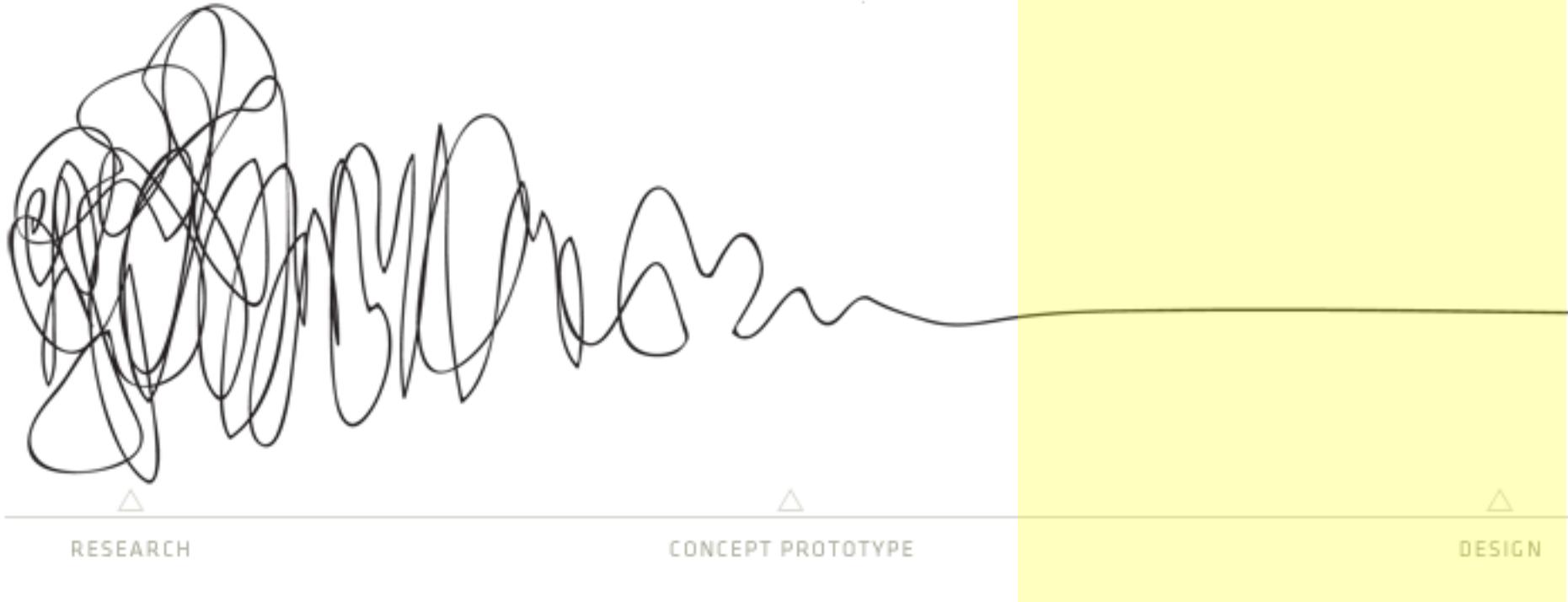
CLARITY / FOCUS



Thursday-Friday

UNCERTAINTY / PATTERNS / INSIGHTS

CLARITY / FOCUS



RESEARCH

CONCEPT PROTOTYPE

DESIGN

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- ✓ Create an original, professional-quality map as part of a collective atlas to boost your design portfolios
- ✓ Provide constructive feedback and learn from your peers' mistakes and breakthroughs
- ✓ Contribute to a model ICA workshop on mapping for a sustainable world



GitHub - uwcartlab/Vienna-DC-2025

+

github.com/uwcartlab/Vienna-DC-2025/tree/main?tab=readme-ov-file

README

2025 SDG Design Challenge / T/U(X)W

Welcome to the 2025 Design Challenge!

The University of Wisconsin Cartography Lab ([Cart Lab](#)) Design Challenge is a day-long mapping workshop that brings together Cartography and Geography students, often in collaboration with campus or community partners, around a curated mapping theme. The Design Challenge is now an institution in the University of Wisconsin–Madison Geography Department, first running in 2015 and wrapping its 10th annual event this past February 2024. This year (officially #DC11) we are taking the Design Challenge international, combining efforts with the Technical University of Wien, the Erasmus Mundus MSc Cartography Programme, and the International Cartographic Association Commission on the User Experience (UX).

Specifically, we will be making a collective atlas of the [United Nations Sustainable Development Goals](#) ... *in less than one week!* The UN adopted the 2030 Agenda for Sustainable Development in 2015, an expansion from the prior 2000–2015 Millennium Development Goals, and defined a global indicator framework for collective transformation and action toward a more equitable and just world. At the time of this writing (October 2024), the SDG global indicator framework comprised 17 goals to address the most pressing problems facing our worlds, 169 targets that prescribe real and actionable outcomes for sustainability, and 231 unique indicators used to measure and monitor progress towards these targets, inform policy at global and local levels, and promote accountability across all stakeholders (<https://unstats.un.org/sdgs/>).

Prompt: SDG Atlas in a Week

Our SDG Atlas in a Week will include at least 17 two page spreads, one covering each of the SDGs plus some from front and back matter. Each spread will include one page for maps, charts, and clarifying annotation about the SDG and a second page providing a title and statement about what the maps are saying and what this means for our planet's future and our individual actions. You will work in a small team to develop one of the atlas spreads, but in addition to mapping SDG indicator data, we also want you to bring your own personal experiences and home locations into the map, telling a *compelling story* about your team's common ground around the SDG and the alternatives that are possible from your different, worldwide perspectives.

Packages

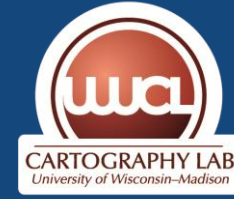
No packages published

Contributors 2

 **reeroth** Robert E. Roth

 **cartobaldrica** Gareth B

https://github.com/uwcartlab/Vienna-DC-2025



Unit 1: Introductions & Inspiration Talks | 12.05.25 10:00-11:15
Welcome to the 2025 T/U(X)W Design Challenge!

Organizers: T/U(X)W



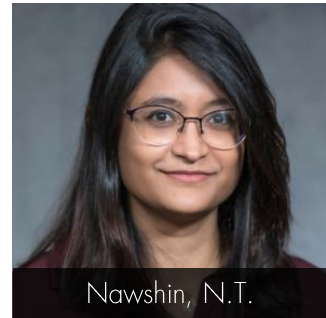
ICA UX

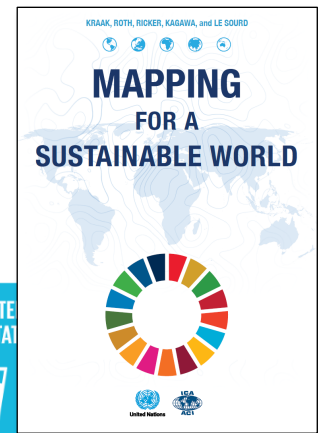


UN



UW Cart Lab



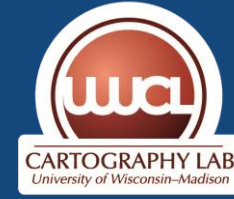


Create a collective atlas of the SDGs telling stories of local & regional inequities from the perspectives of your lived experiences

Unit 1 Objectives

1. Understand the history and purpose of the United Nations Sustainable Development Goals (SDGs)
2. Examine the SDGs as the most pressing geographic challenges facing our planet
3. Use the SDG global indicator framework and the M49 standards
4. List tips and tricks for mapping the SDG global indicator framework
5. Evaluate creative design possibilities for inclusion in a collective atlas



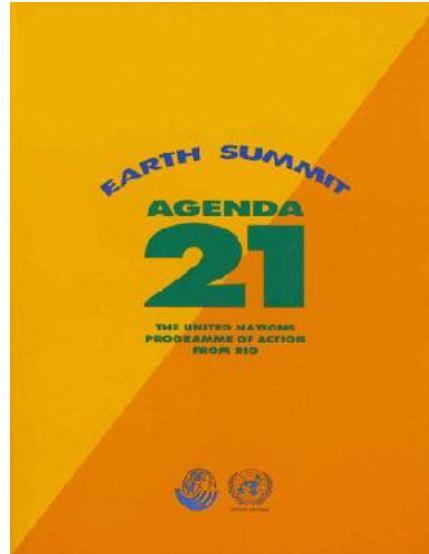


1.1: Lightning Talk 1 | Ayako Kagawa, United Nations

The United Nations Sustainable Development Goals

Where it all began.....

Rio Summit a.k.a. **Earth Summit** (United Nations Conference on Development and Environment - UNCED) in Rio de Janeiro,



... and so was created, Agenda 21

Millenium Development Goals (MDGs)



... and we did not reach our goals



SUSTAINABLE DEVELOPMENT GOALS

17 GOALS TO TRANSFORM OUR WORLD



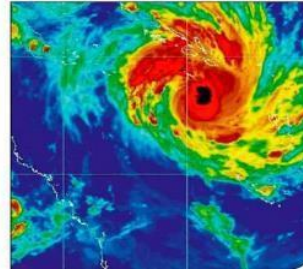
... makes for sobering reading...only 17 percent of the SDG targets are on track, nearly half are showing minimal or moderate progress, and progress over one third has stalled or even regressed.

[SDGs Report 2024](#)



Sustainable Development Goals

How can you measure and monitor sustainable development...



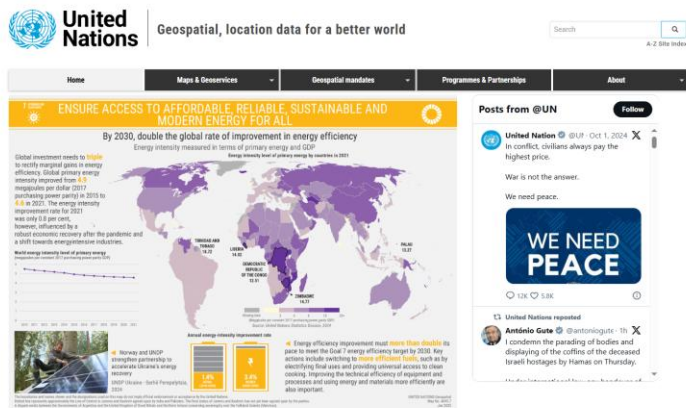
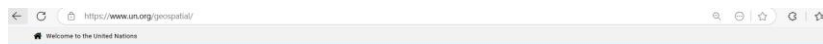
...without geography, place, and location



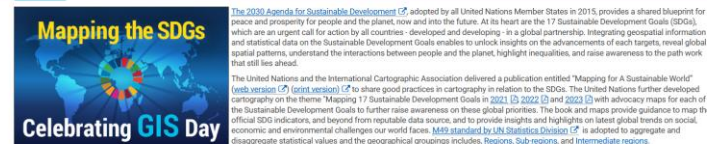
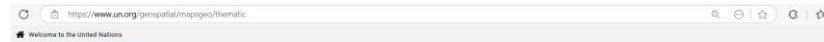
Learning by Doing:



- Mapping the 17 SDGs by the **UN Geospatial team** members
- Sharing lessons learned and best practices through mapping



Map on SDG07 Affordable and Clean Energy



Women in national parliaments 2000-2022

SDG Maps 2024

SDG01: No Poverty	SDG02: Zero Hunger	SDG03: Healthy Lives	SDG04: Quality Education
SDG05: Gender Equality	SDG06: Clean Water and Sanitation	SDG07: Affordable and Clean Energy	

SDG Maps 2023

SDG01: No Poverty	SDG02: Zero Hunger	SDG03: Healthy Lives	SDG04: Quality Education
SDG05: Gender Equality	SDG06: Clean Water and Sanitation	SDG07: Affordable and Clean Energy	SDG08: Decent Work and Economic Growth
SDG09: Industry, Innovation and Infrastructure	SDG10: Reduced Inequalities	SDG11: Sustainable Cities and Communities	SDG12: Responsible Consumption and Production
SDG13: Climate Action	SDG14: Life Below Water	SDG15: Life on Land	SDG16: Peace, Justice and Strong Institutions
SDG17: Partnerships			

UN Geospatial website:

<https://www.un.org/geospatial/>

<https://www.un.org/geospatial/mapsgeo/thematic>

MANY DEVELOPING COUNTRIES ARE FACING A DEBT CRISIS



THE SHARE OF EXPORTS FROM LDCs IN GLOBAL MERCHANDISE TRADE



HAS STAGNATED
AT AROUND **1%**

SINCE 2011



NET ODA REACHED **\$206 BILLION** IN 2022, UP 15.3% FROM 2021

MAINLY OWING TO SPENDING ON REFUGEES
IN DONOR COUNTRIES AND AID TO UKRAINE

NET ODA REACHED 0.37% OF GNI,
STILL BELOW THE TARGET OF 0.7%

2 IN 3 PEOPLE
USED THE INTERNET

IN 2022

259 MILLION MORE MALE THAN FEMALE USERS



ODA FUNDING FOR DATA



DROPPED
BY MORE THAN
20%
(2019-2020)

UN's SDGs website:
<https://sdgs.un.org/goals>

STRENGTHEN THE MEANS OF IMPLEMENTATION AND REVITALIZE THE GLOBAL PARTNERSHIP FOR SUSTAINABLE DEVELOPMENT

Enhance international support for implementing effective and targeted capacity-building in developing countries
Dollar value of financial and technical assistance (including through N-S, S-S and triangular cooperation) committed to developing countries

US\$ 1Trillion

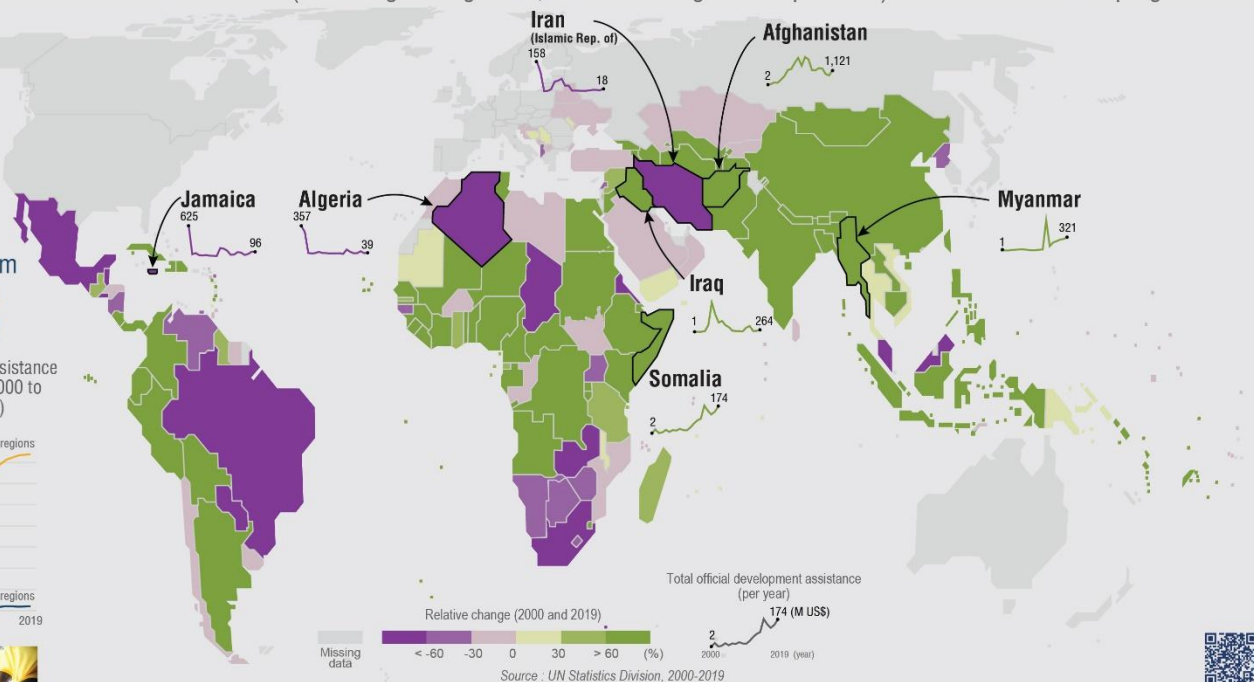
Foreign Direct Investment (FDI)
dropped by **40%** from
\$1.5 Trillion in 2019
to \$1Trillion in 2020

Total of official development assistance
for technical assistance from 2000 to
2019 worldwide (Billion of US\$)



Members of UN Police and the South Sudan Police Services taking part in a rope pulling competition, hold hands in camaraderie, during celebrations to mark the International Day of Peace in Juba.

UN Photo - Isaac Billy, 2011



Strong international partnerships are needed to ensure countries have the means towards the Sustainable Development Goals. North-South, South-South, and triangular cooperation are critical to capacity-building strategies. Values on the maps show indicator 17.9.1 and the relative change of financial and technical contribution in 2000 and in 2019 committed to developing countries, increase in contribution is represented in shades of green, while decrease is in shades of purple. The graph shows the exact trend of contribution throughout the period.

STRENGTHEN THE MEANS OF IMPLEMENTATION AND REVITALIZE THE GLOBAL PARTNERSHIP FOR SUSTAINABLE DEVELOPMENT



Enhance the use of enabling technology, in particular information and communications technology

Proportion of individuals using the Internet



▲ «Learning Crisis
Classroom» installation
by UNICEF

UN Photo-Mark Garten, 2022

Internet users per 100 inhabitants 2015-2021 (percentage)

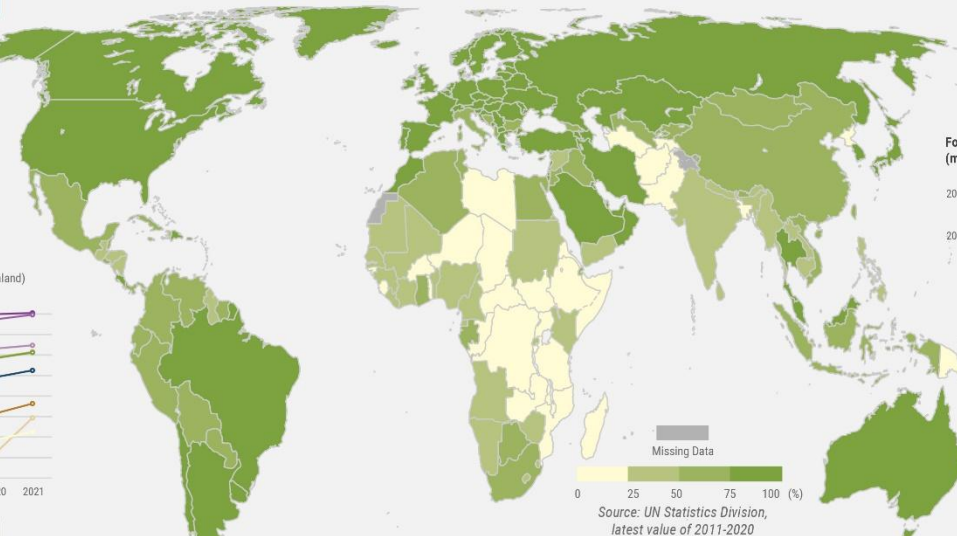


▲ A teenage girl prepares
to take part in a virtual
class at home in Ecuador

UNICEF-Santiago Arcos, 2021

63% of the global population were using the **internet in 2021** and the number of users in the Least Developed Countries increased by 20%. However, the **digital divide**, which is the gap between regions that have or do not have access to modern information and communications technology, still needs to be closed. **Connectivity to the internet** is key to **leave no one behind** and bring the world closer together and improve **digital cooperation**.

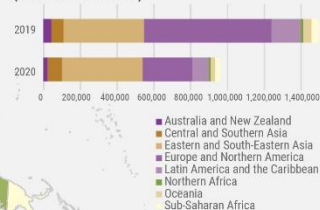
Internet users per 100 inhabitants (percentage)



▲ Next gen, 'lightning' fast
global communication network
on track for 2020 entry

ITU Photo, 2017

Foreign direct investment (FDI) inflows (millions of US dollars)



Source: UN Statistics Division



▼ ITU Photo, 2017



"Digital technologies sustain life, work, health and learning for billions of people. In the face of COVID-19, businesses, governments and the digital community have proven resilient and innovative, helping to protect lives and livelihoods. These challenging times have accelerated the transformation everywhere..."

Secretary-General of the United Nations

The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations.

Dotted line represents approximately the Line of Control in Jammu and Kashmir agreed upon by India and Pakistan. The final status of Jammu and Kashmir has not yet been agreed upon by the parties.

Final boundary between the Republic of Sudan and the Republic of South Sudan has not yet been determined. A dispute exists between the Governments of Argentina and the United Kingdom of Great Britain and Northern Ireland concerning sovereignty over the Falkland Islands (Malvinas).

UNITED NATIONS Geospatial

Map No. 4652.17

Dec 2022

STRENGTHEN THE MEANS OF IMPLEMENTATION AND REVITALIZE THE GLOBAL PARTNESHIP FOR SUSTAINABLE DEVELOPMENT



By 2030, build on existing initiatives to develop measurements of progress on sustainable development in developing countries

Total official development assistance (gross disbursement) for technical cooperation



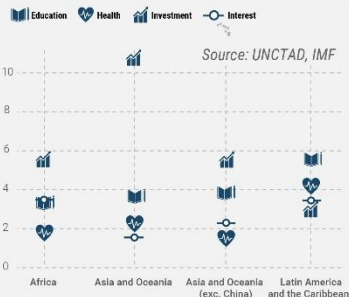
"We need the private sector, civil society, women, youth, scientists, academics, philanthropic foundations – all stakeholders across all fields and disciplines – to come forward, to lend their voices, and their hands, to ensure a transformative process that delivers both for people and planet."

H.E. Mr. Dennis Francis, 78th President of the General Assembly, 2023

▲ General Assembly meets on new partnership for Africa's development

UN Photo- Loey Felipe, 2023

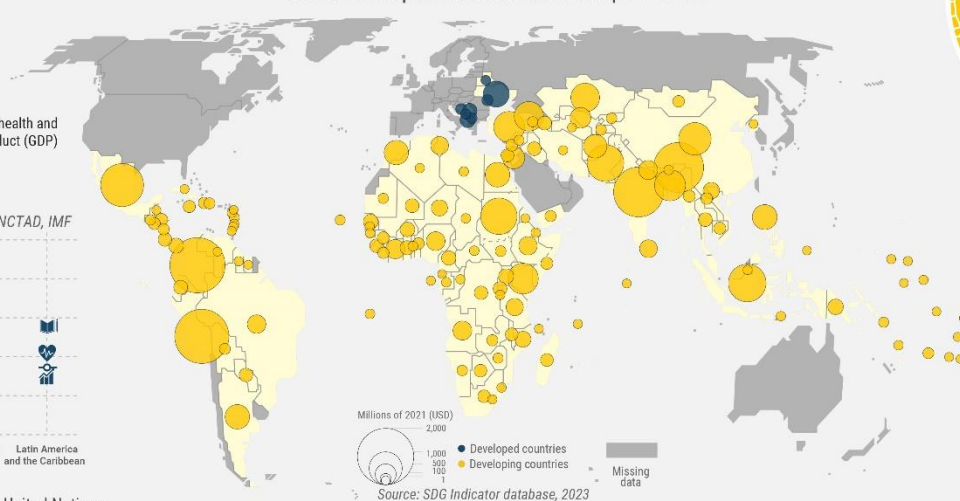
Public expenditure on net interest, education, health and investment as a share of Gross Domestic Product (GDP) in percentage (2019-2021)



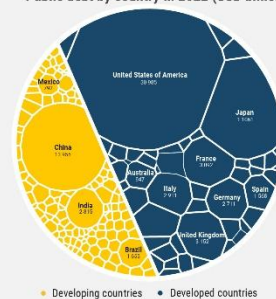
▲ United Nations development projects are helping countries such as Cabo Verde to transform agricultural practices to produce more sustainable products.

UN Photo-Mark Garten, 2023

Total official development assistance for technical cooperation in 2021



Public debt by country in 2022 (USD billion)



Source: UNCTAD, IMF

▲ The public debt by country reached a record of **USD 92 trillion**. Developing countries owe almost **30%** of the total debts, and **70%** attributes to selected countries.

▼ A woman recently impacted by conflict delivering a testimony during a pledging event on the United Nations Central Emergency Response Fund (CERF) for 2024

UN Photo-Mark Garten, 2023



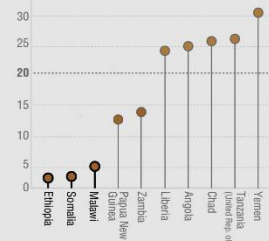
Increasing the availability of high-quality, timely and reliable **data disaggregated** by income, gender, race, ethnicity, migration, disability, **geographic location** and other characteristics relevant in national contexts is key to **leave no one behind** and bring the world closer together and improve the **global partnership** [§ 17.18, 48, 74(g), 76, and 83, A/RES/70/1]

By 2030, provide legal identity for all, including birth registration

Proportion of children under 5 years of age whose births have been registered with a civil authority (% of children under 5 years of age)

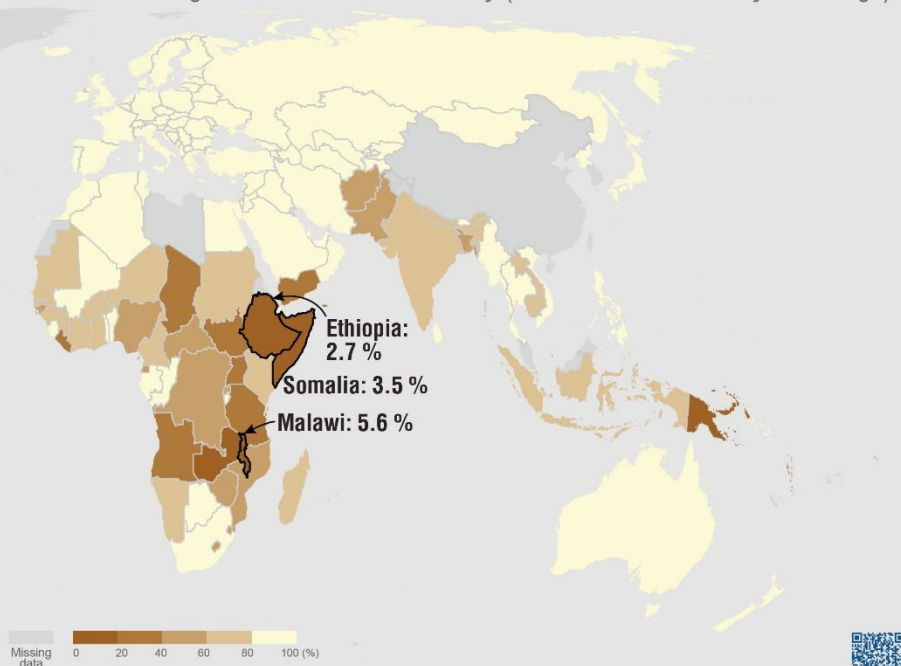
166 Million
children under the
age of 5, or 1 in 4
remain unregistered

Lowest ten country on proportion of
birth registration in 2011-2018 (%)



▶ Displaced children in Aden City from
the city of Taiz due to the conflict, Yemen.

UNICEF - Ahmed Abdulhaleem, 2018



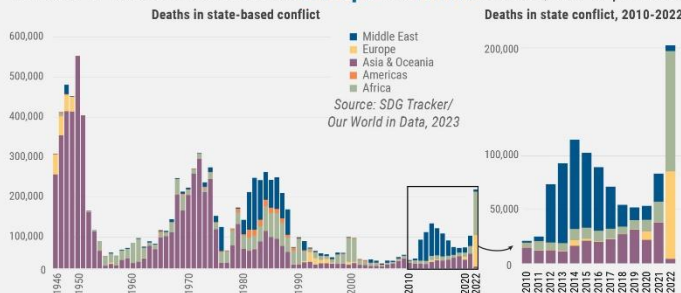
▲ The birth of around one in four children under age 5 worldwide are never officially recorded, depriving them of a proof of legal identity crucial for the protection of their rights and for access to justice and social services. The indicator 16.9.1 (2011-2018) is the proportion of children under 5 years of age whose births have been registered with a civil authority, the countries highlighted in shades of brown show where the gap remains. Oceania, excluding Australia and New Zealand, is the region with the lowest registration rate at 23.1%, followed by Sub-Saharan Africa with 44.8%. War-torn or past conflict countries, such as Somalia, Liberia, Chad or Yemen, exhibit low registration rates.



Significantly reduce all forms of violence and related death rates everywhere

Conflict-related deaths, 1946-2023

▼ Since **1946**, the overall number of deaths in state-based conflict has steadily **decreased** worldwide. Still, **from 2010 to 2022** around **a million people** died through direct violence in wars. During this period, **Middle East** accounts for 50% of total deaths, and tremendous increase of deaths are observed in **Europe** and **Africa** in 2022, due to political crisis.



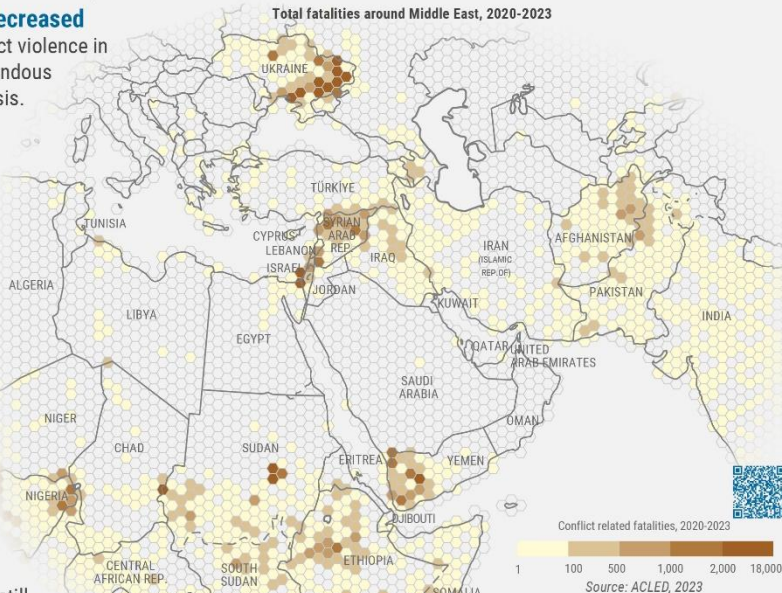
Since **2020**, **incidents** related to battles and conflicts keep on occurring within and at the crossroads of **Africa**, **Asia**, and **Europe** ▼

Battle related incident occurrences, 2020-2023



► Many conflicts are still **ongoing or erupted between 2020 and 2023**, such as in the **Middle East**, the **Horn of Africa** or in countries such as **Israel**, **Lebanon**, **Sudan**, or **Ukraine**.

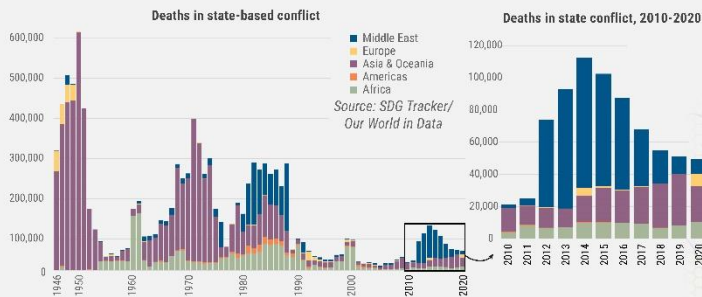
► UN Development Programme and Economic and Social Commission for Western Asia warn that up to 16 years of human development could be wiped out UNDP Photo-PAPP/Abed Zagout, 2023



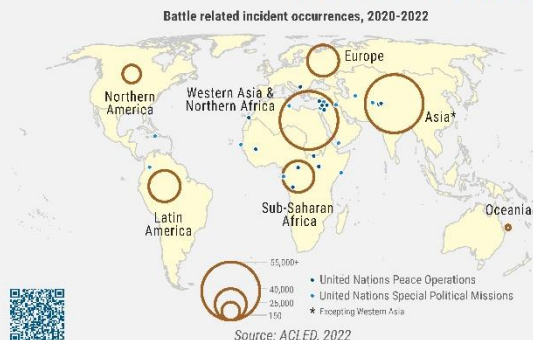
Significantly reduce all forms of violence and related death rates everywhere

Conflict-related deaths, 1946-2020

Since **1946**, the overall amount of deaths in state-based conflict has steadily **decreased** worldwide. Still, **from 2010 to 2020**, around **half a million people** died through direct violence in wars. During this period, the increase is particularly visible in the **Middle East** ▼

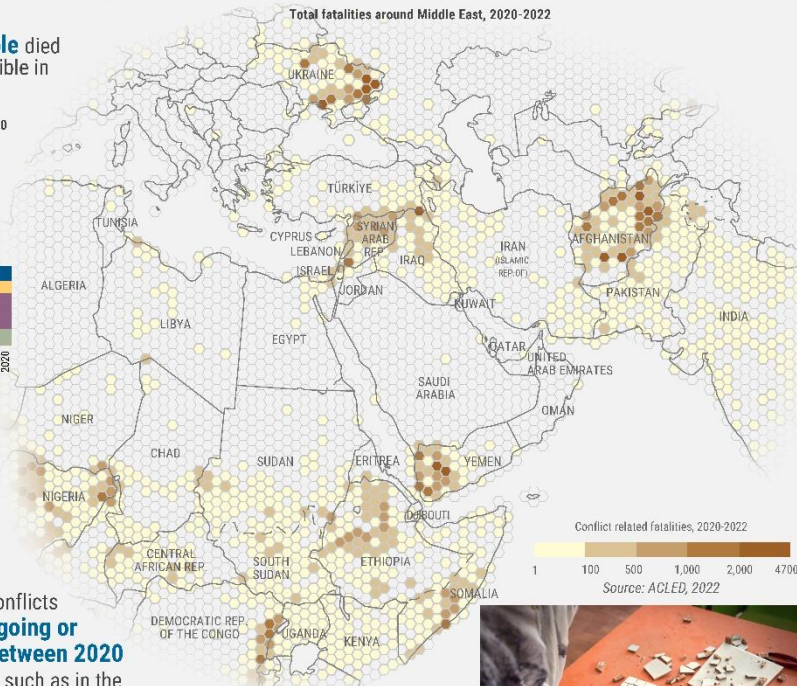


Since **2020**, **incidents** related to battles and conflicts keep on occurring within and at the crossroads of **Africa, Asia**, and **Europe** ▼



► Many conflicts are still **ongoing or erupted between 2020 and 2022**, such as in the **Middle East**, the **Sahel**, the **Horn of Africa** or in countries such as **Afghanistan**, Eastern **D.R.Congo** or **Ukraine**

► UN Women Humanitarian work in Za'atari Refugee Camp in Jordan
UN Women Photo-Christopher Herwig





Upgrade infrastructure and retrofit industries to make them sustainable

Manufacturing value added per capita (constant 2015 US dollars)



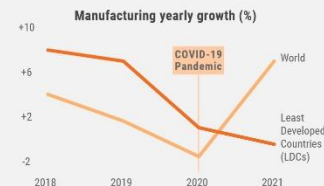
▲ Ladies working in a textile and clothing industry in Bangladesh. Asian countries graduating from the LDC category need to bolster their textile and clothing sectors, particularly in the wake of COVID-19

ILO Photo - Marcel Crozet

In 2021, global manufacturing rebounded from the pandemic, although the **recovery remains incomplete and uneven**. After **dropping 1.3 percentage** in 2020, global manufacturing production **grew by 7.2 percentage** in 2021, surpassing its pre-pandemic level.

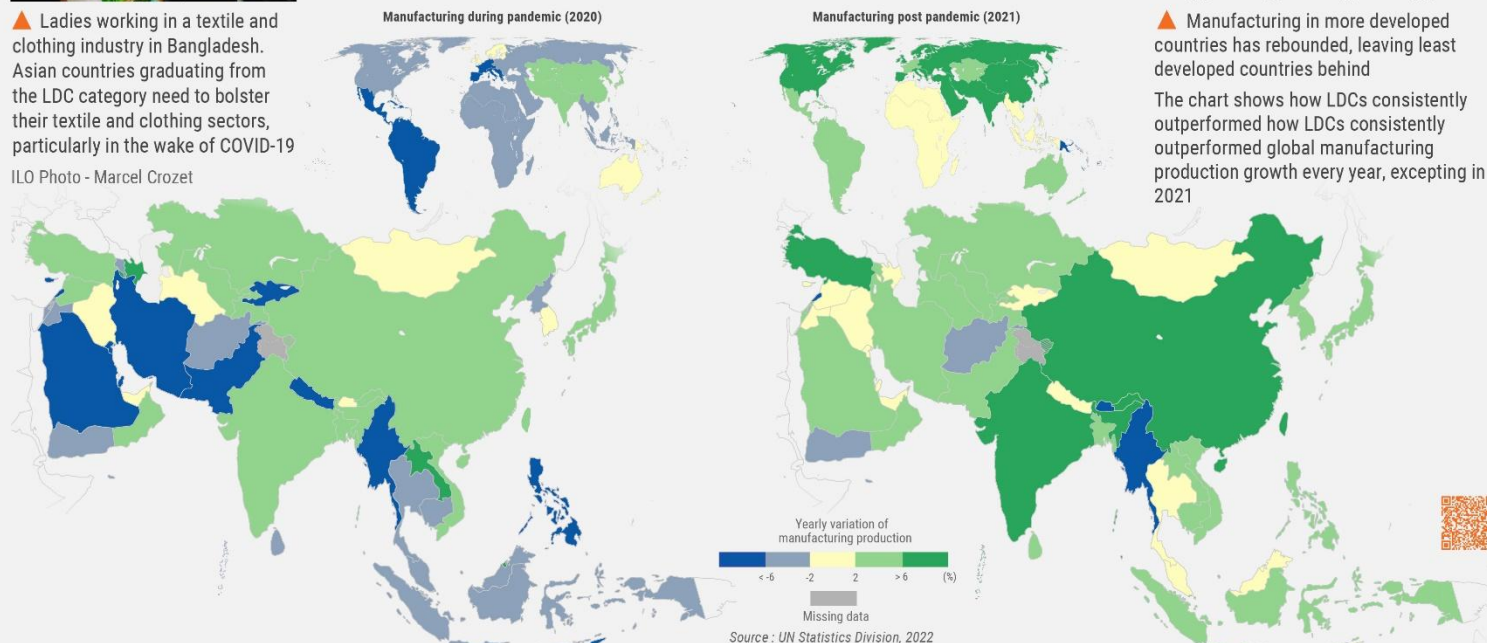
High-income countries benefited from massive policy support to firms and households and the rapid roll-out of effective vaccines.

In contrast, recovery in Least Developed Countries (LDCs) has been sluggish, due to subdued and volatile global demand, global trade disruption and tighter domestic economic policies.



▲ Manufacturing in more developed countries has rebounded, leaving least developed countries behind

The chart shows how LDCs consistently outperformed how LDCs consistently outperformed global manufacturing production growth every year, excepting in 2021

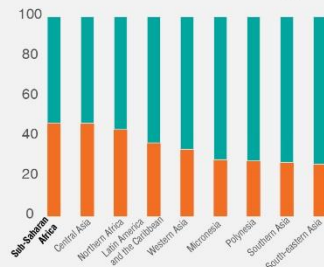


BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE AND SUSTAINABLE INDUSTRIALIZATION & FOSTER INNOVATION



Develop quality, reliable, sustainable and resilient infrastructure, including regional and trans-border infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all

Rural population not served by a all-season road in 2021 (Million)



Chinese engineers with the UN Mission in the South Sudan (UNMISS) opened a road that had been closed for two weeks following heavy rains and flooding.

UNMISS Photo 2013

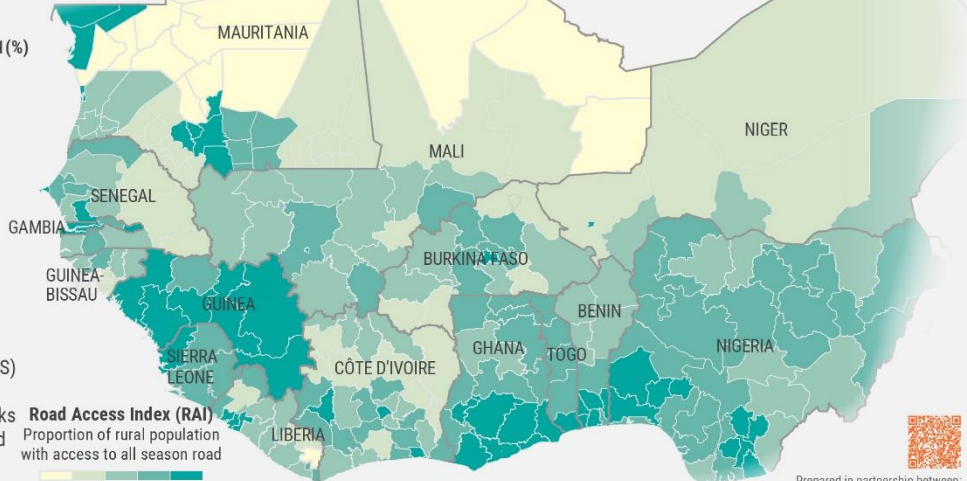
Sub-Saharan Africa is the region with most rural population **not served by all-season roads** worldwide.

Basic infrastructure like **roads**, information and communication technologies, sanitation, electrical power and water remains scarce in many **developing countries**.

Road Access Index (RAI)
Proportion of rural population with access to all season road



Source : UN Geospatial and CIESIN, 2023

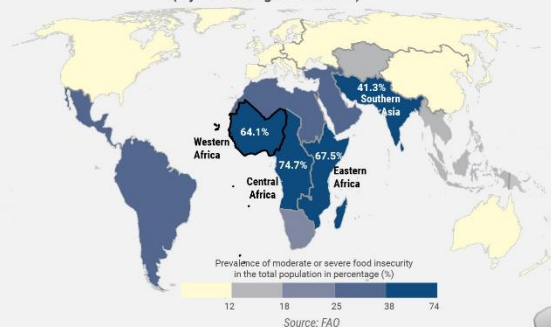


Prepared in partnership between:
Columbia University (CIESIN) and
UNITED NATIONS Geospatial
Map No. 4677.9
Dec 2023



By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses

Prevalence of moderate or severe food insecurity in the total population (3 years average 2020-2022)

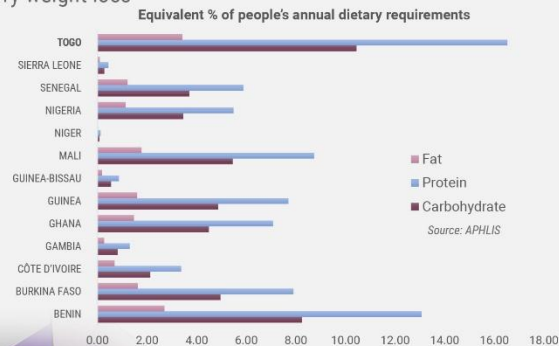


Food loss and waste in **Western Africa** are estimated at up to **250kg per person** and over **\$60 billion dollars annually**. Global economic loss and waste amount to about **US\$940 billion annually**. The reasons for food losses include overproduction, premature harvesting, poor storage, processing facilities, inadequate market systems, overly high standards for the weight, shape and lack of knowledge about postharvest management



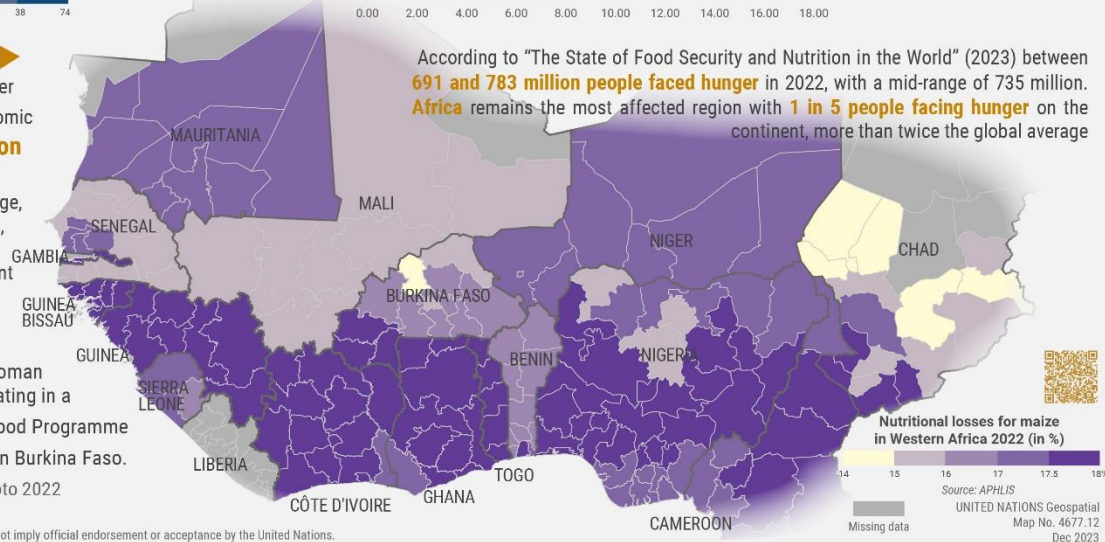
A woman participating in a World Food Programme project in Burkina Faso. WFP Photo 2022

Dry weight loss



Postharvest losses mean that the annual nutritional requirements of millions of people are being lost. In **Togo**, maize postharvest losses were equivalent to the annual carbohydrate requirements over **1.6 millions of children under five years old**. This is the human cost to postharvest in **Sub-Saharan Africa**.

According to "The State of Food Security and Nutrition in the World" (2023) between **691 and 783 million people faced hunger** in 2022, with a mid-range of 735 million. **Africa** remains the most affected region with **1 in 5 people facing hunger** on the continent, more than twice the global average



By 2025, prevent and significantly reduce marine pollution of all kinds

Marine debris count density (size > 200mm)

8 Million tonnes
of **plastic** end up in
oceans every year

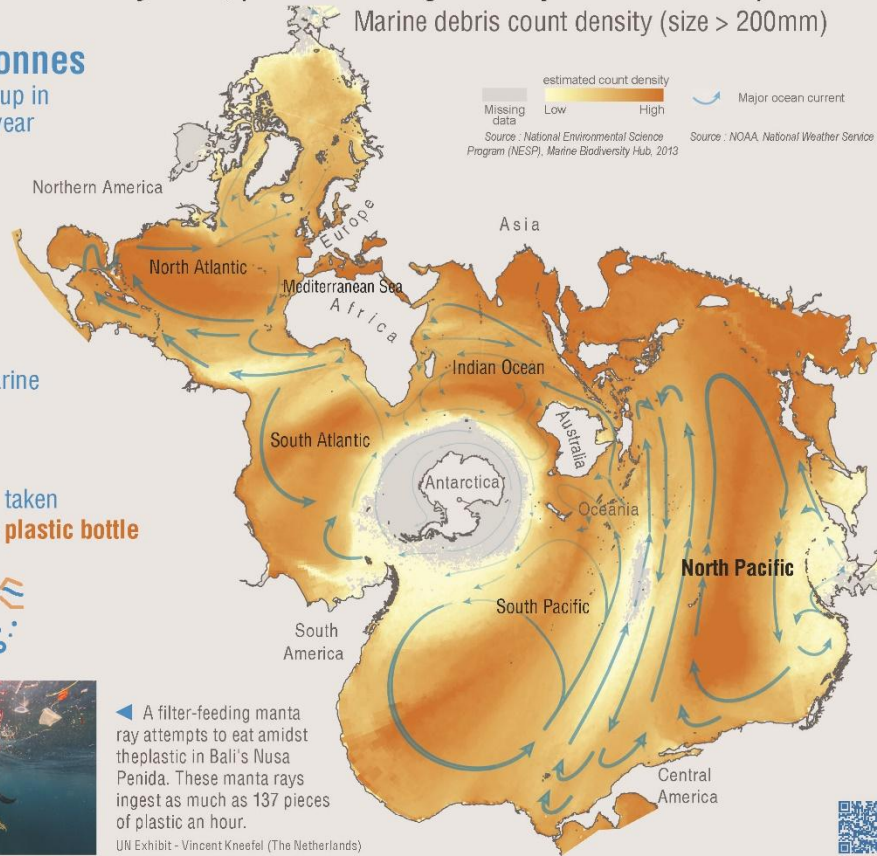
80% of all marine
debris studied
is **plastic**

450 years taken
to disintegrate **plastic bottle**



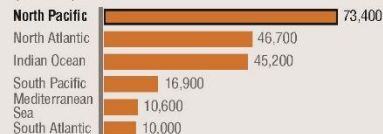
◀ A filter-feeding manta ray attempts to eat amidst the plastic in Bali's Nusa Penida. These manta rays ingest as much as 137 pieces of plastic an hour.

UN Exhibit - Vincent Kneefel (The Netherlands)



A total of **202,800 tonnes**
of **plastic** pieces afloat at global
ocean over 200mm surface

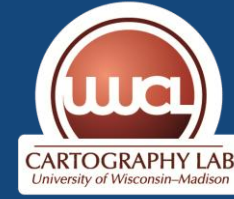
>200mm surface plastic mass by ocean basin in 2013 (Tonnes)



Source : Eriksson M, Lebretton LCM, Carlson HS, Thiel M, Moore CJ, Boreno JC, et al (2014) Plastic Pollution in the World's Oceans: More than 5 Trillion Plastic Pieces Weighing over 250,000 Tons Afloat at Sea. PLoS ONE 9(12): e111913
<https://doi.org/10.1371/journal.pone.0111913>

◀ The sustainability of our oceans is under severe threat as every year an estimated 5 to 12 million metric tons of plastic enter the ocean, costing roughly \$13 billion per year – including clean-up costs and financial losses in fisheries and other industries. About 89% of plastic litter found on the ocean floor are single-use items like plastic bags. The estimated count density of marine debris bigger than 200mm shown on the map (2013) is based on an oceanographic model of floating debris by The National Environmental Science Program (NESP), Australia. Highest values can be observed close to the coastline as the main sources of marine plastics are land-based. The geospatial data on marine debris is shown using the Spilhaus projection which allows to show the ocean as a continuous body of water, and the main ocean currents represented as arrows are intended to emphasize the dynamic nature and movements that contribute to the concentration of the marine plastics in certain location.





1.2: Lightning Talk 2 | Robert Roth, UW Cart Lab, UX Commission

Tips for Mapping the SDG Global Indicator Database

Home

SDG Indicators ▾

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Global SDG Indicators Data Platform

 Select indicators and country or area

SDG Global Database gives you access to data on more than 210 SDG indicators for countries across the globe

by indicator, country, region or time period



Global Database

Country Profile

SDG At

<https://unstats.un.org/sdgs/dataportal/database>

Indicators (Selected 39 of 653 series) Country or Area (Selected 301 of 301)

☒ All

Search

- ▼ ☒ **GOAL 6** Ensure availability and sustainable management of water and sanitation for all
 - ▼ ☒ **TARGET 6.1** By 2030, achieve universal and equitable access to safe and affordable drinking water for all
 - ▼ ☒ **INDICATOR 6.1.1** Proportion of population using safely managed drinking water services
 - Proportion of population using safely managed drinking water services, by urban/rural (%) **SH_H2O_SAFE**
 - ▶ ☐ **TARGET 6.2** By 2030, achieve access to adequate and equitable sanitation and hygiene for all and end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations
 - ▶ ☐ **TARGET 6.3** By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally

Show Results

SDG Global Database
indicators for countries
by indicator, country, region or

Global Database

SDG Indicators

 Official list of SDG indicators

 Metadata Repository

 Hier Classification

<https://unstats.un.org/sdgs/dataportal/database>

Resources

 Regional Groupings

 E-Handbook on SDG Indicators

 UNCT Toolkit

Advanced Access

17 Goals
169 Targets
239 Indicators

• **Data Series** (Selected 2 of 706)

1.1.1 ×

+ Select

• **Countries, areas or regions** (Selected 211 of 211)[All Groupings](#)

Countries or areas

World (total) by SDG regions × World (total) by continental regions × Land Locked Developing Countries (LLDC) × + 3 ...

By default "All" is selected.

+ Select

• **Period** [Range](#) [Years](#) (25 of 62)

2000 × 2001 × 2002 × 2003 × 2004 × 2005 × 2006 × 2007 × 2008 × 2009 × 2010 × 2011 × 2012 × + 12 ...

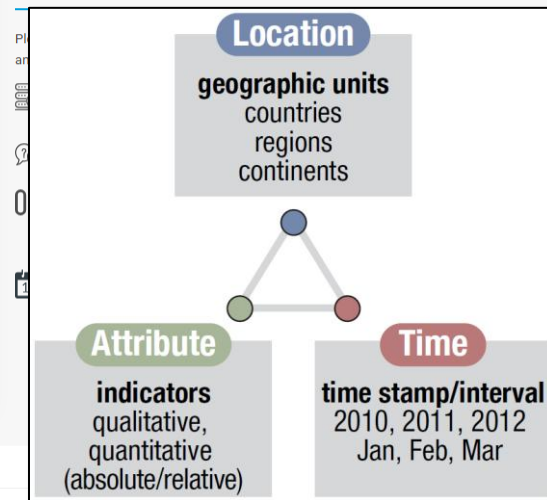
You can select single year or multiple years

• **21,834 observations**[Show Results](#)[SI_POV_DAY1](#)[SI_POV_EMP1](#)Indicator 1.1.1, Series : Proportion of population below international poverty line (%) [SI_POV_DAY1](#)Disaggregated by ☒ Age ☒ Location ☒ Sex[Go to Metadata](#)

Select years to hide ▾

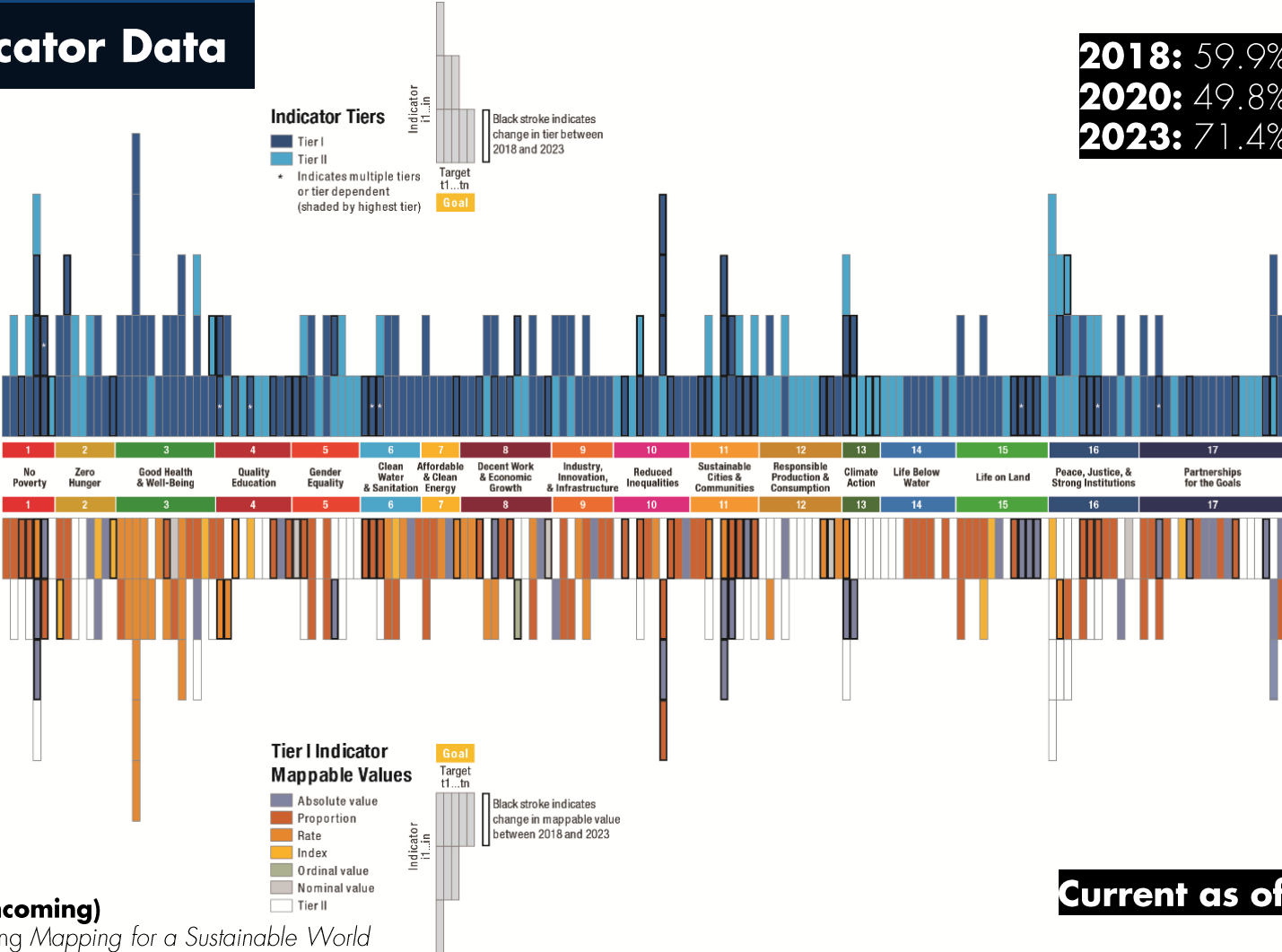
Country	Age	Location	Sex	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Africa	ALLAGE	ALLAREA	BOTHSEX	46.7 ^{fn,N}	46 ^{fn,N}	46 ^{fn,N}	45 ^{fn,N}	42 ^{fn,N}	40.9 ^{fn,N}	40 ^{fn,N}	39 ^{fn,N}	38 ^{fn,N}	37 ^{fn,N}	35.5
Albania	<15Y	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	15-64	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	65+	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	ALLAGE	ALLAREA	BOTHSEX	-	-	1 ^{fn,G}	-	-	0.6 ^{fn,G}	-	-	0 ^{fn,G}	-	-
Albania	ALLAGE	ALLAREA	FEMALE	-	-	-	-	-	-	-	-	-	-	-

Important Information



SDG Indicator Data

2018: 59.9% (132/232)
2020: 49.8% (115/231)
2023: 71.4% (165/231)



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• **Data Series** (Selected 2 of 706)

1.1.1 ×

+ Select

• **Countries, areas or regions** (Selected 211 of 211)

All Groupings

Countries or areas

World (total) by SDG regions × World (total) by continental regions × Land Locked Developing Countries (LLDC) × + 3 ...

By default "All" is selected.

+ Select

• **Period** [Range](#) [Years](#) (25 of 62)

2000 × 2001 × 2002 × 2003 × 2004 × 2005 × 2006 × 2007 × 2008 × 2009 × 2010 × 2011 × 2012 × + 12 ...

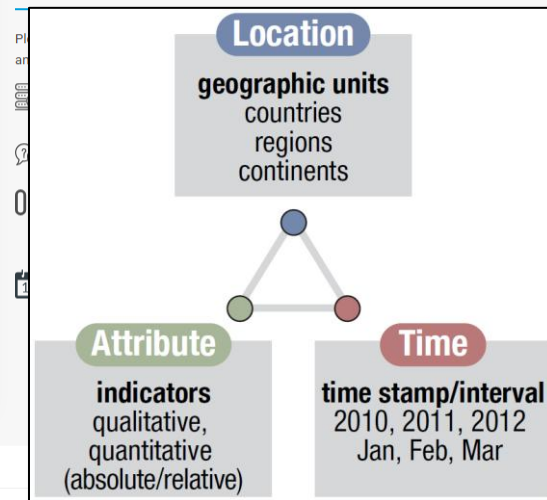
You can select single year or multiple years

• **21,834 observations**[Show Results](#)[SI_POV_DAY1](#)[SI_POV_EMP1](#)Indicator 1.1.1, Series : Proportion of population below international poverty line (%) [SI_POV_DAY1](#)Disaggregated by ☒ Age ☒ Location ☒ Sex[Go to Metadata](#)

Select years to hide ▾

Country	Age	Location	Sex	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Africa	ALLAGE	ALLAREA	BOTHSEX	46.7 ^{fn,N}	46 ^{fn,N}	46 ^{fn,N}	45 ^{fn,N}	42 ^{fn,N}	40.9 ^{fn,N}	40 ^{fn,N}	39 ^{fn,N}	38 ^{fn,N}	37 ^{fn,N}	35.5
Albania	<15Y	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	15-64	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	65+	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	ALLAGE	ALLAREA	BOTHSEX	-	-	1 ^{fn,G}	-	-	0.6 ^{fn,G}	-	-	0 ^{fn,G}	-	-
Albania	ALLAGE	ALLAREA	FEMALE	-	-	-	-	-	-	-	-	-	-	-

Important Information



"M49" Regional Groupings

SDG Intermediate Regions



For data model, as nodes com- (7) coordinates nodes. Vector (nodes only), s), polygons (ents), and volumes (ation 2.8). The on for capturing tion data depends

Finally, "(geo)spatial" is a "special" case in data science due to *spatial autocorrelation*: near locations are more likely to be similar in both attribute and time than distant locations. Thus, common techniques for data capture and statistical analysis do not always apply to location data given their topological relationships, therefore requiring special considerations.

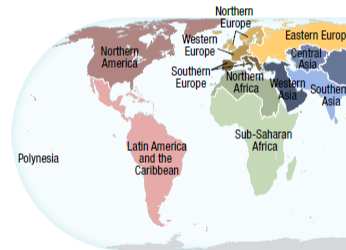
M49 *regions* are major continental regions that are divided into *sub-regions* (Figure 1.3-1), with some sub-regions further divided into *intermediate regions* (Figure 1.3-2). These geographic regions are drawn to obtain greater homogeneity in sizes of population, demographic circumstances, and accuracy of demographic statistics. Finally, a *country* is a sovereign political and territorial entity acknowledged in the M49 standard.

For the purpose of presentation, the SDG regional groupings combine specific M49 geographic regions and sub-regions (Figure 1.3-3).

In addition, specific country group lists are also defined for Least Developed Countries (LDC), Landlocked Developing Countries (LLDC), and Small Island Developing States (SIDS). Lists for SDG regional groupings and specific country groups are detailed at: <https://unstats.un.org/>

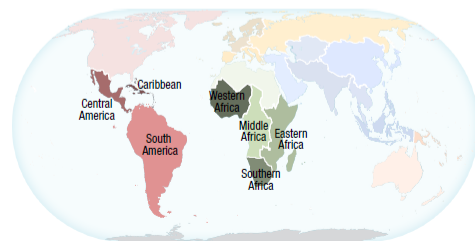
SDG regional groupings used to compile and report indicator data.

and data formats before mapping.



SDG Sub-regions

Figure 1.3-2: M49 intermediate regions. Additional M49 intermediate regions used for greater specificity within some sub-regions. Colour hue again is used for the broader M49 regions for comparison to Figure 1.3-1.



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Data Series (Selected 2 of 706)

1.1.1 ×

+ Select

Countries, areas or regions (Selected 211 of 211)

[All Groupings](#)

Countries or areas

[World \(total\) by SDG regions](#) × [World \(total\) by continental regions](#) × [Land Locked Developing Countries \(LLDC\)](#) × [+ 3 ...](#)

By default "All" is selected.

+ Select

Period [Range](#) [Years](#) (25 of 62)
[2000](#) × [2001](#) × [2002](#) × [2003](#) × [2004](#) × [2005](#) × [2006](#) × [2007](#) × [2008](#) × [2009](#) × [2010](#) × [2011](#) × [2012](#) × [+ 12 ...](#)

You can select single year or multiple years

21,834 observations

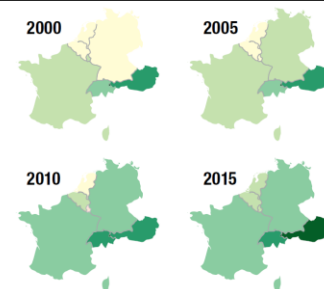
[Show Results](#)[SI_POV_DAY1](#)[SI_POV_EMP1](#)Indicator 1.1.1, Series : Proportion of population below international poverty line (%) [SI_POV_DAY1](#)Disaggregated by ☒ Age ☒ Location ☒ Sex[Go to Metadata](#)

Select years to hide ▾

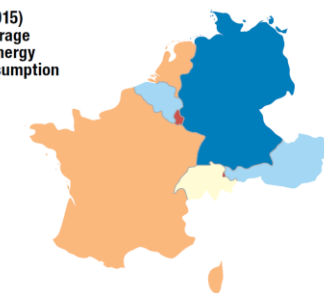
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Albania	<15Y	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	15-64	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	65+	ALLAREA	BOTHSEX	-	-	-	-	-	-	-	-	-	-	-
Albania	ALLAGE	ALLAREA	BOTHSEX	-	-	1 ^{fn,G}	-	-	0,6 ^{fn,G}	-	-	0 ^{fn,G}	-	-
Albania	ALLAGE	ALLAREA	FEMALE	-	-	-	-	-	-	-	-	-	-	-

**Western Europe (2000–2015)
Renewable Energy Share of
Total Final Energy Consumption**

35–56%
20–35%
10–20%
5–10%
0–5%
Missing data


**Western Europe (2000 versus 2015)
Per Cent Change Relative to Average
Increase (5.6%) in Renewable Energy
Share of Total Final Energy Consumption**

3–5%
1–3%
-1–1%
-1–3%
-3–5%
Missing data



Data Series (Selected 2 of 706)

1.1.1 ×

+ Select

- Countries, areas or regions (Selected 211 of 211)

All Groupings

Countries or areas

World (total) by SDG regions × World (total) by continental regions × Land Locked Developing Countries (LLDC) × + 3 ...

By default "All" is selected.

+ Select

- Period Range **Years** (25 of 62)

2000 x 2001 x 2002 x 2003 x 2004 x 2005 x 2006 x 2007 x 2008 x 2009 x 2010 x 2011 x 2012 x + 12 ...

You can select single year or multiple years

- 21,834 observations

 [Show Results](#)



Important Information

Please select parameters from each of the sections on the left, and click on 'Show Results' to perform a search.



 Database last updated on Tuesday, April 1, 2025. [See History](#)



 If you need help using this site, [Read FAQs](#)



For latest reference metadata information for the Tier I and II indicators in the global indicator framework, explore the [Metadata Repository](#)



 This new database interface has been launched on 15 September 2021. Please use the link below for questions and feedback. The old interface is still available for limited time at <https://unstats.un.org/sdgs/indicators/database/legacy>.

SI_POV_DAY1

SI_POV_EMP1

Indicator 1.1.1. Series : Proportion of population below international poverty line (%) SI_POV_DAY1

Disaggregated by ☒ Age ☒ Location ☒ Sex[Go to Metadata](#)

Q Search this table...

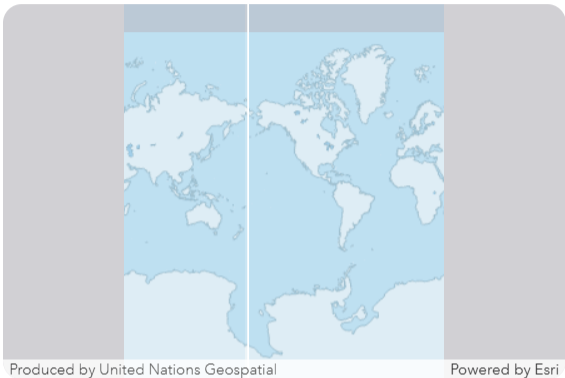
Select years to hide ▾

[illegible]

Tips for Mapping the SDGs

- 1. Generalization:** Use the UN Geodata Simplified boundary file for world and regional maps
- 2. Projections:** Use a recentered, equal area projection for most thematic maps
- 3. Typography:** Visually accent critical places ("characters") to your message
- 4. Symbolization:** Consider the appropriate visual metaphor evoked by your thematic map
- 5. Normalization:** Ensure choropleth maps use normalized indicators
- 6. Classification:** Consider the attribute distribution when selecting a classification scheme
- 7. Layout:** Use the grid to inform layout, seeking balance but asymmetry





UN Geodata simplified



United Nations Geospatial
UN GeoPortal

Download

✓ Use the UN Geodata Simplified boundary file for world and regional maps

Summary

UN Geodata simplified in GeoJSON

The United Nations Geospatial Data, or Geodata, is a worldwide geospatial dataset of the United Nations.

The United Nations Geodata is provided to facilitate the preparation of cartographic materials in the United Nations includes geometry, attributes and labels to facilitate the adequate depiction and naming of geographic features for the preparation of maps in accordance with United Nations policies and practices.

The geospatial datasets here included are referred to as UN Geodata simplified and are generalized based on UNGeodata 25 million scale.

The feature layers include polygons/areas of countries (BNDL_simplified), lines for international boundaries and limits (BNDL_simplified), and major water body (WBYA_simplified). In addition, aggregated regional areas are available following M49 methodology (GEOA_simplified, SUBA_simplified, INTA_simplified) and SDG regional grouping (SDGA_simplified).

Details



Content
GeoJson



June 12, 2023
Info Updated



June 12, 2023
Data Updated



June 12, 2023
Published Date



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Goal	Target	Indicator	SeriesCod	SeriesDes	GeoAreaC	GeoAreaM	Reporting Sex	Units	2000	2001	fid	objectid	iso3cd	m49_cd	nam_en	lbl_en	georeg	geo_cd	sub_cd	int_cd	
2	5	5.5	5.5.1	SG_GEN_F	Number o	4	Afghanistan	G	FEMALE	NUMBER	1	1	1	ABW	533	Aruba	Aruba (Neth.)	AME	19	419	29
3					umber o	8	Albania	G	FEMALE	NUMBER	8	8	2	AFG	004	Afghanistan	AFGHANISTAN	ASI	142	34	m4
4					umber o	12	Algeria	G	FEMALE	NUMBER	12	13	3	AGO	024	Angola	ANGOLA	AFR	2	202	
5					umber o	20	Andorra	G	FEMALE	NUMBER	2	2	4								
6	5	5.5	5.5.1	SG_GEN_F	Number o	24	Angola	G	FEMALE	NUMBER	34	34	4	AIA	660	Anguilla	Anguilla *	AME	19	419	
7	5	5.5	5.5.1	SG_GEN_F	Number o	28	Antigua ar	G	FEMALE	NUMBER	1	1	5	ALA	248	Åland Islands	Åland Islands (F...	EUR	150	154	
8	5	5.5	5.5.1	SG_GEN_F	Number o	32	Argentina	G	FEMALE	NUMBER	72	68	6	ALB	008	Albania	ALBANIA	EUR	150	39	NULL
9	5	5.5	5.5.1	SG_GEN_F	Number o	51	Armenia	G	FEMALE	NUMBER	4	4	7	AND	020	Andorra	ANDORRA	EUR	150	39	NULL
10	5	5.5	5.5.1	SG_GEN_F	Number o	36	Australia	G	FEMALE	NUMBER	33	34	7								NULL
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12	5	5.5	5.5.1	SG_GEN_F	Number o	31	Azerbaijan	G	FEMALE	NUMBER	13	13	9	ARG	032	Argentina	ARGENTINA	AME	19	419	5
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19	5	5.5	5.5.1	SG_GEN_F	Number o	84	Belize	G	FEMALE	NUMBER	2	2	16	BDI	108	Burundi	BURUNDI	AFR	2	202	14
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21	5	5.5	5.5.1	SG_GEN_F	Number o	64	Bhutan	G	FEMALE	NUMBER	3	14	18								
22	5	5.5	5.5.1	SG_GEN_F	Number o	68	Bolivia (PlG		FEMALE	NUMBER	15	15	19	ATF	260	French Souther...	French Souther...	AFR	2	202	14
23	5	5.5	5.5.1	SG_GEN_F	Number o	70	Bosnia and	G	FEMALE	NUMBER	3	3	18	ATG	028	Antigua and Ba...	ANTIGUA AND ...	AME	19	419	29
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34	5	5.5	5.5.1	SG_GEN_F	Number o	140	Central Af	G	FEMALE	NUMBER	8	8	27	BIH	070	Bosnia and Her...	BOSNIA AND H...	EUR	150	39	NULL
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43	5	5.5	5.5.1	SG_GEN_F	Number o	191	Croatia	G	FEMALE	NUMBER	0	31	34	BRB	052	Barbados	BARBADOS	AME	19	419	29
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47	5	5.5	5.5.1	SG_GEN_F	Number o	408	Democrat	G	FEMALE	NUMBER			37	CAF	140	Central African ...	CENTRAL AFRIC...	AFR	2	202	77

Tutorials

- Choropleth Tutorial From [Mapping SDGs Technical Supplement](#)
 - [1.1 - Overview and Project Planning](#)
 - [1.2 - Downloading QGIS](#)
 - [1.3 - Downloading Location and Attribute Data](#)
 - [1.4 - Cleaning Attribute Data](#)
 - [1.5 - Creating a Project and Setting the Projection](#)
 - [1.6 - Adding and Joining Attribute Data](#)
 - [1.7 - Editing the Attribute Table](#)
 - [1.8 - Classifying Attribute Data](#)
 - [1.9 - Visual Accenting](#)
 - [1.10 - Labeling and Leader Lines](#)
 - [1.11 - Adding Elements to Layout View](#)
 - [1.12 - Exporting the Map](#)
 - [1.13 - Advanced Layout Styling](#)
- Proportional Symbol Tutorial From [Mapping SDGs Technical Supplement](#)
 - [2.1 - Overview and Setup](#)
 - [2.2 - Projecting Map and Adding Data](#)
 - [2.3 - Extracting European Countries and Centroids](#)
 - [2.4 - Setting Proportional Symbols](#)
 - [2.5 - Adjusting Style](#)
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- Dot Density
 - [1 - Overview and Setup](#)
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 - [3 - Creating Dot Layer](#)
- Exporting from QGIS to Illustrator

Organizers

- TUW

Projections

Projection Centring

The **central meridian** of a map projection is the longitude along which the map surface of a projection is focused (see [Section 1.2](#)). A 0° central meridian at Greenwich, United Kingdom, often is selected for world maps since it is the zero reference, or **prime meridian**, for east-west notations of longitude. The **equator** is the zero reference for north-south

✓ Use a recentered, equal area projection for most thematic maps

compare two perspectives on the same SDG indicator. These views tailor the map to more local audiences, placing China and Canada in the centre of the map, respectively, by changing the central meridian of the projection.

Choosing an alternative central meridian also clarifies map features and spatial relationships. For instance,

Figure 2.5-2 splits Canada, with part of the country at the far left of the map and part at the far right. Adjusting the central meridian slightly eastward into the Pacific Ocean maintains a general focus on China while also avoiding broken linework in other continents, improving map interpretation. The Eckert IV world map projection in this

Earth's axis of rotation, which results in standard parallels. A **transverse** aspect rotates the map surface 90° from normal, instead resulting in standard meridians. An **oblique** aspect describes other rotations between normal and transverse, such as the orthographic projection giving the appearance of a view from space (e.g., the “globes” in **Figure 2.4-1**).

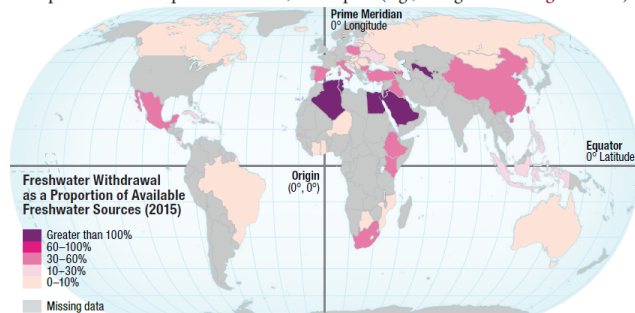
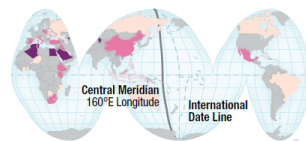
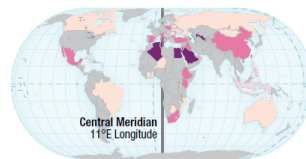
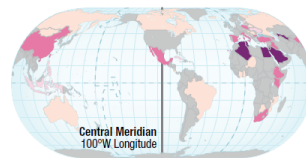
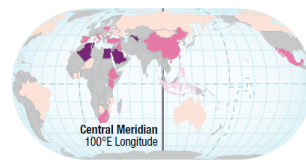


Figure 2.5-1: Projection Centring. The map of Indicator 6.4.2 (2015) depicts freshwater withdrawal by country. The map uses the prime meridian, or 0° longitude, as the central meridian. The equator, or 0° latitude, also is marked.



Rather than touching at a single **tangent** point (for planar projections) or line (for conic and cylindrical), the map surface also can slice into the globe. Such **secant** projections are common because they result in one standard line (for planar) or two standard lines (for conic and cylindrical), reducing distortion overall.

Finally, **interruptions** also can be

Figure 2.5-2: Centring on China. The map reprojects the **Figure 2.5-1** map of Indicator 6.4.2 (2015) using a central meridian of 100° E. Such map centring tailors the map to a local audience, here China, with the tradeoff of making the values of countries in North America, like Canada, more difficult to interpret.

Figure 2.5-3: Centring on Canada. The map reprojects the **Figure 2.5-1** map of Indicator 6.4.2 (2015) using a central meridian of 100° W, changing the local perspective to Canada. Because **Figures 2.5-1, 2.5-2, and 2.5-3** use the equivalent Eckert IV projection, the sizes of China and Canada (and all other enumeration units) remain the same across the three maps, despite the shape of these countries changing dramatically when recentered.

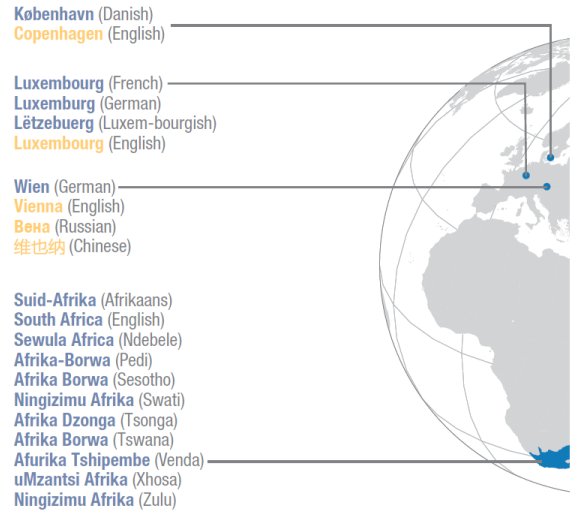
Figure 2.5-4: Centring on 11° E. When world SDG maps are made for a global audience, or when a local audience does not impact the purpose of the map, a normal, equivalent projection centred on 11° E is recommended. An 11° E centring splits the Pacific Ocean, resulting in improved visual balance of negative space for continental datasets. While an 11° E centring avoids breaking many country boundaries, some Small Island Developing States are impacted.

Figure 2.5-5: Interrupted projection centred on 160° E. Rather than breaking the map at a single meridian, an interrupted projection slices the map surface into lobes, with each lobe having its own central meridian. The figure depicts the **Figure 2.5-1** map of Indicator 6.4.2 (2015) using the Interrupted Goode Homolosine projection. Lobe centring is selected to focus on oceans rather than continents, making it a useful projection for depicting Small Island Developing States.

added to the projection to reduce local distortions in each resulting map lobe. Interruptions help to centre the map primarily on land versus ocean phenomena. **Figure 2.5-5** shows the world in the Interrupted Goode Homolosine projection, updating the **Figure 2.4-2** with a central meridian of 160° W to emphasize the Small Island Developing States.

Toponymy is the study of place names, their origins, meanings, and uses. **Toponyms**, or geographic place names, reflect relationships among people and cultures across geography and history. A toponym may originate from within the named place (an **endonym**) or from outside of the named place (an **exonym**), with endonyms requiring **transliteration** to alternative languages and alphabets for global identification (Figure 2.12-1). Accordingly, there may be multiple accepted names for a single place, and Indigenous groups and local communities may

contest exonyms derived from colonial, racist, or otherwise culturally insensitive geopolitical and historical practices. The United Nations leads national and international standardization and management of geographic names. Its mission on toponymy was articulated in 1959 by United Nations Economic and Social Council resolution 715A (XXVII), which established the **United Nations Group of Experts on Geographical Names (UNGEGN)** and led to the first meeting on toponyms in 1960. Today, the UNGEGN assists national place-name



efforts—particularly to involve Indigenous groups and local communities in the process—and facilitates discussion of

ical and historical implications of using exonyms versus endonyms. However, as cartography is transnational, selection

✓ Use English / Latin characters, but consider endonyms

tent of map labels (see Section 2.11) and, thus, serve as important reference information for clarifying locations significant to the map subject and purpose. Cartographers need to be aware of the geopolit-

ic names endorsed by the respective national authoritative bodies are recommended for labelling national-level maps.

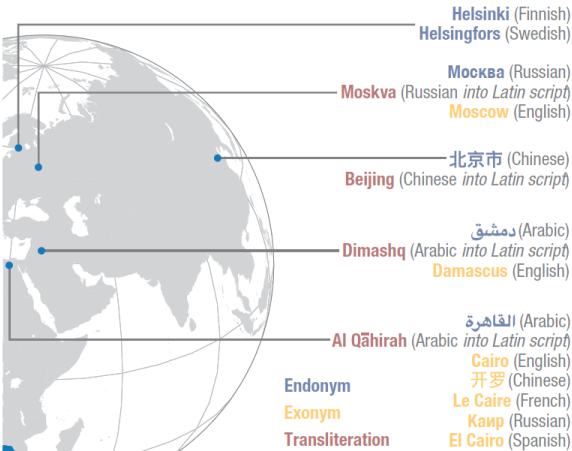
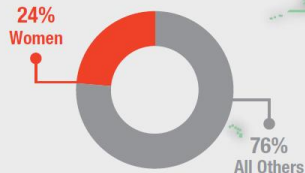


Figure 2.12-1: Toponymy. Endonyms (blue) are provided in local alphabets for several places in Africa, Asia, and Europe. Many places have multiple accepted endonyms. Exonyms (yellow) and transliteration (red) to Latin script are listed beneath each endonym.

GOAL 5: ACHIEVE GENDER EQUALITY AND EMPOWER ALL WOMEN AND GIRLS

- Only 24% of parliamentary leaders were women in 2020



- Only four national parliaments have 50% or greater representation by women

Proportion of Seats held by Women in National Parliaments (Per cent of Total Seats; 2020)



Bolivia: 53.1%

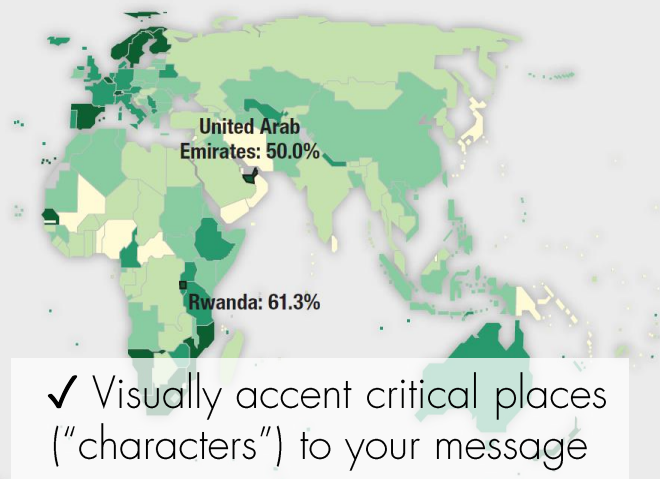
Cuba: 53.2%



◀ Wide view of the opening meeting of the sixty-fourth session of the Commission on the Status of Women (CSW). Member States adopted a political declaration in which they pledged to step up action to fully implement the landmark Beijing Declaration and Platform for Action on gender equality, agreed to 25 years ago. (Source: UN Photo/Loey Felipe, 2020)

SDG Target 5.5

Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making



- ▲ The map depicts Indicator 5.5.1 (2020) on the proportion of seats held by women in national parliaments as a choropleth map. Countries for the choropleth map are highly generalized to show only the overall thematic patterns, simplifying the message. This style also increases the visual weight of smaller nations. UN Women promotes this basemap for its publication on "Women in Politics."

Although simplified, the map remains projected in the Eckert IV equivalent projection used throughout the book, allowing for comparison of areas in the choropleth. The choropleth map uses an equal interval classification for the uniform attribute distribution and a sequential colour scheme that crosses yellow to green colour hues but primarily relies on the ordered visual variable colour value.

5 GENDER EQUALITY



3.1 Thematic Maps

A *thematic map* depicts the variation of one or sometimes several (see [Section 3.7](#)) geographic phenomena, mapping location and attribute information together. Meeting the SDGs

typically are mapped at an ordinal or numerical level of measurement, as enumeration results in quantitative counts or frequencies. Nominal differences can be represented in thematic maps, such

✓ Consider the visual metaphor evoked by your thematic map

attribute information that is enumerated within polygonal geographic units (see [Section 1.4](#)). Enumerated attributes

ation units by their attribute values, primarily relying on colour value. A *proportional symbol map* (see [Section](#)

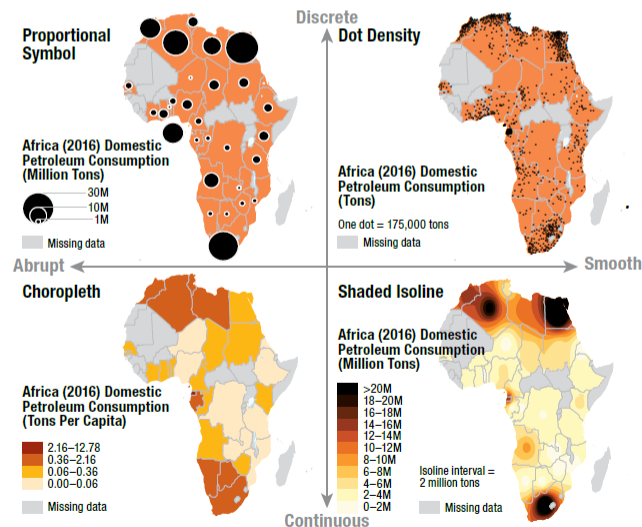


Figure 3.1-1: Thematic map types. The four maps depict Indicator 12.2.2 (2016) on domestic petroleum consumption. **Top-left:** Proportional symbol. **Top-right:** Dot density. **Bottom-left:** Choropleth. **Bottom-right:** Shaded isoline. 58 Mapping for a Sustainable World

Common Thematic Map Types

Figure 3.1-2: Choosing a thematic map type. [Section 3.2](#) treats nominal maps, [Section 3.3](#) choropleth maps, and [Section 3.4](#) proportion

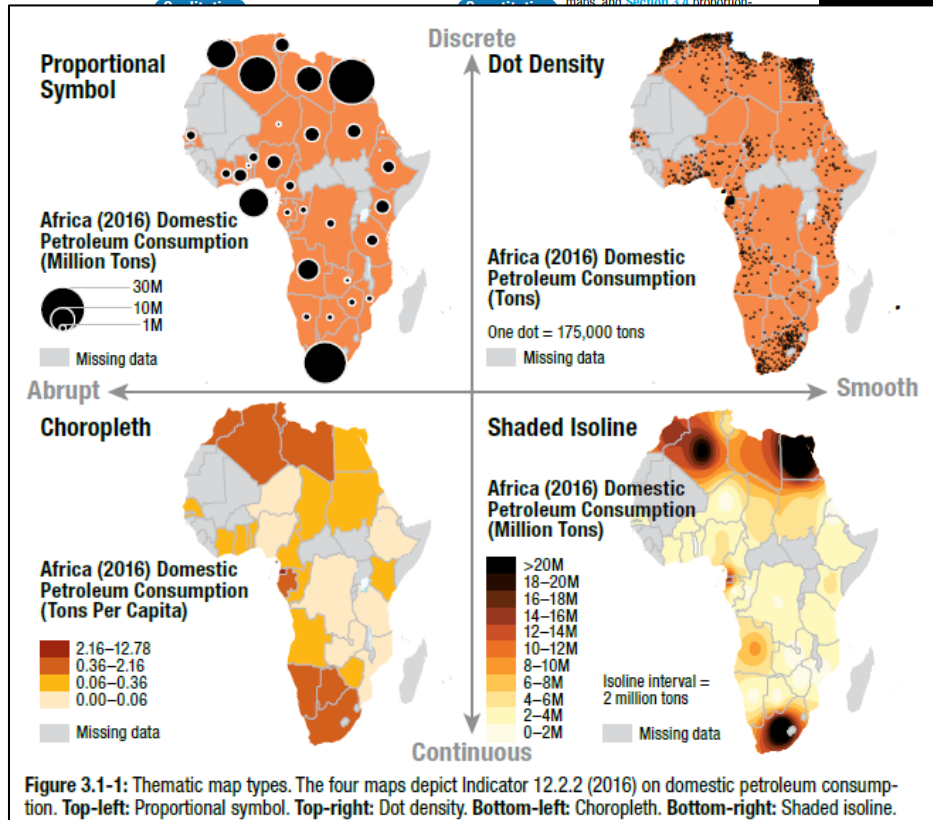


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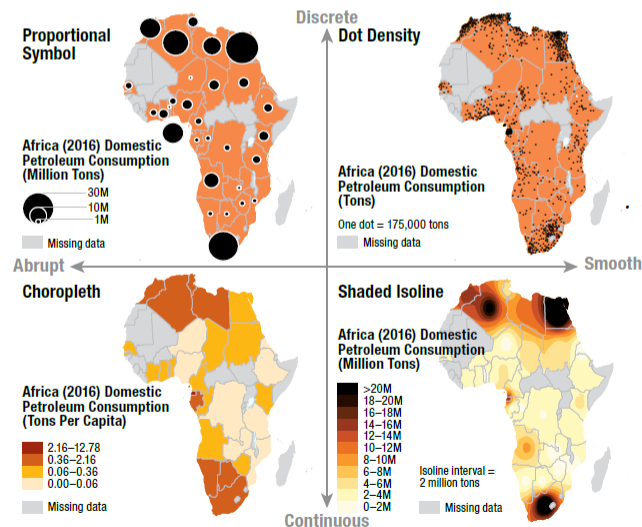
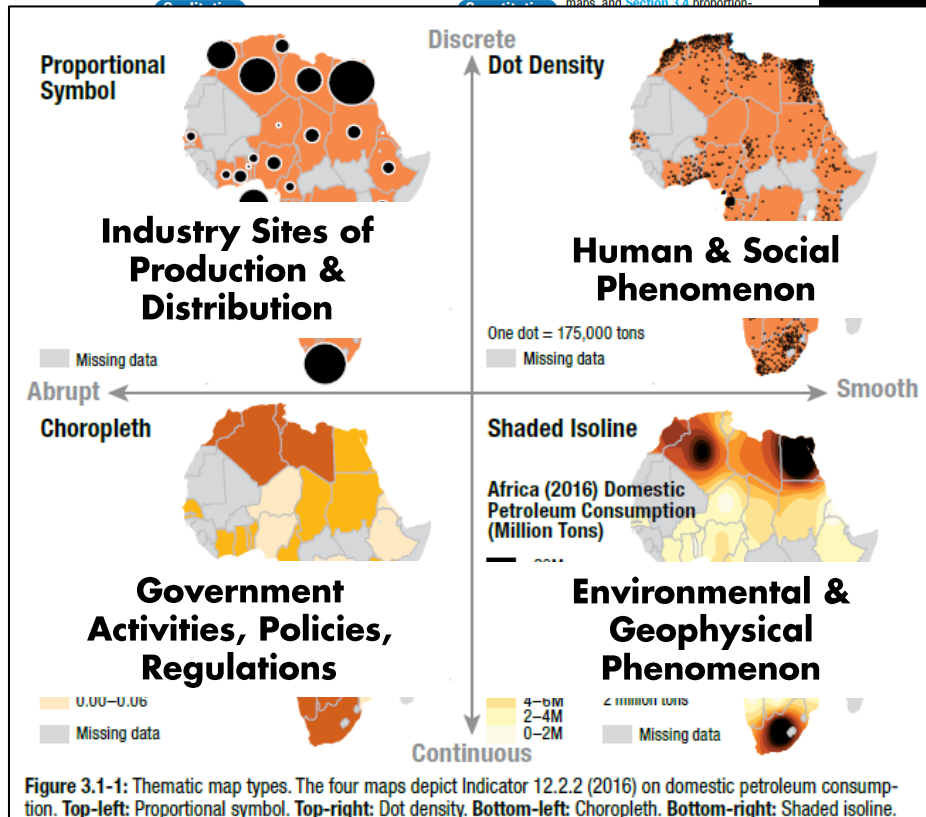


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Common Thematic Map Types

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1.8 The Modifiable Areal Unit Problem &

As introduced in [Section 1.4](#), most of the SDG indicator datasets contain enumerated attributes by country, making data transformation and normalization particularly important for mapping

individual-level data. Enumeration units sometimes follow regularly gridded boundaries (e.g., lines of latitude or longitude) in North America and South America but mostly have irregular

✓ Ensure choropleth maps use normalized indicators

areal unit problem (MAUP). In other words, the same underlying individual-level data when enumerated by different boundaries may lead to different findings, conclusions, or decisions derived from the map ([Figure 1.8-1](#)).

The two active factors in MAUP are the zone of aggregation (i.e., the arrangement and shape of the enumeration units) and the geographic scale (i.e., the size of the enumeration unit).

The **zone of aggregation** describes the arrangement and shape of the polygonal boundaries used for enumerating

varying in arrangement, shape, and size.

The **geographic scale** refers to the size and extent of a geographic phenomenon, such as the political boundaries used as enumeration units for the SDG indicators (see [Section 2.6](#) for comparison of geographic to cartographic scale). The United Nations encourages collection of SDG indicator data at a local level, or **small geographic scale**, with fine enumeration units having a detailed zone of aggregation. The SDG indicator data then is aggregated upward to a national level, or **large geo-**

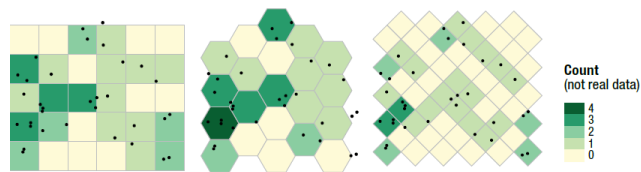


Figure 1.8-1: Understanding the modifiable areal unit problem (MAUP). The three images show the same, hypothetical point distribution enumerated by three sets of boundaries that differ in their arrangement, shape, and size. The shading shows the unnormalized count for each polygon unit, producing different visual patterns. **Left:** The pattern leans to the left. **Centre:** A clear cluster emerges in the bottom-left. **Right:** The pattern spreads more to the centre and right. Notably, choropleth maps require normalization (see [Section 3.3](#)) to reduce the visual impact of MAUP when shading the enumeration unit boundaries. Other thematic maps like dot density, isoline, and proportional symbol do not require normalization because their symbols are not tied directly to the enumeration unit boundaries. Example maps do not depict SDG indicators.

the Ecologic Normalization

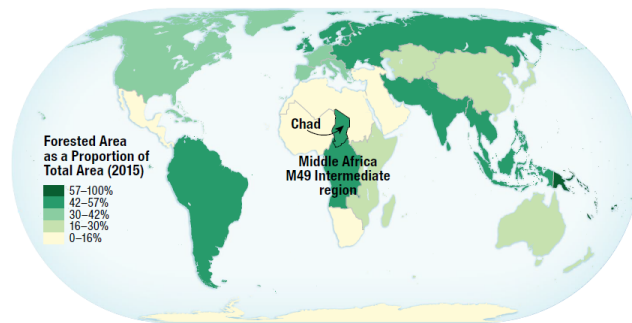


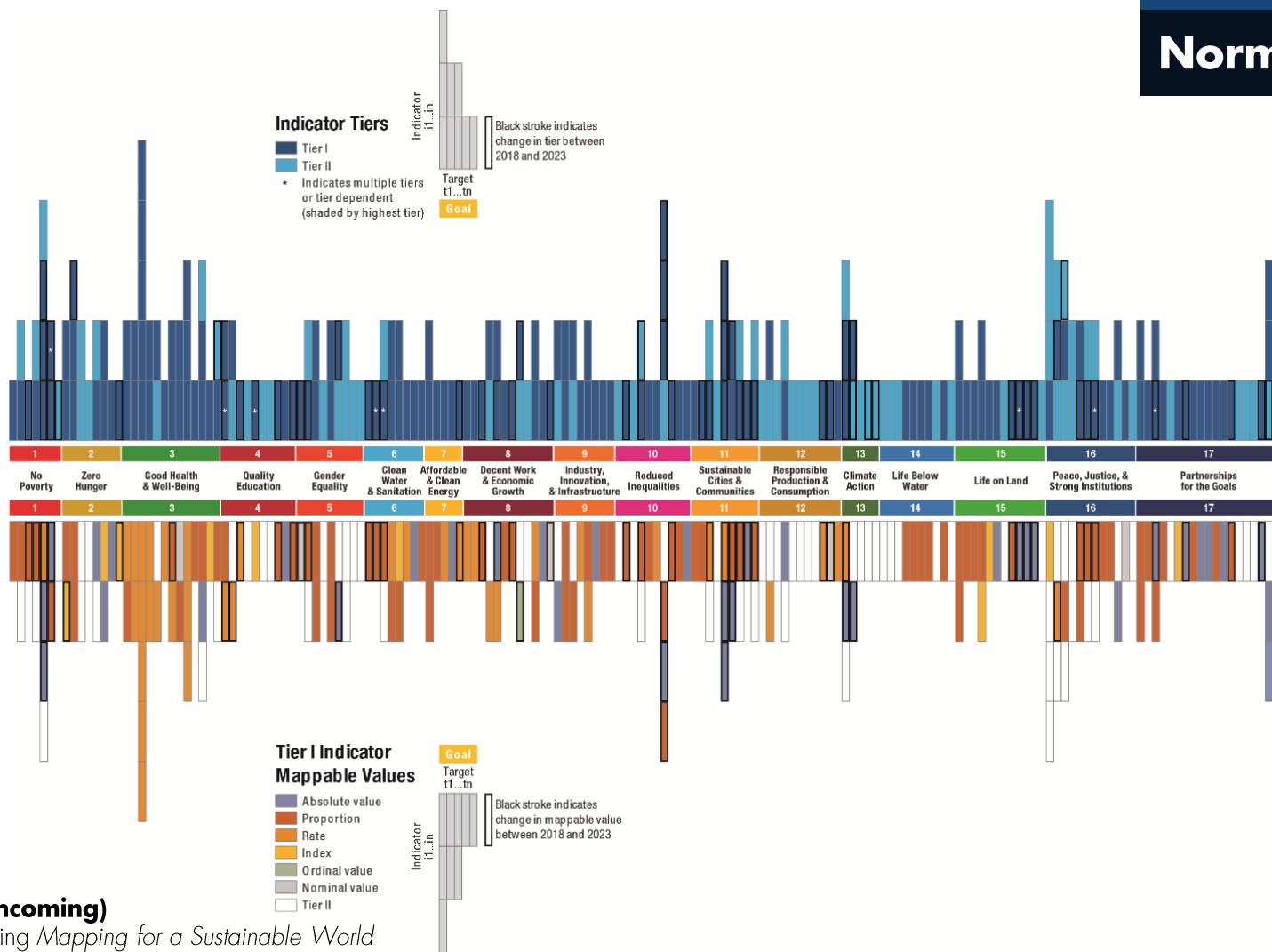
Figure 1.8-2: The effects of the modifiable areal unit problem and the ecological fallacy. Mapping Indicator 15.1.1 (2015) on forested area as a proportion of total area of the M49 sub-regions and intermediate regions rather than countries depicts an incorrect pattern of forestation in Middle Africa. Chad, an arid Sub-Saharan African country highlighted in the map, is only 5% forested but appears dark green in the resulting map given the tropic regions to the south also included within the “Middle Africa” intermediate region. In this case, the coarser resolution may lead to an incorrect understanding of forestation (and therefore deforestation) in Chad and the biogeography of Africa broadly. Unlike many of the SDG indicators, Indicator 15.1.1 has a multimodal rather than skewed distribution due to the high variability of precipitation by latitude. The resulting map uses an optimal breaks classification (see [Section 1.9](#)).

graphic scale, with coarse enumeration units combining adjacent boundaries from the detailed zones of aggregation. Disaggregation from national- to local-level enumeration requires sophisticated mapping techniques, such as dasymetric mapping (see [Section 3.5](#)), making it inappropriate to infer assumptions about local-level conditions from SDG indicator data on their own.

MAUP and geographic scale may result in the **ecological fallacy**: the same individual-level data enumerated at different boundary resolutions (i.e., geographic scales) results in different statistical relationships in the enumer-

ated attributes. Regarding the SDG indicators, perceived patterns across space for a given indicator and correlations between indicators may be an artefact of the arrangement, shape, and size of the enumeration units used to process and transform the data, not the actual geographic phenomenon the SDG indicator is designed to capture. For example, when aggregating SDG indicators from individual countries to M49 geographic regions, the resulting attribute data tells a different story about the region that some constituent countries may debate ([Figure 1.8-2](#)).

Normalization



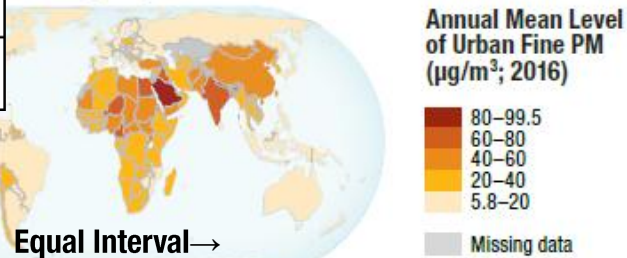
Roth et al. (forthcoming)

Making and Remaking Mapping for a Sustainable World

1.9 Data Classification

Classification describes the process of organizing map features into groups

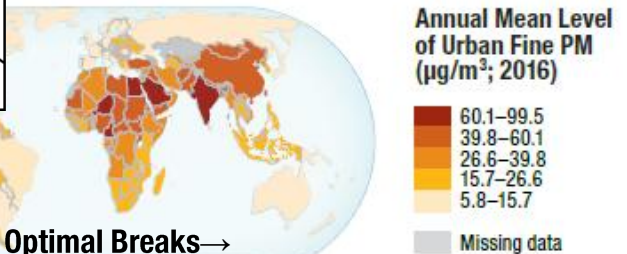
to arrive at an appropriate classification scheme, such as the portion of



Equal Interval → Uniform Distributions

Figure 1.9-2 Ordinal map reclassifies all unique rankings to show only the ranking above and below Norway's value). Classification grows in difficulty when applied to numerical data. As with all

geometric for skewed attribute distributions, *equal interval* for uniform attribute distributions (Figure 1.9-2), *optimal breaks* for multimodal attribute distributions (Figure 1.9-3), and



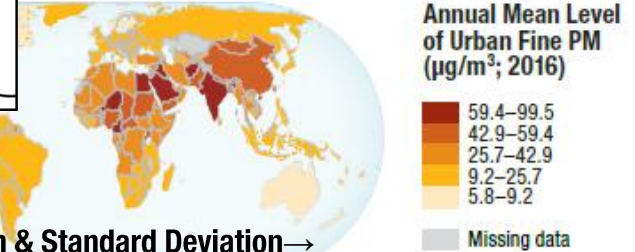
Optimal Breaks → Clustered Distributions

Annual Mean Level of Urban Fine PM ($\mu\text{g}/\text{m}^3$; 2016)

Figure 1.9-1: Data distributions and classification. **Left:** The histogram depicts the left-skewed attribute distribution for indicator 11.6.2 (2016) on the annual mean levels of urban fine particulate matter. **Right:** The resulting arithmetic scheme increases distances between class breaks in a regular progression, here expanding each class width by 5 $\mu\text{g}/\text{m}^3$ to provide more detail for features in the clustered side of the distribution rather than emphasizing outliers.

for ordinal level

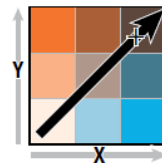
of the scale with several extreme outliers—the arithmetic scheme currently is



Mean & Standard Deviation → Normal Distribution

Annual Mean Level of Urban Fine PM ($\mu\text{g}/\text{m}^3$; 2016)

Figure 1.9-3: Optimal breaks. Optimal breaks treats classes like clusters, minimizing differences within the class while maximizing differences



Annual Mean Level of Urban Fine PM ($\mu\text{g}/\text{m}^3$; 2016)

Annual Mean Level of Urban Fine PM ($\mu\text{g}/\text{m}^3$; 2016)

Quantile → Attribute Comparisons

✓ Consider the attribute distribution when selecting a classification scheme

1.9 Data Classification

Classification describes the process of organizing map features into groups to improve legibility in the representation. Classification is one of the ways that cartographers generalize thematic

maps, reducing attribute data to a limited number of classes (Section 2.7). It adds uncertainty (Section 2.13), can be heavily influenced by the selection of class

boundaries between classes. Classification of all levels of ordinal and nominal data (Figure 1.7-2) can be applied to numerous aspects of cartographic classification. Cartographers

use various methods to arrive at an appropriate **classification scheme**, such as the portion of the distribution the cartographer wishes to emphasize, the total number of classes (with most schemes using four to seven

classes on different scales or for ordinal level attributes (Figure 1.9-5). Figures 1.9-2, 1.9-3, 1.9-4, and 1.9-5 reclassify Figure 1.9-1 data, showing how classification impacts the resulting

map of the scale with several extreme outliers—the arithmetic scheme currently is

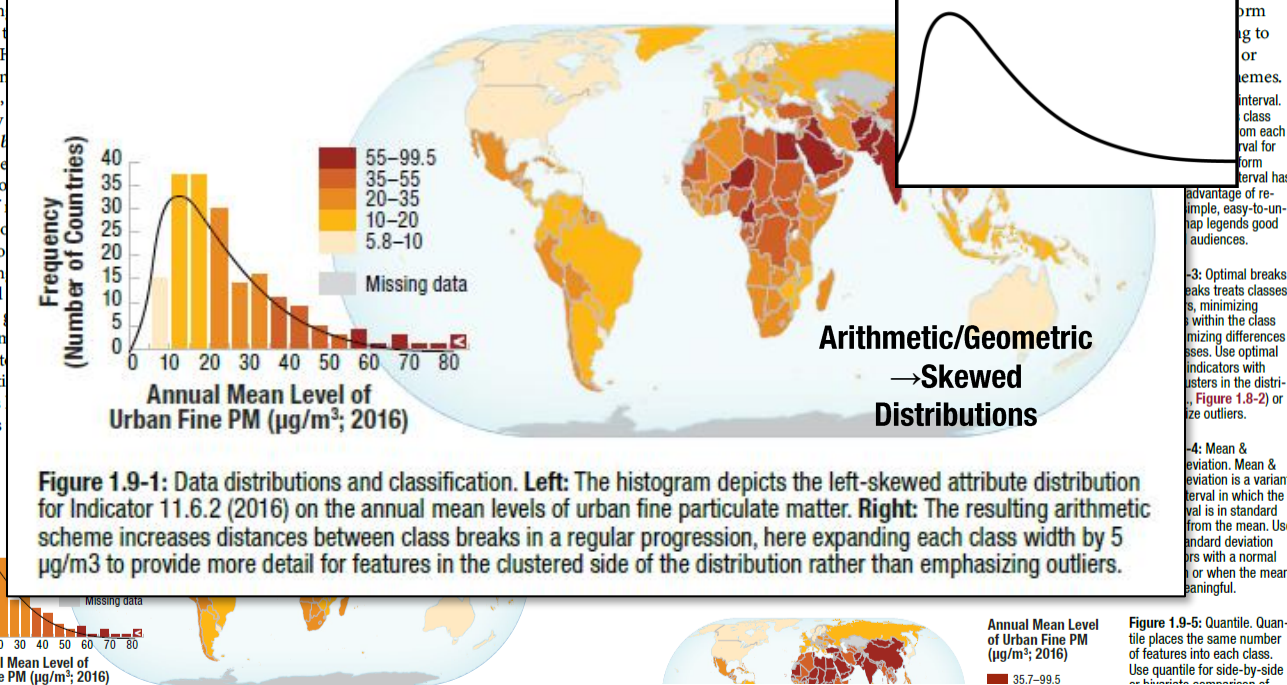


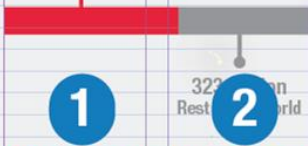
Figure 1.9-1: Data distributions and classification. Left: The histogram depicts the left-skewed attribute distribution for Indicator 11.6.2 (2016) on the annual mean levels of urban fine particulate matter. Right: The resulting arithmetic scheme increases distances between class breaks in a regular progression, here expanding each class width by 5 µg/m³ to provide more detail for features in the clustered side of the distribution rather than emphasizing outliers.

✓ Consider the attribute distribution when selecting a classification scheme

GOAL 1: END POVERTY IN ALL ITS FORMS EVERYWHERE

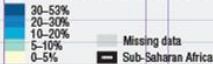
► 736 million people lived in extreme poverty in 2015

413 Million
Sub-Saharan Africa



► Most people that live on less than \$1.90 a day live in Sub-Saharan Africa

Proportion of Population Living Below the International Poverty Line (most current value 2012–2018)



The UN Secretary-General meets people living in a camp for internally displaced persons (IDPs) in the town of Bangassou, Central African Republic. (Source: UN Photo/Eskinder Debebe, 2017)

72pt Gutter

12pt rule

12pt margin

78pt columns

✓ Use the grid to inform layout, seeking balance but asymmetry



▲ The map depicts Indicator 1.1.1 (most current value 2012–2018) on the proportion of population living below the international poverty line (set at \$1.90 per day) as a choropleth by SDG groupings. The M49 standard is a multi-level, global set of region, sub-region, and intermediate region groupings for obtaining greater homogeneity in sizes of demography. The SDG groupings are derived from the M49 methodology and use a combination of regions and sub-regions.

Indicator 1.1.1 is a ratio level, relative value (a proportion), and thus is normalized for choropleth mapping to mitigate effects from the modifiable areal unit problem. The choropleth map uses an arithmetic classification for the left-skewed attribute distribution and a sequential colour scheme for an apparent increase from low-to-high.



GOAL 1: END POVERTY IN ALL ITS FORMS EVERYWHERE

► 736 million people lived in extreme poverty in 2015

413 Million
Sub-Saharan Africa

1

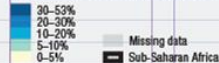
2

3

4

► Most people that live on less than \$1.90 a day live in Sub-Saharan Africa

Proportion of Population Living Below the International Poverty Line (most current value 2012–2018)



The UN Secretary-General meets people living in a camp for internally displaced persons (IDPs) in the town of Bangassou, Central African Republic. (Source: UN Photo/Eskinder Debebe, 2017)

72pt Gutter

12pt rule

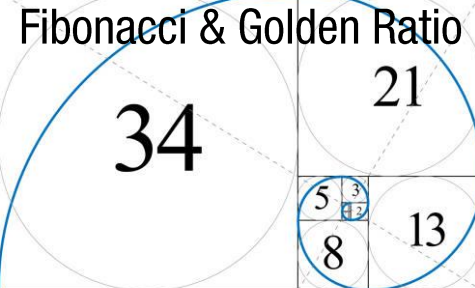
12pt margin

78pt columns

✓ Use the grid to inform layout, seeking balance but asymmetry

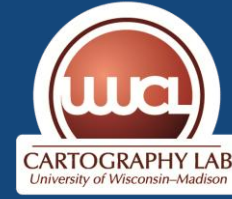


8-column grid:



▲ The map depicts Indicator 1.1.1 (most of the proportion of population living below the standard is a multi-level, global set of intermediate region groupings for obtaining sizes of demography. The SDG grouping methodology and use a combination of

Indicator 1.1.1 is a ratio level, relative value is normalized for choropleth mapping to a modifiable areal unit problem. The choropleth classification for the left-skewed attribute sequential colour scheme for an apparent

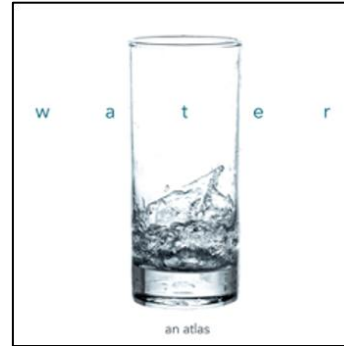


1.3: Lightning Talk 3 | Alicia Cowart, UW Cart Lab
Making an Atlas in a Day

Collective Atlas Creation

Inspired by
Guerrilla Cartography

guerrillacartography.org



GUERRILLA CARTOGRAPHY

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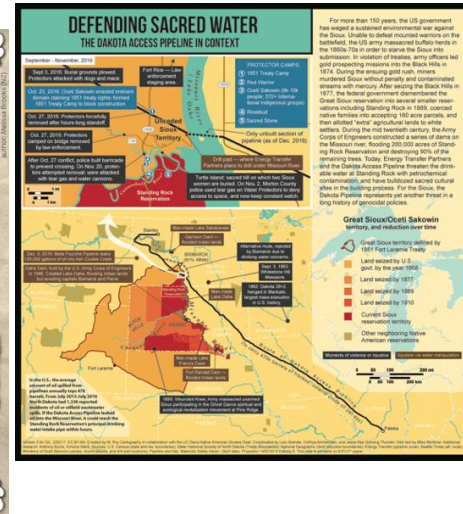
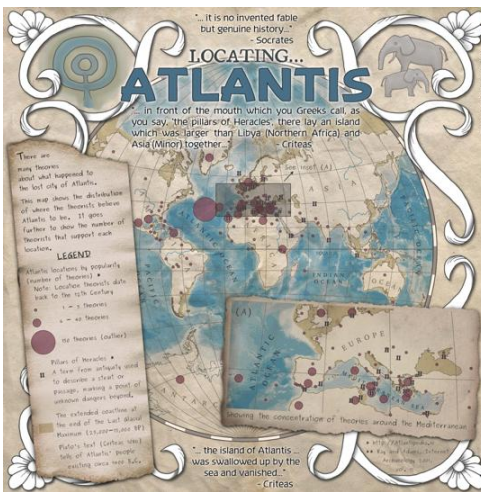
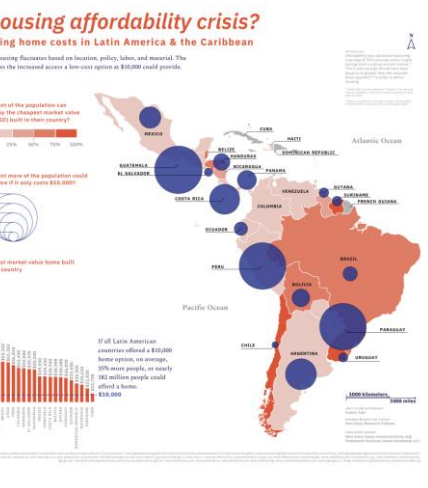
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[BLOG](#)

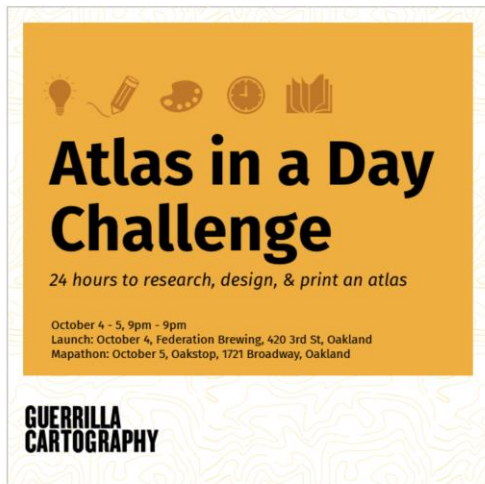
[DONATE](#)

We are a loose band of mapmakers, researchers, and designers intent on widely promoting the cartographic arts and facilitating an expansion of the art, methods, and thematic scope of cartography, through collaborative projects and disruptive publishing.

SHELTER
an atlas



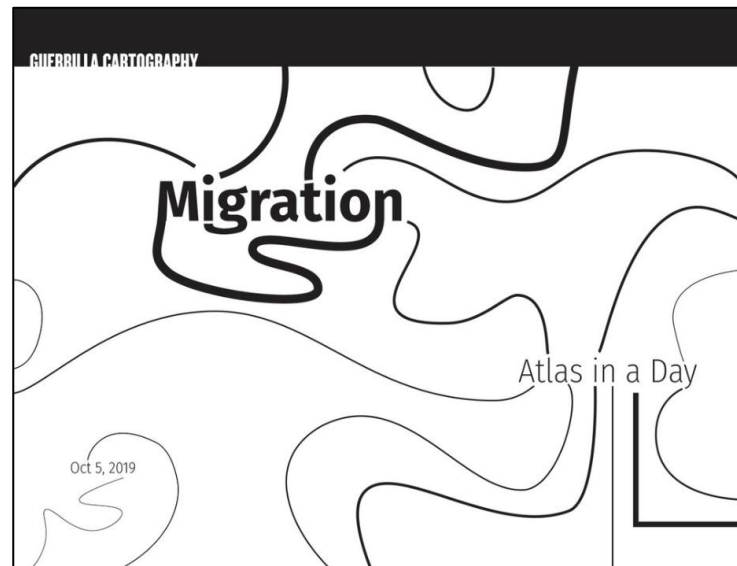
What is an Atlas in a Day?



ATLAS IN A DAY CHALLENGE #1

The theme of this challenge is ...

MIGRATION.



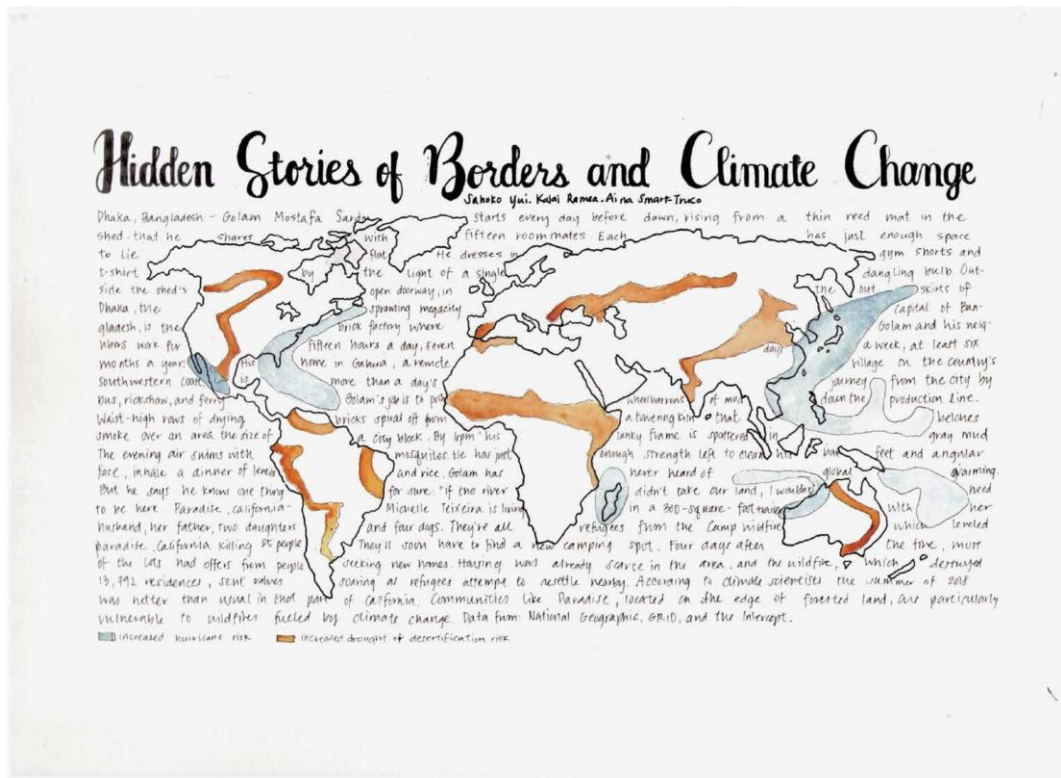
On Saturday, October 5, 2019, more than 50 guerrilla cartographers gathered, in person and online, to collaborate in producing an atlas in a single day on the subject of migration. Researching and designing an atlas on a complex theme in the span of 24 hours is indeed a challenge. This volume of maps is a concrete demonstration of the power of crowdsourcing and collaboration.

The subject of the atlas—migration, broadly defined—was only announced to the Guerrilla Cartography Collaborative via live stream at 9 pm PDT, Friday night. The following day, some 35 collaborators met at Oakstop in downtown Oakland, California, and began sharing ideas and working on their maps. Another 28 challenge participants worked on their maps remotely, representing 19 other locations in four countries. Creating on computers, paper, and boards; using crayons, watercolors, embroidery thread, and other media, the challenge collaborators submitted 43 maps on the theme of migration by the 7:30 pm deadline. A single electronic file was compiled and one rough print on a scrappy printer was attempted by 9 pm, fulfilling the challenge.



Why?

- ✓ Encourages creative collaboration
- ✓ Combines diverse styles & ideas into a cohesive whole
- ✓ Low stakes opportunity to experiment, learn and try new things
- ✓ Time constraints emphasize process over product



The Spread of Emerald Ash Borer Across the Northeast

Emerald ash borer (EAB), *Agrilus planipennis* Fairmaire, is an invasive Asian beetle that was discovered in south-eastern Michigan near Detroit in the summer of 2002. Since then it has spread throughout much of the eastern portion of the country. This map shows the spread in the Northeast over the past decade. Darker areas represent more recent occurrences. The path of the beetle can be inferred as you move from lighter to darker areas. Some of the occurrences in the northern counties likely originated from Canada.



Map prepared by Rick Lederer-Barnes, Cambridge, NY - @UpstateGIS

Data Sources: EAB occurrences USDA, Counties
US Census Bureau, Water features ESRI

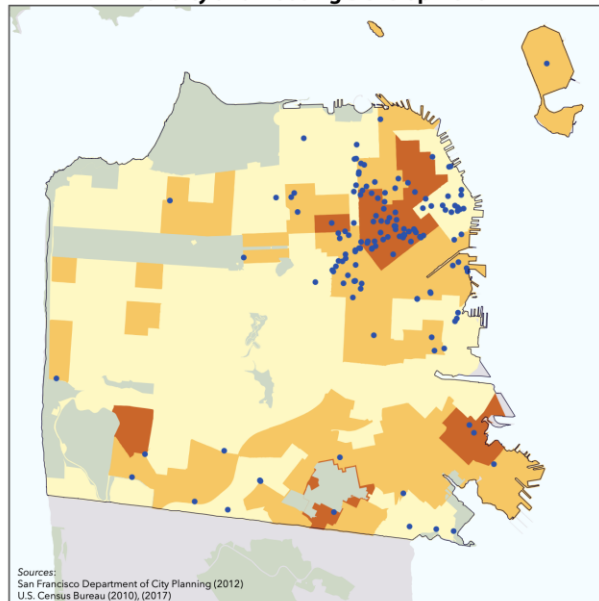
WHO'S IN AND WHO'S OUT

Over the past decade, housing prices in San Francisco have skyrocketed. High-poverty areas with low rents are attractive for new market-rate real estate development, as the disparity between current rents and the potentially achievable rent can lead to a high return on investment. Indeed, in San Francisco in the early 2010s, large new residential developments were concentrated in areas with high

Poverty, New Housing Development, and Residential Turnover in San Francisco, CA

poverty rates (quantified as those earning less than two times the federal poverty rate). By 2017, as new housing developments were completed, some of those areas also saw an influx of new residents (i.e. people who had moved into the area within the past two years). The city has seen high rates of residential turnover, as some low-income residents have been forced to leave for places with cheaper housing.

Poverty and Housing Development

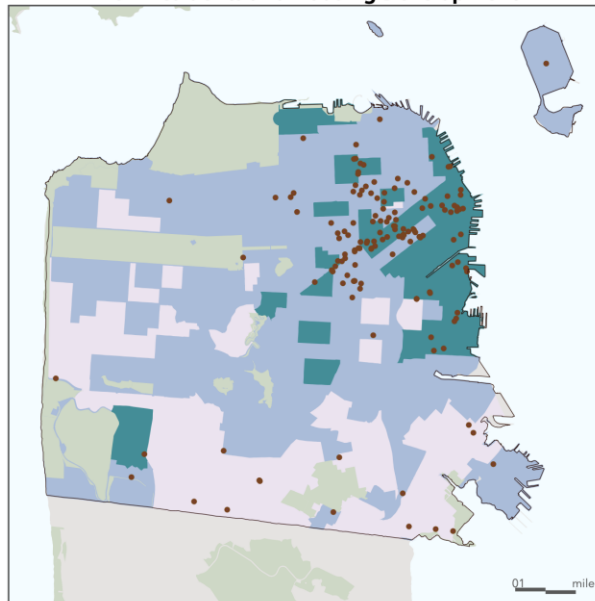


Population in Poverty (2010)
0-25%
26-50%
51% or more

**Development Pipeline
(2012, 20 units or more)**



New Residents and Housing Development



Moved In After 2014 (2017)
0-5%
6-15%
16% or more

**Development Pipeline
(2012, 20 units or more)**

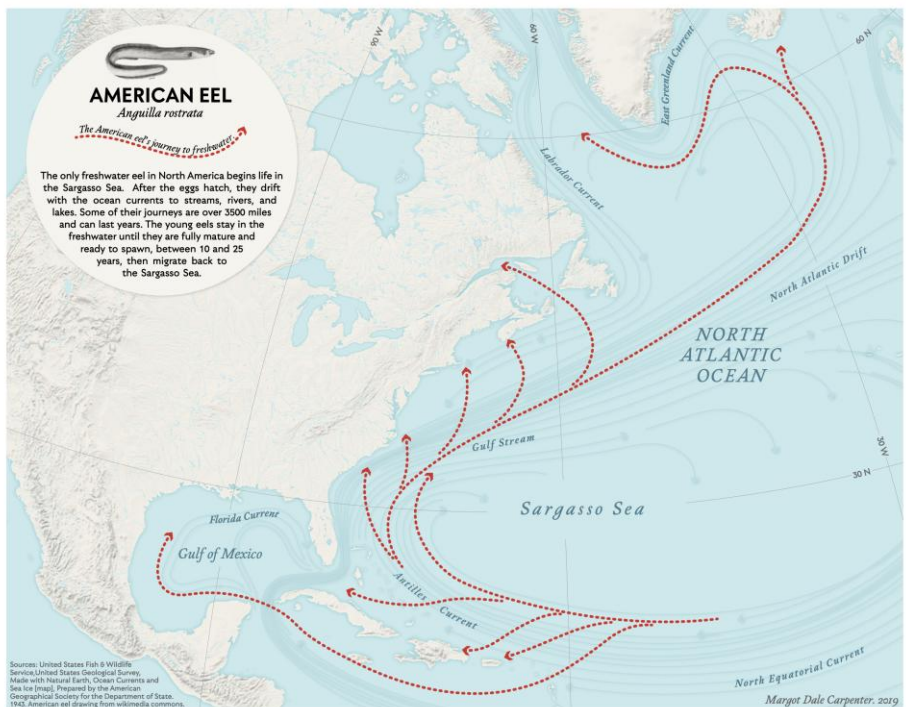


AMERICAN EEL

Anguilla rostrata

The American eel's journey to freshwater

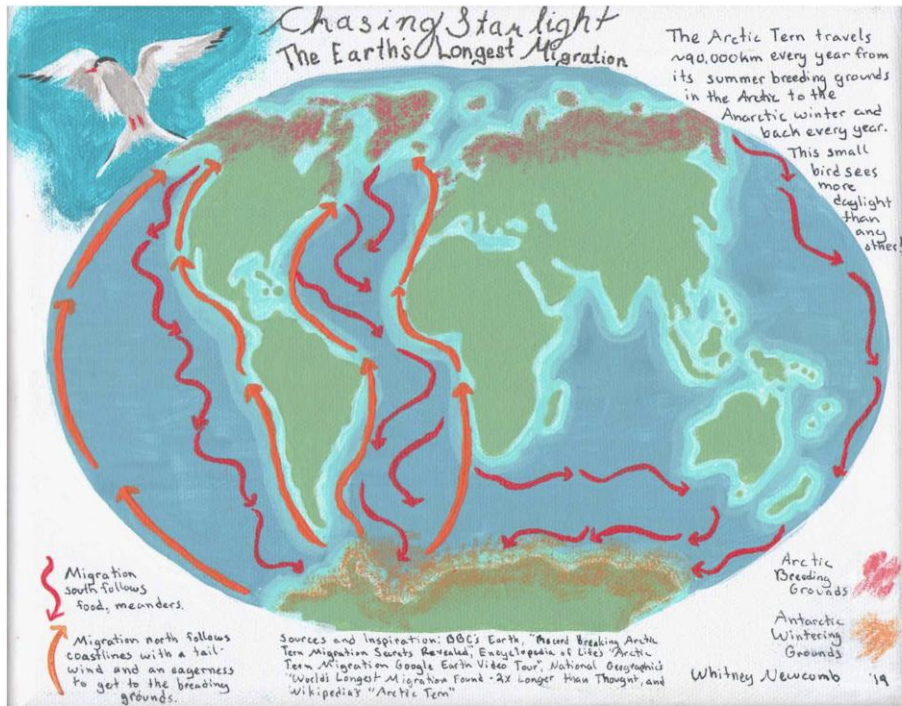
The only freshwater eel in North America begins life in the Sargasso Sea. After the eggs hatch, they drift with the ocean currents to streams, rivers, and lakes. Some of their journeys are over 3500 miles and can last years. The young eels stay in the freshwater until they are fully mature and ready to spawn, between 10 and 25 years, then migrate back to the Sargasso Sea.

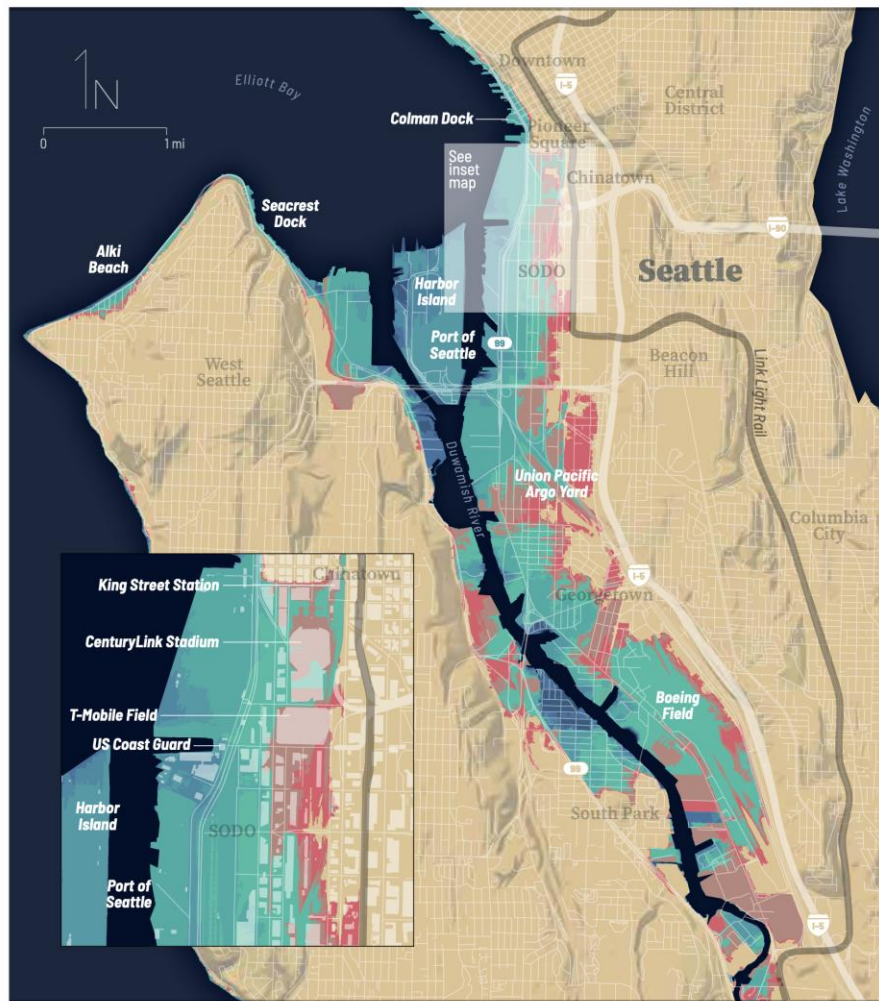


Sources: United States Fish & Wildlife Service, United States Geological Survey, Made with Natural Earth, Ocean Currents and Sea Ice (img). Prepared by the American Geographical Society for the Department of State, 1943. American eel drawing from Wikimedia commons.

Margot Dale Carpenter, 2019

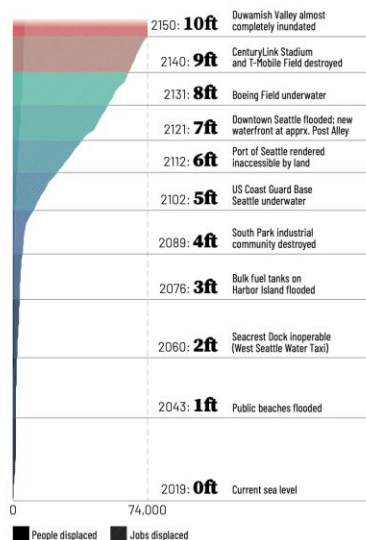
Chasing Starlight The Earth's Longest Migration





Migration away from flooded land in Seattle

How many people and jobs would be displaced by rising sea levels in the Duwamish River Valley?

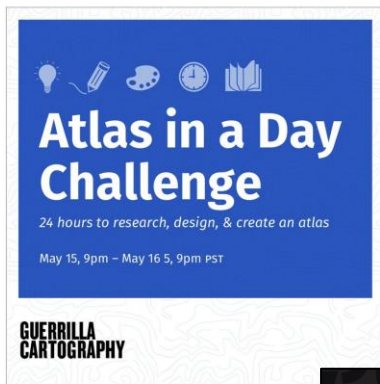


Sources

City of Seattle, ESRI, King County, Microsoft, NOAA, Sound Transit, US Census Bureau.
Sea level rise estimates from 1% probability of exceedance, average of 10-year period.
Estimates are approximations of outcomes under a high-emissions scenario and assuming no major mitigative infrastructure is built.

<http://www.watcoastalnetwork.com/files/theme/wcrp/SLR-Report-Miller-et-al-2018.pdf>, p. 14

What is an Atlas in a Day?



ATLAS IN A DAY CHALLENGE

The theme of this challenge is ...

COMMUNITY.



• **9:15am PDT: Welcome** by **Darin Jensen**, GC founder and President of Board of Directors
Darin will speak about the global community formed by **Guerrilla Cartography** since its first atlas, *Food: An Atlas*, was published in 2012.

• **9:30am PDT: Speaker session #1**

- **Daniel Raven-Ellison**, Guerrilla Geographer, London, UK:
 - *Mapping a 90K network of Slow Ways walking routes while in Lockdown*
- **Sahoko Yui, Kalai Ramea, Aina Smart-Truco**, Oakland, California:
 - *Hidden stories of Atlas in a Day*
- **Martina Neuburger, Katharina Schmidt, Katrin Singer**, Institute of Geography, University of Hamburg, Hamburg, Germany:
 - *Questioning own mapping practices: academic feminist communities, positionality and historical legacies*
- **Annie Brulé**, Brulé Illustration & Design, Vashon, Washington:
 - *Decolonizing the Map: Indigenous-led Mapping Initiatives*
- O/A begins at 10:30am PDT

• **12pm - 1:15pm PDT: Speaker session #2**

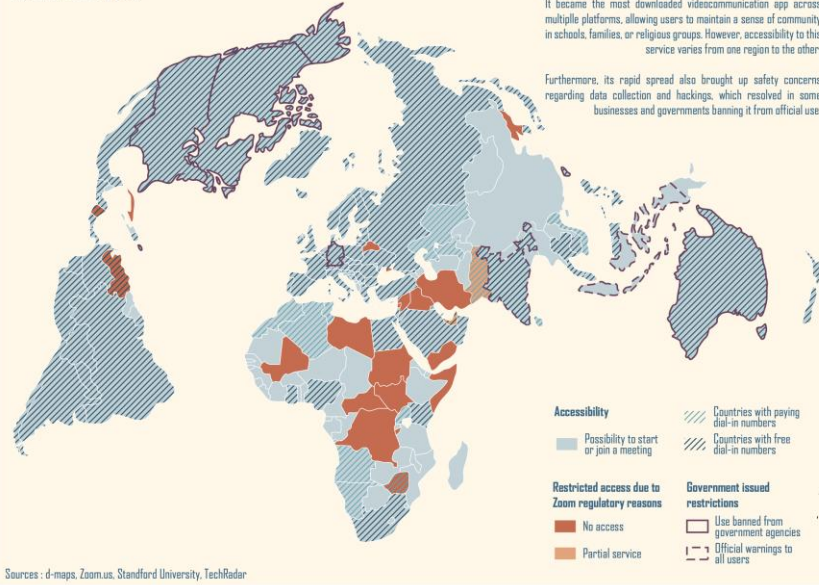
- **Anirvan Chatterjee and Barnali Ghosh**, Berkeley South Asian Radical History Walking Tour, Berkeley, California:
 - *The 1940 Singh Census: Visualizing the story of the Punjabi/Mexican community*
- **Alex Tarr**, Assistant Professor of Geography, Worcester State University, Worcester, Massachusetts:
 - *Cartographers are not Computers*
- **Annita Lucchesi**, University of Arizona, Tucson, Arizona:
 - *Mapping for Justice: Creating Cartographies of Social Change*
- **Melissa Brooks**, GIS Analyst/Map UX, New Plymouth, New Zealand:
 - *Quick tips to review and refine your map*
- **Speaker session #2 Q/A** begins at 1pm PDT

• **4:00pm - 5:15pm PDT: Speaker session #3**

- **Sapna E. Thottathil**, PhD, Associate Director of Sustainability, University of California, Office of the President and Guerrilla Cartography Board Member, Oakland, California:
 - *Institutions as Anchors in Communities*
- **Iconoclastas**, Mapping Collective, Argentina:
 - *Collective mapping and collaborative research from Argentina* (this talk will be presented in Spanish)
- **Steven R Holloway**, toMake™ Press & Editions, Missoula, Montana:
 - *The deLight of Slow Map-making & Responding*
- **Amber J. Bosse**, PhD Candidate, University of Kentucky, Provo, Utah:
 - *Map Anyway*
- **Speaker session #3 Q/A** begins at 5pm PDT

Creating community in a time of social distancing ? Zoom.us : Degrees of accessibility and caution

Abigail Rabinovitch, Paris (2020)

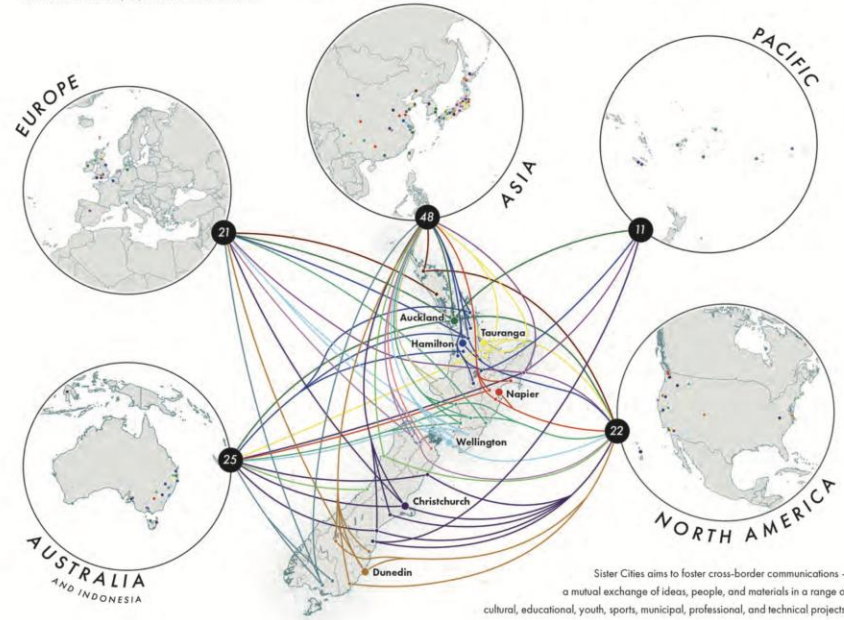


Sister City Connections of New Zealand

Made in 24hr for Adas in a Day May 2020 in Christchurch, New Zealand

Andrew Douglas-Clifford (@themagkwi)
and Andrew Tyrrell (@southarrowmaps)

Date: ESRI, LINZ
sistercities.org.nz



THE ICHETUCKNEE SILTSNAIL *Floridobia mica*

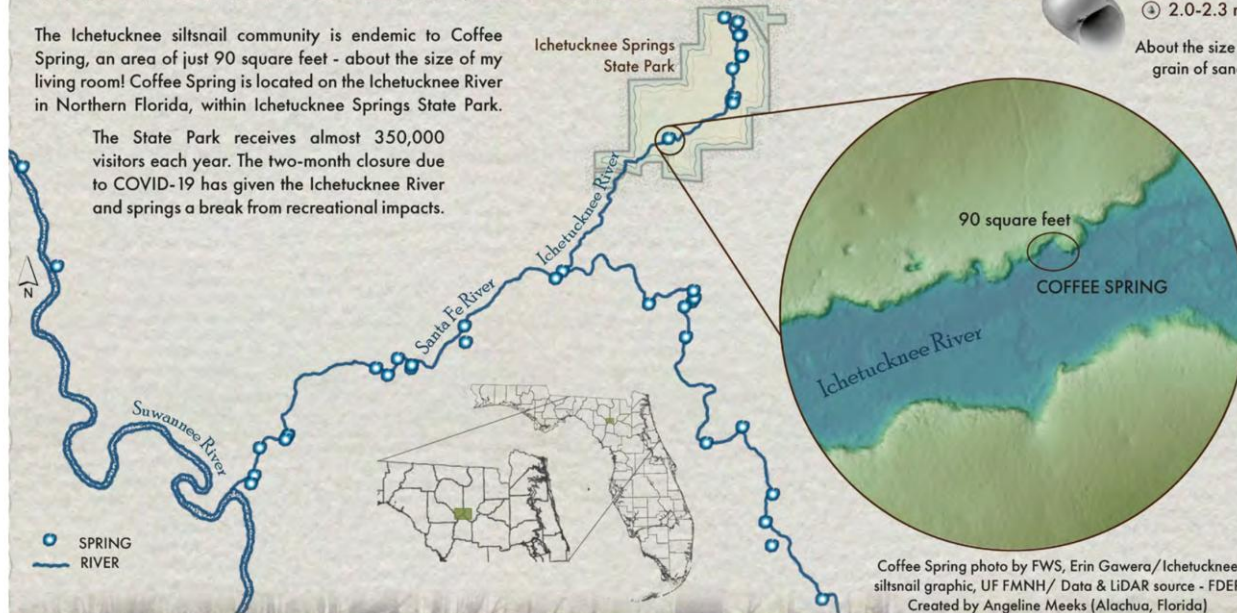


ACTUAL SIZE:
④ 2.0-2.3 mm

About the size of a
grain of sand!

The Ichetucknee silt snail community is endemic to Coffee Spring, an area of just 90 square feet - about the size of my living room! Coffee Spring is located on the Ichetucknee River in Northern Florida, within Ichetucknee Springs State Park.

The State Park receives almost 350,000 visitors each year. The two-month closure due to COVID-19 has given the Ichetucknee River and springs a break from recreational impacts.



Coffee Spring photo by FWS, Erin Gawera / Ichetucknee
silt snail graphic, UF FMNH / Data & LiDAR source - FDEP
Created by Angeline Meeks (Alachua, Florida)





The Yahara River

The Yahara River in Madison, Wisconsin has been part of my community for five years providing open space for gathering, kayaking, and time outdoors. I began the map further left over three years ago and as I revisited it, I found several map layers missing. This series highlights taken-for-granted layers that acknowledge a settler colonial history as well as a space marked by social and environmental crises. These layers are the layers that we typically choose not to see. Just like invisibilities in maps, my own sense of community and positionality allow me to see or unsee the many layers around the Yahara River.

Tenney Park Locks

In Madison, the Yahara River is a controlled waterway that connects two lakes, Lake Mendota to the north and Lake Monona to the south. The Lock at Tenney Park serves as a reminder of altered landscapes and the continued occupation of Ho-Chunk ancestral lands. In an 1832 treaty, the Ho-Chunk were forced to cede this territory, including the Yahara River. Decades of ethnic cleansing followed when both the federal and state governments repeatedly, but unsuccessfully, sought to forcibly remove the Ho-Chunk from Wisconsin.

2018 Flooding

In August 2018, Madison experienced a massive rain event receiving between 11 and 13 inches of rain in one afternoon. The banks of the Yahara River spilled over as Lake Mendota reached capacity (dark blue), while flash flooding due to current surface pooled along local roads and neighborhoods (light blue). Many lives and homes were directly impacted. This event will not be the last and serves as a glimpse of what climate change will look like in Madison and in the Midwest.

¹and acknowledge based on University of Wisconsin-Madison's own acknowledgment of occupying Ho-Chunk lands.

Housing 2020

COVID-19 public health crisis has exposed structural systems of oppression and privilege that have shaped and continue to shape access to testing, employment, food, transportation, healthcare, childcare, education, and housing, to name a few. Housing issues have become even more pronounced as states and cities across the country are laid that its "Sale at Home" leaving many people experiencing homelessness in extraordinary circumstances. The City of Madison's emergency response permitted temporary encampments in city parks and greenways. This map layer depicts a recent encampment along the Yahara River in green juxtaposed with neighborhood buildings in grey.

Fishing communities of the Tonlé Sap are under threat



A FISHY TALE

It's raining hard and we've waited for this, confined in this refuge of ours. For months we have waited. Like we have, so many times before.

Rivers rush forward, and the current takes us in. It's strong and we are weak. It takes us where it will. Like it has, so many times before.

Here are the four faces, and the water keeps flowing in. The current takes a turn, and sweeps towards North. We're on a pilgrimage to the promised waters. Once again, like so many times before.

The water breaks the barriers and flood enters the forest. Here we feast, and here we make. Here we thrive, and the fishermen too, under the shade of the mangrove trees. Like so many times before.

The rains are gone, and the flood recedes. It's our time to go, like so many times before. The current's grown weak, and we strong. It's time to head for shelter.

Here in the gentle waters of our tributary, we've completed the annual cycle. Here in our precious deep pool, growing ever shallower, we wait. Like countless times before.

Here in our precious little pond, I ponder: will the waters come, like they always have, so many times before. The flooded forest each year smaller, our refuge never quite so shallow, will they release the water, behind their towering walls.

¹Tonlé Sap Lake is the most productive freshwater fishery in the world, supplying up to 80% of Cambodian's protein. Fishing and farming communities in Cambodia, particularly dependent on the Tonlé Sap Lake, are under threat from rampant dam construction in the Mekong Basin.

The fish depend on the annual flood pulse for their feeding and reproduction. Dam building in the basin has already affected the fish catch and the extent of the floods.

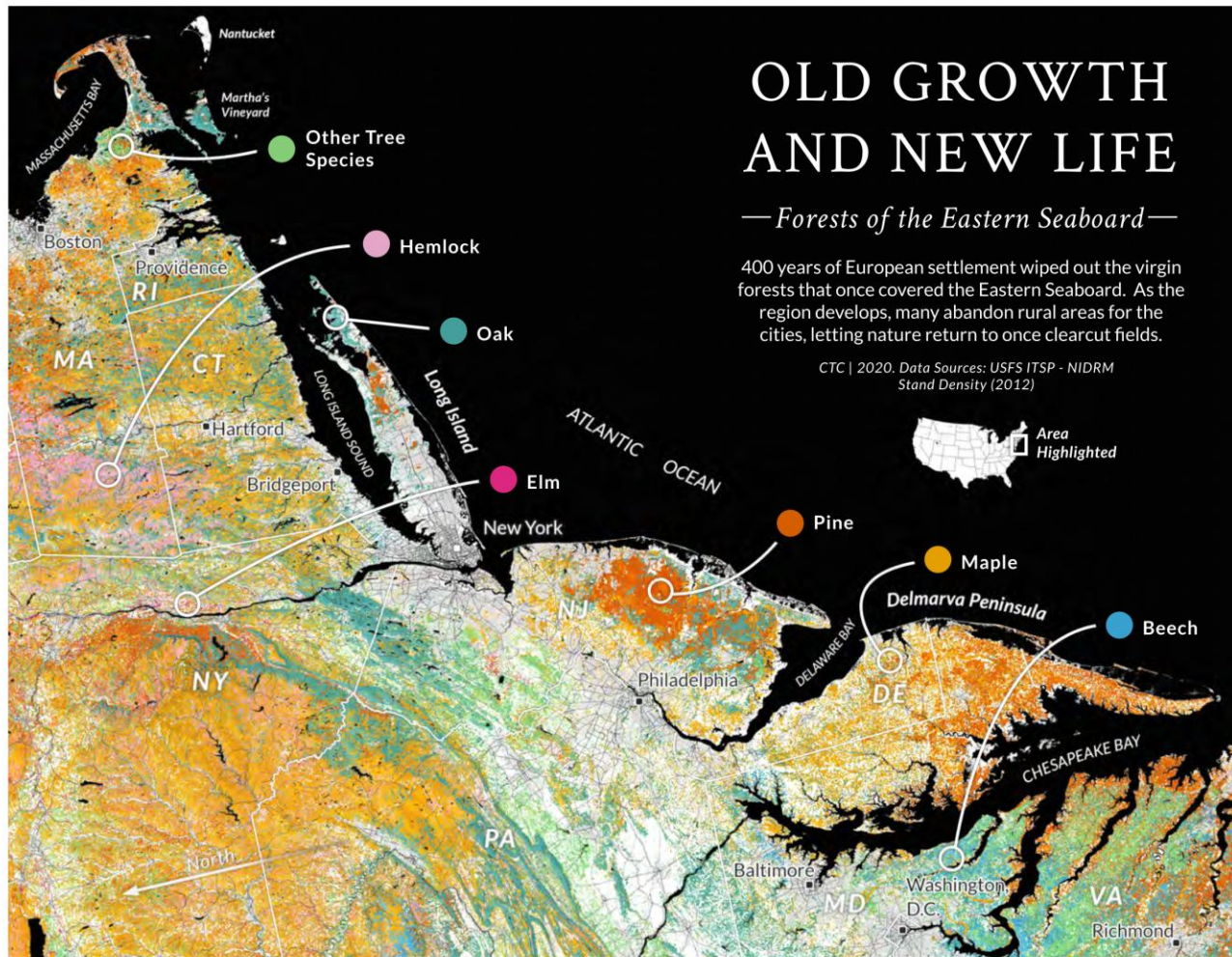
The age old annual cycle of life, for both the fish and the communities which evolved around the cycle, is under risk of disappearing entirely.

OLD GROWTH AND NEW LIFE

—Forests of the Eastern Seaboard—

400 years of European settlement wiped out the virgin forests that once covered the Eastern Seaboard. As the region develops, many abandon rural areas for the cities, letting nature return to once clearcut fields.

CTC | 2020. Data Sources: USFS ITSP - NIDRM
Stand Density (2012)



What is an Atlas in a Day?

Concept adopted and adapted for UW
Design Challenge 2024

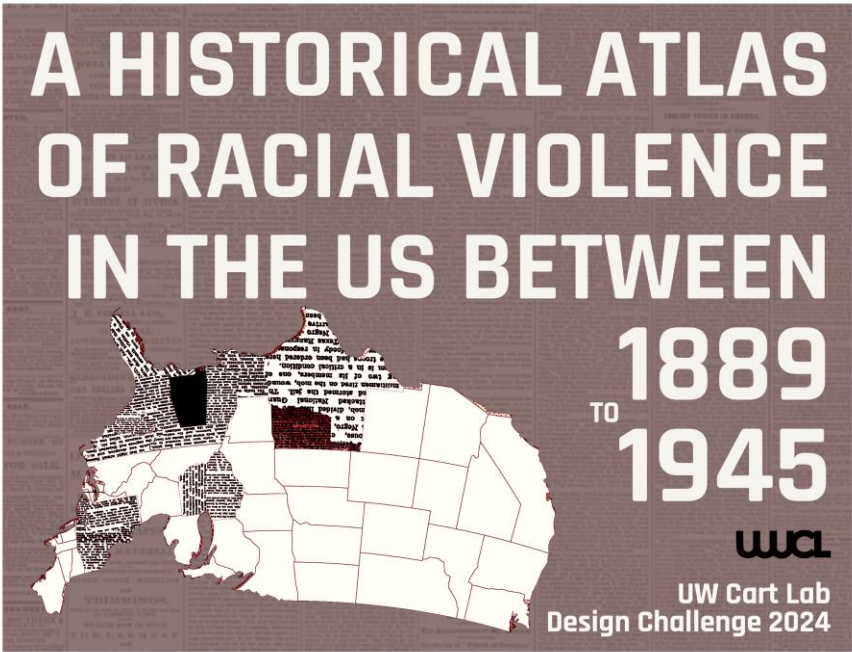




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The Making of Southern Horrors

"...the dark and bloody record of the South shows 728 Afro-Americans lynched during the past eight years"

In writing the pamphlet *Southern Horrors: Lynch Law in All Its Phases*, anti-lynching activist and journalist Ida B. Wells was influenced by the sheer number of incidents of white mob violence against black communities in the South and the pervasive stereotypes of black men in the media.

However, the pamphlet would not have been possible without her personal encounters with white mob violence in Memphis in 1892 and the love and labor of black women in her community in New York City during her exile.

All quotes from *Southern Horrors*.
Nigam Ramachandran 2024

Lynching at the Curve
"On March 9, 1892, there were lynched in this same city three of the best specimens of young since-the-war Afro-American manhood. They were peaceful, law-abiding citizens and energetic business men...They owned a flourishing grocery business in a thickly populated suburb of Memphis.

Three of these men, the president, the manager and clerk of the grocery—"the leaders of the conspiracy"—were secretly taken from jail and lynched in a shockingly brutal manner. "The Negroes are getting too independent," they say, "we must teach them a lesson."

the Curve
"People's Grocery Store"

White Mob Destroys the Memphis Free Press Office
On May 27th, 1892, a white mob destroyed the office of Ida B. Wells' newspaper, forcing her into exile in New York City. The move North, and the lynching of the three black men at the Curve pushed Wells to research lynchings in the South and write *Southern Horrors*.

1892: Memphis, TN

"To the Afro-American women of New York and Brooklyn, whose race love, earnest zeal and selfless effort at Lyric Hall, in the City of New York, on the night of October 5, 1892—made possible its publication, this pamphlet is gratefully dedicated by the author.

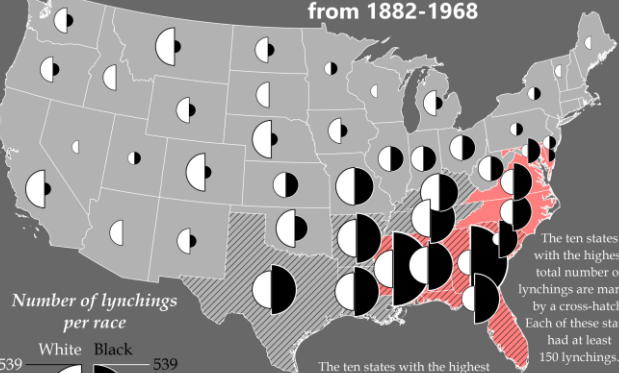
The Afro-American is not a bestial race. If this work can contribute in any way toward proving this, and at the same time arouse the conscience of the American people to a demand for justice to every citizen...I shall feel I have done my race a service"

—Ida B. Wells

- White mob violence in Memphis
- White mob violence recorded in *Southern Horrors*

Lynching data provided via the Tuskegee Institute
Research provided via NCHRS
Map Created By: Mei Mei Kohane

In Black and White: A Map of Lynchings Across the United States from 1882-1968



Just Another Day in America

Yanbing Chen & Brynn Patrello: 2024



Tangible and Digital Flow Map of Historical Lynchings in United States

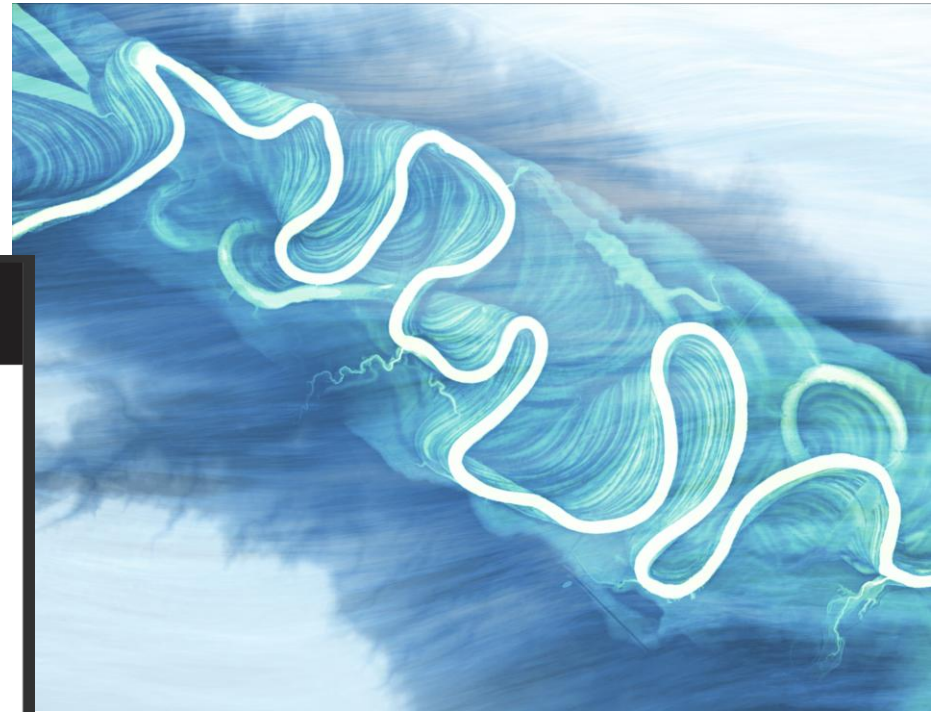
by Atlas Guo

(UW Cartography Lab Design Challenge 2024)



Atlas in a Day Evolves

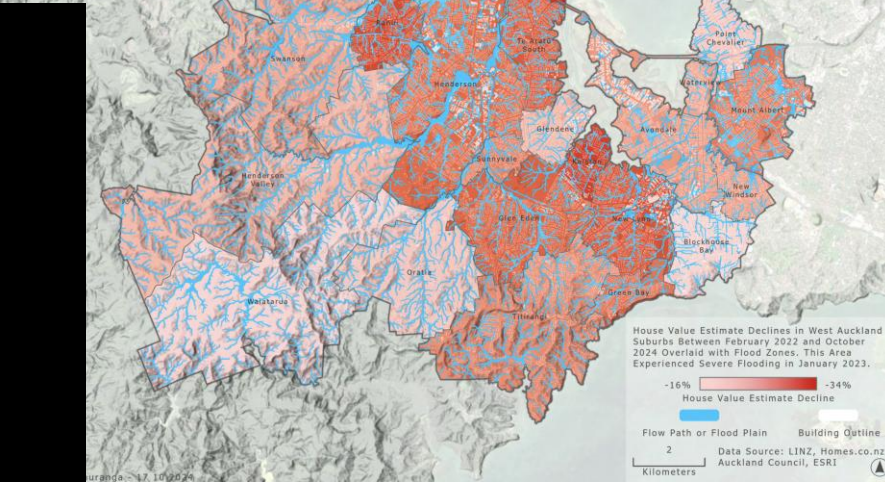
Atlas at a Conference



Atlas in a Day Evolves

A Flow of Disadvantage

Where floodwater flows, it's more than just water that goes. There's a flow of hope and security leaving the region, followed by disinvestment from institutions only eyeing returns. Over time, these areas become relegated to those with no choice but to remain, communities trapped by rising insurance costs and shrinking opportunities. As a flowing flood deepens a valley so to it deepens the divides in society, where the most disadvantaged are left behind.



What if Native Americans still lived on their lands?
What would a hydrography map of Texas look like?

River names from Texas State Historical Assn, Wikipedia, and the Texas Almanac. I welcome your additions and corrections to these names of tribes and rivers. Its purpose is to inspire creation of maps that highlight the languages of Native Americans. Where a name could not be located, I labeled the major river with the current name with an outline of letters and no fill color.

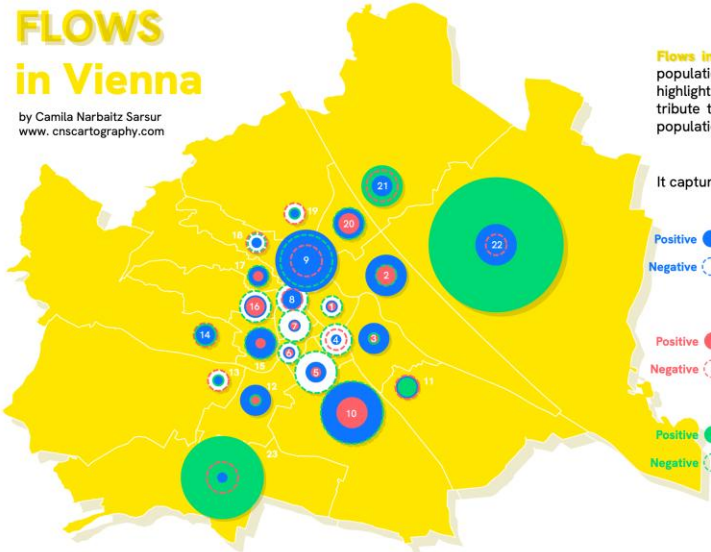
Sally Holt, October 19, 2024

Names of tribes from Indian tribes, cultures & languages map by National Atlas (1967).
Hydrography from 1:1,000,000-Scale Streams of Texas by Nat. Atlas (2014).
Color shaded relief from Natural Earth v3.2.0 (2024).

Atlas in a Day Evolves

Flows in Vienna

by Camila Narbaitz Sarsur
www.cns-cartography.com



Flows in Vienna map visualizes the population dynamics within the city, highlighting the movements that contribute to changes in each district's population.

It captures three types of migration:

- Positive** **International Migration**
Residents moving abroad and newcomers arriving from other countries.
- Negative**
- Positive** **Domestic Migration**
Residents moving to more suburban or rural areas in Austria and newcomers from this areas to Vienna.
- Negative**
- Positive** **Internal City Migration**
Relocations within Vienna.
- Negative**

The 2nd, 9th, and 10th districts are receiving the highest levels of international migration, with more people arriving from abroad compared to those leaving for other countries. Notably, no district shows a negative balance in international migration, indicating that Vienna, as a whole, is attracting more international residents than it is losing to emigration.

The 9th, 21st, and 23rd are the districts experiencing net losses from domestic migration, as people seek opportunities, quieter lifestyles, or different living conditions elsewhere in

The 21st, 22nd, and 23rd districts are experiencing the most significant population growth due to internal relocations within Vienna, as they attract many residents from other parts of the city. In contrast, the central districts are seeing the lowest levels of inward migration and are losing residents to outer districts as more people move away from the city center.

Data source: City of Vienna - <https://data.wien.gv.at>

Scheduled domestic flights

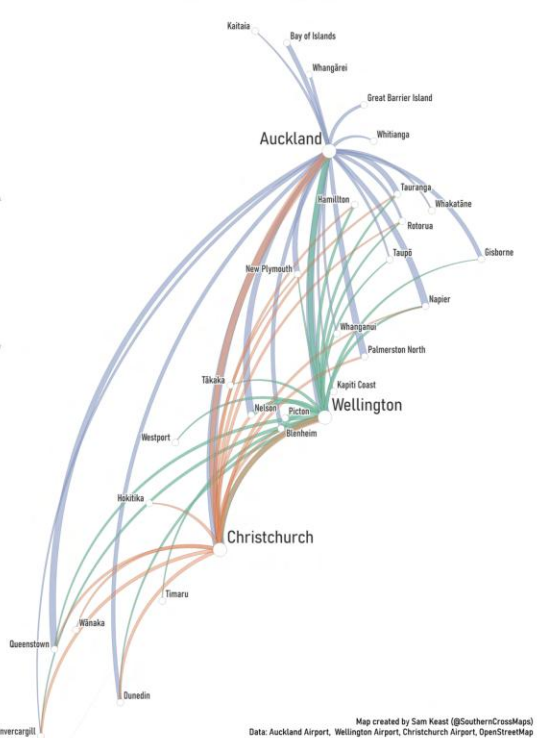
for the 18th October, 2024, from the three largest airports in New Zealand

Flights departing from

Auckland
Wellington
Christchurch

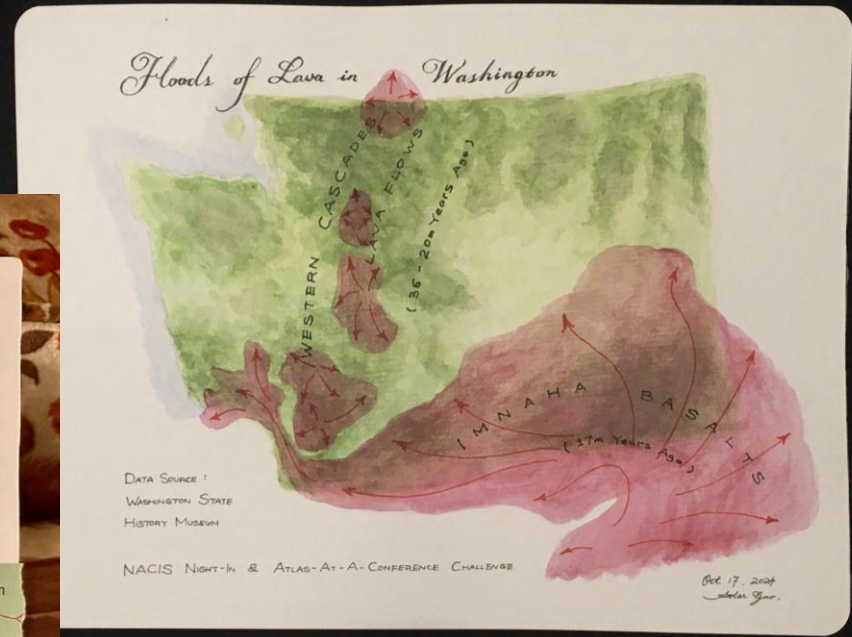
Frequency of flights

5 15 30



Map created by Sam Keast (@SouthernCrossMaps)
Data: Auckland Airport, Wellington Airport, Christchurch Airport, OpenStreetMap

Atlas in a Day Evolves



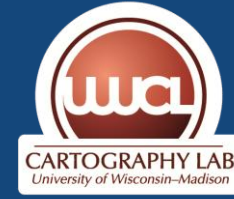
Atlas in a Day Evolves

Atlas in a Week – Design Challenge 2025

Enjoy the process!

- ✓ Creative collaboration
- ✓ Combine diverse styles & ideas into a cohesive whole
- ✓ Low stakes opportunity to experiment, learn and try new things
- ✓ Time constraints emphasize process over product





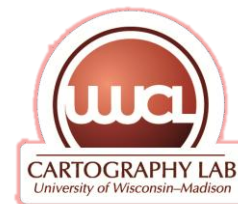
Unit 2: Team Formation & Brainstorming | 12.05.25 11:30–13:00
Picking Your SDG & Brainstorming Your Topic!

Objectives for the Day

- 1. Connect data to stories:** Combine official SDG indicators with personal or community-based experiences
- 2. Practice design as an argument:** Use visual form—map style, symbols, layout—as part of the message
- 3. Reflect on information complexity:** Highlight contradictions, uncertainties, and/or alternatives
- 4. Collaborate across contexts:** Discover shared experiences within your team, particularly if you come from different regions



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- 13:00 to 14:00 // Lunch

• Unit 3: Design Sprint 1, Obtaining Your Data, Researching Your Topic

- 14:00 to 15:30 // Team Work (organizers available for questions)

- 15:30 to 16:00 // Creative Break, the Marshmallow Challenge (facilitated by Robert Roth)

• Unit 4: Design Sprint 2, Sketching & Organizing Your Visual Story

- 16:00 to 17:30 // Team Work (organizers available for questions) *The goal is to have cleaned, mappable data in a GIS by day's end*
- 17:30+ // Continue Work As Needed (Optional)

Day 2 // Tuesday 13.05.25

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 - 14:30 to 15:00 // Adobe Illustrator Demo
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Day 3 // Wednesday 14.05.25

- Initial Drafts Due in Google Slides by 16:00
- **Unit 6: Team Presentations & Draft Feedback**
 - 16:00 to 18:00 // Presentations
- 18:00+ // Social (!) *Recommendations appreciated!*

Day 4 // Thursday 15.05.25

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GitHub - uwcartlab/Vienna-DC-2025

+

github.com/uwcartlab/Vienna-DC-2025/tree/main?tab=readme-ov-file

README

2025 SDG Design Challenge / T/U(X)W

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The University of Wisconsin Cartography Lab ([Cart Lab](#)) Design Challenge is a day-long mapping workshop that brings together Cartography and Geography students, often in collaboration with campus or community partners, around a curated mapping theme. The Design Challenge is now an institution in the University of Wisconsin–Madison Geography Department, first running in 2015 and wrapping its 10th annual event this past February 2024. This year (officially #DC11) we are taking the Design Challenge international, combining efforts with the Technical University of Wien, the Erasmus Mundus MSc Cartography Programme, and the International Cartographic Association Commission on the User Experience (UX).

Specifically, we will be making a collective atlas of the [United Nations Sustainable Development Goals](#) ... *in less than one week!* The UN adopted the 2030 Agenda for Sustainable Development in 2015, an expansion from the prior 2000–2015 Millennium Development Goals, and defined a global indicator framework for collective transformation and action toward a more equitable and just world. At the time of this writing (October 2024), the SDG global indicator framework comprised 17 goals to address the most pressing problems facing our worlds, 169 targets that prescribe real and actionable outcomes for sustainability, and 231 unique indicators used to measure and monitor progress towards these targets, inform policy at global and local levels, and promote accountability across all stakeholders (<https://unstats.un.org/sdgs/>).

Prompt: SDG Atlas in a Week

Our SDG Atlas in a Week will include at least 17 two page spreads, one covering each of the SDGs plus some from front and back matter. Each spread will include one page for maps, charts, and clarifying annotation about the SDG and a second page providing a title and statement about what the maps are saying and what this means for our planet's future and our individual actions. You will work in a small team to develop one of the atlas spreads, but in addition to mapping SDG indicator data, we also want you to bring your own personal experiences and home locations into the map, telling a *compelling story* about your team's common ground around the SDG and the alternatives that are possible from your different, worldwide perspectives.

Packages

No packages published

Contributors 2

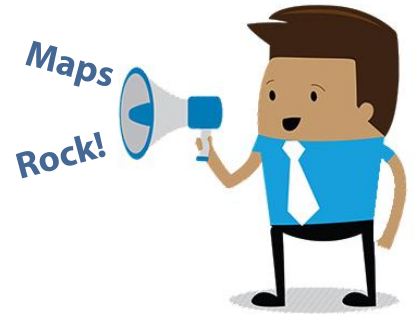
 **reeroth** Robert E. Roth

 **cartobaldrica** Gareth B

https://github.com/uwcartlab/Vienna-DC-2025

Unit 2 Objectives

1. Appraise the 17 SDGs based on experiences and interests
2. Organize into small groups according to mutual interests in the SDGs
3. Discuss professional and research interests in the SDGs
4. Formulate potential data-driven stories about an SDG
5. Apply unique experiences and interests to SDG stories



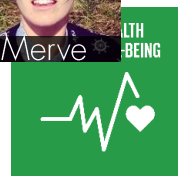
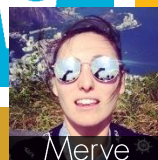
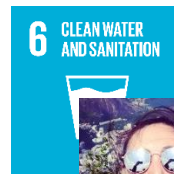
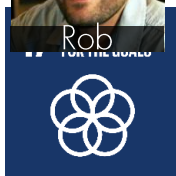




Alicia



physical
natural



Alicia



Storytelling Prompts:

1. What is the main point of your proposed project?

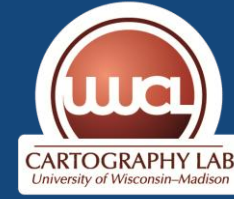
- What is necessary **background context** for understanding your SDG map? i.e., What is the most relevant indicator?
- What is the key **problem** driving your story? i.e., Why should we care?
- What is main **conclusion** of your story? i.e., What do you want audiences to remember?

2. Which place(s) are important regarding your topic?

- i.e., Who are the **character(s)** of your story? What is their cartographic scale? i.e., regions, countries, cities, etc.
- Begin looking at indicator data patterns: What **outliers** (high and low) or **changes** make interesting juxtapositions?
- From your experiences, what **alternatives or solutions** do these places represent? How should they change in the future?

3. What will your atlas spread include?

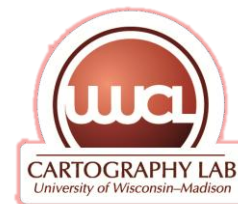
- What **maps, charts, images, and commentary** is important to tell your story?
- What do the SDG indicators show and **what do the data not show**?
- How do you want audiences to **feel** while reading the map and **behave** differently after seeing your map?



Unit 3: Design Sprint 1 | 12.05.25 14:00–15:30

Obtaining Your Data & Researching Your SDG

Schedule at a Glance



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• **Unit 4: Design Sprint 2, Sketching & Organizing Your Visual Story**

- 16:00 to 17:30 // Team Work (organizers available for questions) *The goal is to have cleaned, mappable data in a GIS by day's end*
- 17:30+ // Continue Work As Needed (Optional)

Day 2 // Tuesday 13.05.25

- **Unit 5: Design Sprint 3, Polishing Your Visual Story**
 - 14:30 to 15:00 // Adobe Illustrator Demo
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- 18:00+ // PhD Student Social (Alicia and Rob will crash!) *Recommendations appreciated!*

Unit 3 Objectives

1. Identify the main point of your story
2. Assess SDG indicator data availability and quality for an SDG story
3. Execute joins of SDG indicator attributes to geospatial boundary files in GIS software
4. Analyze patterns, trends, and uncertainties in SDG indicator data
5. Assemble UN and alternative geospatial datasets for an SDG story



GitHub - uwcartlab/Vienna-DC-2025

github.com/uwcartlab/Vienna-DC-2025/tree/main?tab=readme-ov-file

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Packages

No packages published

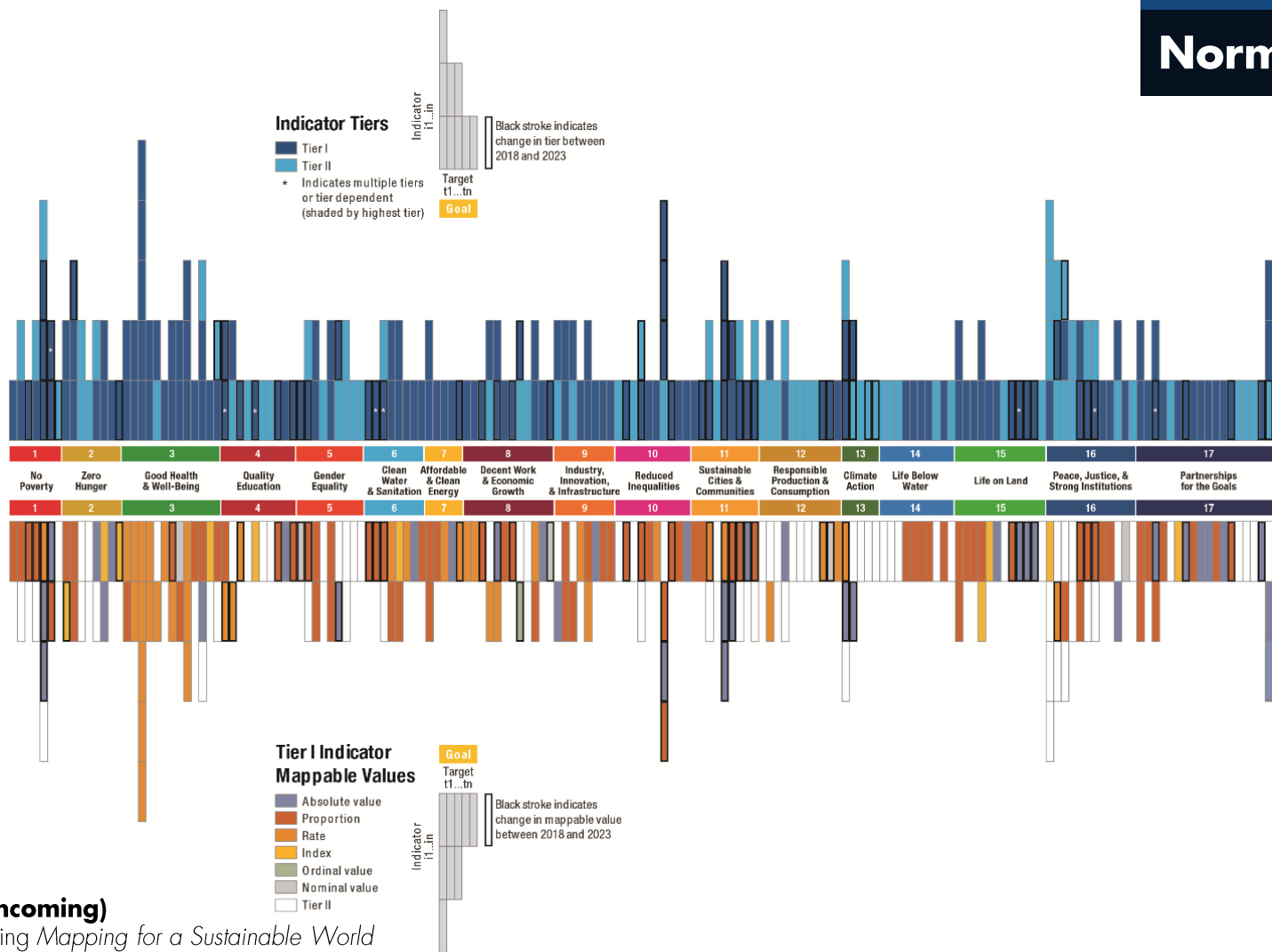
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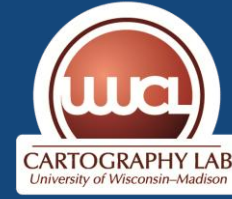
https://github.com/uwcartlab/Vienna-DC-2025

Normalization



Roth et al. (forthcoming)

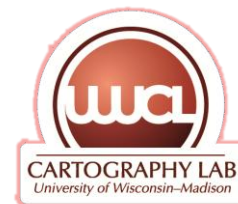
Making and Remaking Mapping for a Sustainable World



Unit 4: Design Sprint 2 | 12.05.25 16:00–17:30+

Sketching & Organizing Your Visual Story

Schedule at a Glance



Day 1 // Monday 12.05.25

- **Unit 1: Introductions & Inspiration Talks**
 - 10:00 to 10:15 // Arrival (and coffee!)
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- 13:00 to 14:00 // Lunch
- **Unit 3: Design Sprint 1, Obtaining Your Data, Researching Your Topic**
 - 14:00 to 15:30 // Team Work (organizers available for questions)
- 15:30 to 16:00 // Creative Break, the Marshmallow Challenge (facilitated by Robert Roth)

- **Unit 4: Design Sprint 2, Sketching & Organizing Your Visual Story**

- 16:00 to 17:30 // Team Work (organizers available for questions) *The goal is to have cleaned, mappable data in a GIS by day's end*
- 17:30+ // Continue Work As Needed (Optional)

Day 2 // Tuesday 13.05.25

- **Unit 5: Design Sprint 3, Polishing Your Visual Story**
 - 14:30 to 15:00 // Adobe Illustrator Demo
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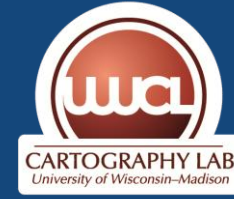
Unit 4 Objectives

1. Locate important places (i.e., key characters) to an SDG story
2. Compare and contrast how different places represent alternative solutions for an SDG
3. Sketch different layouts for an SDG atlas spread
4. Select an optimal layout based on the grid
5. Design appropriate thematic maps and supporting charts for an SDG story in GIS and statistical software

Important: The goal is to have cleaned, mappable data in a GIS by day's end!

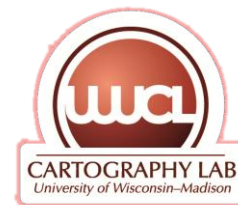
Important: Remember to take a screenshot of your Day 1 progress to Google Slides!





Unit 5: Design Sprint 3 | 13.05.25 14:30–17:00+
Polishing Your Visual Story

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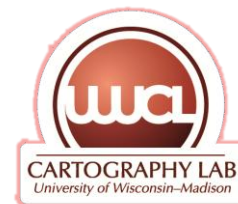
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 - 12:00 to 13:00 // Brainstorming Your Story (breakouts with Alicia Cowart, Merve Keskin, N.T. Nawshin, Robert Roth, Zdenek Stachon, and available TUW staff)
- 13:00 to 14:00 // Lunch
- **Unit 3: Design Sprint 1, Obtaining Your Data, Researching Your Topic**
 - 14:00 to 15:30 // Team Work (organizers available for questions)
- 15:30 to 16:00 // Creative Break, the Marshmallow Challenge (facilitated by Robert Roth)
- **Unit 4: Design Sprint 2, Sketching & Organizing Your Visual Story**
 - 16:00 to 17:30 // Team Work (organizers available for questions) *The goal is to have cleaned, mappable data in a GIS by day's end*
 - 17:30+ // Continue Work As Needed (Optional)

Day 2 // Tuesday 13.05.25

- **Unit 5: Design Sprint 3, Polishing Your Visual Story**
 - 14:30 to 15:00 // Adobe Illustrator Demo
 - 15:00 to 17:00 // Team Work (organizers available for questions) *The goal is to be out of GIS and into Adobe Illustrator by day's end*
 - 17:00+ // Continue Work As Needed (Optional)

Day 3 // Wednesday 14.05.25

- Initial Drafts Due in Google Slides by 16:00
- **Unit 6: Team Presentations & Draft Feedback**
 - 16:00 to 18:00 // Presentations
- 18:00+ // Social (!) *Recommendations appreciated!*

Day 4 // Thursday 15.05.25

- Second Drafts Due to Rob by 16:00
- PhD Student Show and Tell (N.T. Nawshin, Gareth Baldrice-Franklin) *Contact Rob if you are interested in giving a 5-10 minute lightning talk*
- 18:00+ // PhD Student Social (Alicia and Rob will crash!) *Recommendations appreciated!*

Vienna DC Submission

Use this form to submit your draft designs!

What is your map's title? *

Short answer text

Please list all of your map's authors in alphabetical order by last name. *

Long answer text

In one paragraph, introduce your SDG and how did your team want to approach it at the start of the Design Challenge? This can include why you selected the SDG as one of your top choices. Please make your answer roughly 200 words. *

Long answer text

In one paragraph, describe your design process during the event. What worked, what did not work. What did you learn along the way? Please make your answer roughly 200 words. *

Long answer text

In one paragraph, tell the story your final design tells about your SDG. What is the background context, the key problem or challenge, and the main conclusion of take-home point you want to make about the SDG? Please make your answer roughly 200 words. *

Long answer text

Please attach your .ai Adobe Illustrator file *

 Add file

 View folder

Please attach all of your design files including datasets and GIS project files as a .zip. This is to have a back-up copy in case you lose your work.

 Add file

 View folder

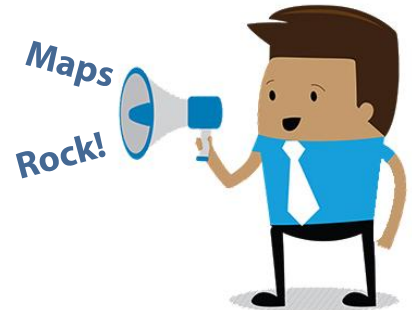


Unit 5 Objectives

1. Demonstrate tips and tricks for polishing mapping in graphic design software
2. Critique cartographic design decisions that are difficult to change outside of GIS
3. Explore different visual styles that reinforce an SDG story according to the form, color, type, and texture
4. Describe the creative design process behind an SDG atlas spread as an artistic statement
5. Develop an SDG atlas spread in graphic design software using the grid

Important: The goal is to be out of GIS and into Adobe Illustrator by day's end!

Important: Take a Day 2 progress screenshot; final draft screenshot should be added to Google Slides by **16:00 tomorrow** (Wednesday)



Tips for Mapping the SDGs

- 1. Generalization:** Use the UN Geodata Simplified boundary file for world and regional maps
- 2. Projections:** Use a recentered, equal area projection for most thematic maps
- 3. Typography:** Visually accent critical places ("characters") to your message
- 4. Symbolization:** Consider the appropriate visual metaphor evoked by your thematic map
- 5. Normalization:** Ensure choropleth maps use normalized indicators
- 6. Classification:** Consider the attribute distribution when selecting a classification scheme
- 7. Layout:** Use the grid to inform layout, seeking balance but asymmetry

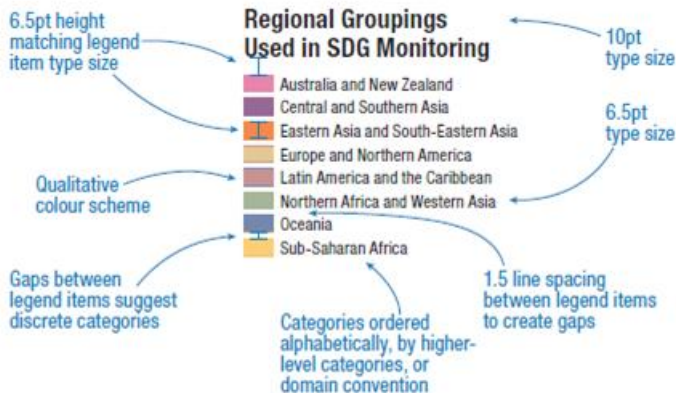
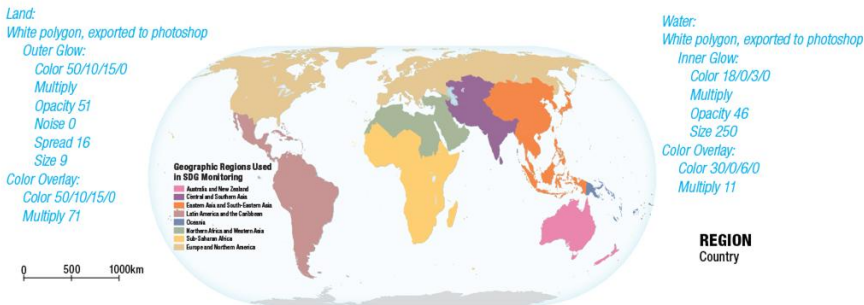


STROKES & CORNERS

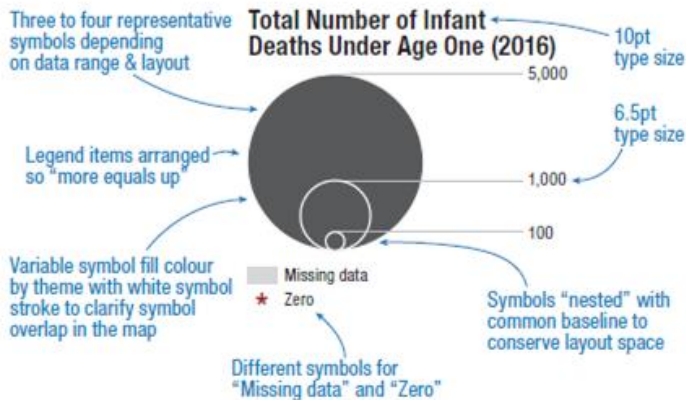
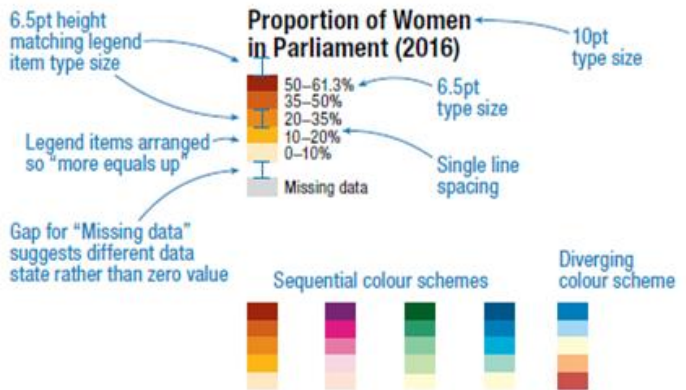
- country boundary (30K, 0.4 stroke width)
- figure arrowheads (100K, arrow 2, scale 80%, rounded cap)
- figure rounded rectangles (3pt corner radius)
- figure rounded rectangles outline (30K, 0.4 stroke width)

EXPORTS

- Place .AI files rather than image exports into layout
- Export a copy of maps/figures as .PDF on "high quality print" setting



Visual Style



Cartography often is described as both an art and a science. As an inter-governmental organization, the United Nations serves global *humanitarian* efforts, and its treatment of map design as an artistic process enables new ways to *humanize* the otherwise abstract statistical SDG indicator data.

As *visual art*, maps help us share our experiences of the world, promote empathy and compassion about uneven social and environmental conditions, and inform policy and politics for forging a sustainable future. As *visual culture*, maps reflect our interests and values, make us to confront our failures and prejudices, and reveal poten-

tial alternatives and opportunities.

The potential for artistic perspectives on cartography are numerous and ever expanding. One important way that artist elements enter every map design is through its *visual style*, or a cohesive set of design characteristics and qualities that reinforce rather than undermine the purpose of the map.

A visual style can be deconstructed into its elemental forms, colours, typefaces, and textures (Figure 2.14-1). *Form* describes the variable aspects of the geospatial linework, including its generalized detail (see Section 2.7) and line weights, cap and joint styles, and tapering. Colour and typography—de-

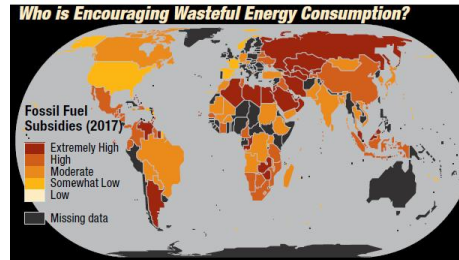
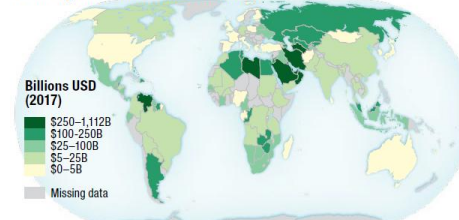


Figure 2.14-1: Visual styles in four web basemaps. **Left:** OpenStreetMap Mapnik uses detailed linework, complex colours and textures, and dense type, activating the audience and producing a sense of accuracy and trustworthiness in the basemap. **Left-centre:** Esri World Light Gray Canvas uses simplified linework and visual hierarchy along with muted colours, textures, and labels to deactivate the audience, making it a good basemap when additional data overlays should rise to figure (see Section 4.6). **Right-centre:** CARTO Dark Matter similarly uses simplified linework and sparse type and textures but with a darkened colour palette, setting an ominous, tense tone in the map. **Right:** Stamen Watercolor includes whimsical linework, bright colours with no type, and textures mimicking the spread of watercolours over paper, evoking a pleasant almost serene sense of beauty with the map.

Figure 2.14-2: Visual styles and emotion. Indicator 12.c.1 (2017) on per capita fossil fuel subsidies in USD is mapped in two different visual styles.

Top: The minimalist, authoritative style includes detailed linework and a soft colour palette. The map uses an arithmetic classification to emphasize low values in the left-skewed data distribution. Such a minimalist style likely evokes a pleasant, but deactivated emotional experience. **Bottom:** The alternative, creative style includes angular linework and bold colours against a black background, with the title framing the map purpose. Ordinal data are mapped with a quantile classification, placing more countries into the higher classes. Such a sensationalist style likely evokes an activated, unpleasant emotional experience, which may or may not be appropriate for the mapping context.

Per Capita Fossil Fuel Subsidies

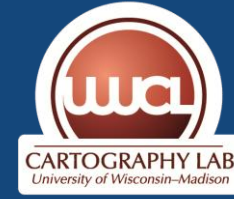


fined in Section 2.10 and Section 2.11 respectively—greatly influence the map's visual style. Both colour and typography evoke strong but different connotations cross-culturally. Texture, a visual variable (see Section 2.9), describes the additional pattern fills and overlays that add visual complexity and often suggest a physical materiality or historical pastiche.

The visual style of the map has a major impact on the audience's emotional experience: both setting the mood as a pleasant or unpleasant topic while also engaging or boring the reader. Understanding the relationship between style and emotion helps cartographers identify

a visual style that matches the gravity of the mapped phenomena while avoiding blatant map propaganda. A minimalist, authoritative style often is applied when mapping the SDG indicators, such as many of the maps in this book (Figure 2.14-2, top), to assert neutrality and suggest data accuracy. However, no map design is perfect nor free from subjectivity (see Section 2.3), and alternative, creative visual styles may better promote awareness, generate dialogue, and stimulate action (Figure 2.14-2, bottom).

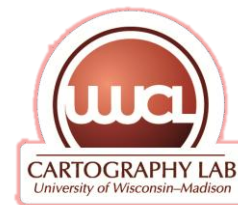
A visual style that works well in one mapping context may be illegible, misleading, or even harmful in others.



Unit 6: Presentations | 14.05.25 16:00–18:00

Sharing Your Squiggles!

Schedule at a Glance



Day 1 // Monday 12.05.25

- **Unit 1: Introductions & Inspiration Talks**
 - 10:00 to 10:15 // Arrival (and coffee!)
 - 10:15 to 10:30 // Introductions from Georg Gartner and Robert Roth
 - 10:30 to 10:45 // Inspiration Talk 1: The United Nations Sustainable Development Goals (Ayako Kagawa)
 - 10:45 to 11:00 // Inspiration Talk 2: Tips for Mapping the SDGs (Robert Roth)
 - 11:00 to 11:15 // Inspiration Talk 3: Making an Atlas in a Day (Alicia Cowart)
- 11:15 to 11:30 // Break
- **Unit 2: Team Formation**
 - 11:30 to 12:00 // Picking Your SDG (rank your top three SDG themes, narrow into small groups, facilitated by Robert Roth)
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Day 2 // Tuesday 13.05.25

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UNCERTAINTY / PATTERNS / INSIGHTS

CLARITY / FOCUS

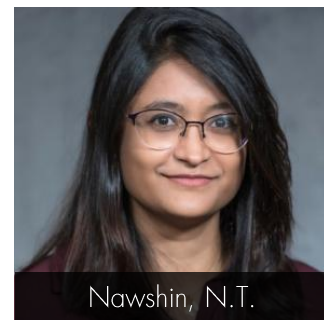
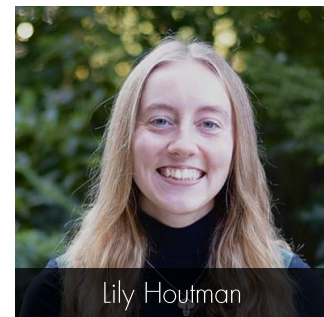
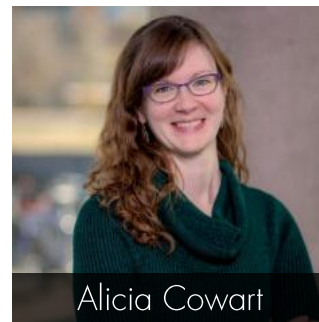


RESEARCH

CONCEPT PROTOTYPE

DESIGN

Organizers: T/U(X)W



Unit 6 Objectives

1. Deconstruct the creative design process into the most relevant milestones using visual screenshots
2. Report on the creative design process behind the SDG atlas spread
3. Recommend design revisions based on cartographic principles as constructive peer feedback
4. Implement peer feedback into an SDG spread
5. Produce a collective atlas of the SDGs using desktop publishing software

Important: Draft designs are due to Rob on Thursday 16:00!

